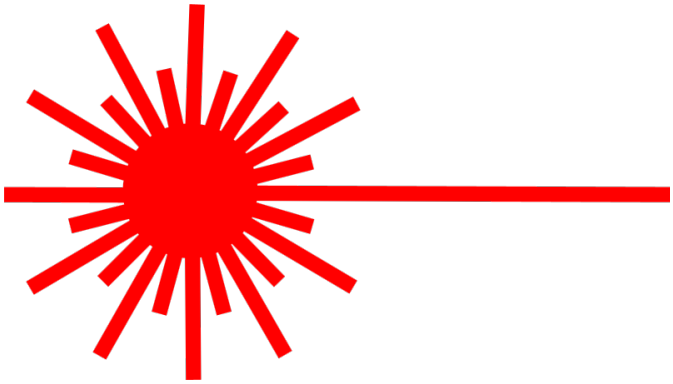
# Fiat Lux:
# A Physics Survey Course Focusing on Light

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# Chapter 1

# Light is Knowledge: Introduction

I see skies of blue / And clouds of white
The bright blessed day / The dark sacred night
And I think to myself / What a wonderful world

---

*What a Wonderful World*, Louis Armstrong
lyrics by George David Weiss and Robert Thiele

Our evolving models of light reflect our growing understanding of the physical universe. By focusing on this one phenomenon, I hope to illuminate many of the principles of modern physics.

Did you notice, in that first paragraph, how light metaphors ("reflect", "focusing", "illuminate") infuse our language? Light represents knowledge. **Fiat Lux** ("Let there be light", from Genesis 1:3), the motto of the University of California, describes a mandate to educate. Of our five classical senses, four can only tell you about your immediate environment (Fig 1.1), but sight allows you to experience the entire universe. Just by looking up on a clear night, you can see there's iron oxide on the surface of Mars. The redshift of distant galaxies tells us about the large-scale structure of spacetime. Look in the right places, and you see events that occurred long before the age of man and on the other side of the universe.



Figure 1.1: A dog's keen sense of smell does not help with astronomy.

As our insights deepen, our descriptions of light become more mathematically complex. So this book necessarily gets more difficult as it progresses, and I do not shy away from math, which is the true language of science. After Appendix A on physical constants and units, Appendix B reviews some relevant math, including complex numbers, calculus, and linear algebra. In colored

fonts throughout the book, I've included some parenthetical material which you can skip on first reading: expert-level comments, speculative physics, and philosophical diversions. I also suggest many links to **Wikipedia** articles, which are not necessarily stable but are often very informative, and can lead to much enjoyable Wiki-surfing.

This is not a popular science book. Though I have the greatest respect for authors (Carl Sagan, Brian Greene, Lisa Randall, Janna Levin, among many others) who convey key ideas in physics without math, to me that's like studying French Literature in English. Nor is it a textbook on electromagnetism, about which many much deeper books have been written.[1–4] My target audience is a physics undergrad with a strong math background and a desire to survey the landscape. When I was such a student entering MIT, I had taken solid high school courses covering ladders sliding down walls (mechanics) and electric and magnetic fields from charges and currents (electromagnetism). In a sense I expected college physics to be just more complicated sliding ladders and charge configurations. I didn't really understand that you can spend a semester (or a lifetime) studing solid state physics, cosmology, thermodynamics, fluid dynamics, low-temperature physics, special relativity, gravity, nuclear physics, particle theory, or many other topics which were unknown to me. This book is an attempt to introduce some of these subfields through their approaches to the single phenomenon of light.

Roughly speaking, the first five chapters are high-school level, chapters 6-11 are college level (for a physics major!), and chapters 12-17 are graduate level. The book is intended as a sort of survey course, but unavoidably it gravitates towards my own interests of cosmology (chapter 15) and particle physics (chapter 16), and therefore focuses on the prerequisites that take us there. For big-bang cosmology we will need special and general relativity, and thermodynamics. For quantum field theory (Feynman diagrams), we will need special relativity, Lagrangian mechanics, and quantum mechanics (including Heisenberg picture and path integrals). To study the particles and forces of the Standard Model, we need gauge theory and Yang-Mills.

I wish that I had the expertise to better cover the other subfields mentioned above. To keep myself honest, I have tried to confine speculative physics (which will have a very 90's vibe) to the penultimate chapter. For both of those reasons, string theory gets only a passing nod in chapter 17, even though it has driven most of the progress in particle theory for decades.

I want to acknowledge the invaluable assistance of my grad-school colleague Joshua W. Burton, who has remained a mentor and friend all these years. I have been fortunate to learn from many amazing teachers/professors, including (not limited to) Br. Robert Sullivan, CFX (SJP), Prof. Alan Guth (MIT), Prof. Bob Jaffe (MIT), Prof. Henry Kendall (MIT), Prof. Gene Commins (UCB), Prof. J.D. Jackson (UCB), Prof. Mahiko Suzuki (UCB), and Prof. Larry Hall (UCB). Also Prof. Barton Zwiebach (MIT), who created an excellent high-level edX course on quantum mechanics[5], as well as a college-level string theory textbook[6].

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# Chapter 2

# Light is Geometry: Astrometrics

Light travels in straight lines, so some simple observations tell us a lot about the geometry of the sun, Earth, and moon. With just a little more work, we can derive Newton's Law of Gravity.

## 2.1 Eratosthenes: How Far is the Moon?

The ancient Greeks didn't know much about light, but they knew it travels in straight lines and casts shadows. Plato famously compared our world to the shadows on the walls of a cave, a pale reflection of some unseen ideal world. Curiously, the Greeks were able to perform some impressive astrometrics (the measure of astronomical distances) armed just with this simple knowledge and some clever observations. **Eratosthenes** of Cyrene (276 BC - 195 BC), the chief librarian of the Library of Alexandria, calculated the distance to the moon by observing a lunar eclipse. Let's go through his reasoning.

While observing the moon, extend your arm to full length and compare the apparent size of your finger to the apparent size of the moon. I find the moon is about half the width of my little fingernail. That's about 5 mm at 60 cm distance, or 1/2 a degree of angle. By similar triangles (Fig 2.1), the ratio of the moon's distance to the moon's diameter is about 120:1.

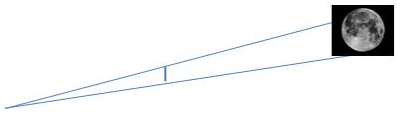


Figure 2.1: Similar triangles

How do we know the diameter of the moon? During a lunar eclipse, the Earth's shadow falls on the moon, and the shadow is approximately the same size as the Earth itself. That claim assumes the sun is much further away than the moon, which we know because when the sun and moon are 90° apart, the moon is almost exactly half full. Also, the finite size of the sun means the shadow is a fuzzy line; the innermost region is the umbra, where ***all*** of the sun's light is blocked, while the outermost region is the penumbra, where ***some*** of the sun's light is blocked. For this analysis, by "shadow" I mean the imaginary line halfway between the umbra and the penumbra. That being said, the Earth's shadow describes an arc on the face of the moon which is about 4x the size of the moon (Fig 2.2), and thus the moon's diameter is about 1/4 of the Earth's diameter.

How large is the Earth? Eratosthenes knew the Earth is a sphere, because it always casts a circular shadow on the moon during lunar eclipses. He also knew that in the Egyptian city of



Figure 2.2: Lunar Eclipse 8/16/2008, composite photo by Anthony Ayiomamitis

Syene, on one day a year, the sun was directly overhead. In today's language, Syene (now called Aswan) sits at the intersection of the Nile River and the Tropic of Cancer, and the sun reaches the zenith on the summer solstice. Since the level of the Nile was so critical to farming, the Syene government had built a deep well, called a Nilometer, down to the water table. On the summer solstice, if you hung your head over the edge of the Nilometer, you would see your reflection in the water surrounded by a halo of light, from the sun directly over your head. Eratosthenes worked in Alexandria, also on the Nile (and thus directly north of Syene). The highest the sun ever got in Alexandria was about 7.2° off the vertical, easily measured by looking at the shadow of a vertical stick in the ground. Eratosthenes thus concluded that (in modern language) Alexandria is 7.2° of latitude north of Syene, or 0.125 radians. (To convert degrees to radians, multiply by $\pi/180$.) The distance between Alexandria and Syene, then, is $0.125\,R_E$, and Eratosthenes measured this distance (by pedestrian means, perhaps literally) to be 800 km (500 miles). Thus $R_E$ is 6,400 km (4,000 miles), which is remarkably accurate! (You should check out Carl Sagan's version of this story from his TV series "Cosmos", which you can find on YouTube – just Google "Sagan Eratosthenes".)

Putting these results together, the moon is $120 * \frac{1}{4} * 2 * 6,400 = 384,000\,\text{km}$ (240,000 miles) away. Eratosthenes is believed to have gotten it right to within a few percent.

For a sense of scale, that's about how far you can drive a car in its lifetime. You could imagine driving that far in a few years, if not for the fact that it's straight up, and there's no air, and there are no gas stations along the way. But we did send men in a rocket in 1969, and they left behind Lunar Laser Ranging Reflectors, which reflect light back the way it came (same geometry used in reflective street signs). So now, by reflecting lasers off the moon, we can measure its distance to an accuracy of centimeters, and also see that it's moving away from Earth at about 4 cm/yr. That's a known consequence of tidal drag; as the ocean tides slowly rob energy from the spinning Earth and make the day grow longer, that energy goes into flinging the moon further away. There will some day be a Last-Ever Total Eclipse of the Sun (but don't worry, it's a half-billion years away).



Figure 2.3: The author visits the Aswan Nilometer

## 2.2 Columbus

You might be surprised to know that Eratosthenes knew the Earth is a sphere of radius 6,400 km. Why, then, did Columbus think he could reach India by sailing west? Sadly, Eratosthenes's accurate analysis was later overridden by Posidonius, who came up with a smaller value, and Ptolemy published this result. With the fall of the Roman Empire and the ensuing Dark Ages, nobody repeated (and published) the analysis for a thousand years. We're not sure why Posidonius got it wrong; he may have had the wrong distance between Alexandria and Rhodes, or used different units (they used "stadia", not kilometers or miles). But one theory is that Posidonius viewed starlight (Canopus) down a narrow opening, and did not account for the effect of diffraction (which the Greeks didn't know about). If so, we could say that Columbus's discovery of America was also due to a property of light!

## 2.3 Classical Interlude I: Newton's Laws of Motion

For the next section, we need some basic classical mechanics. Newton's Second Law is

$$\boldsymbol{F} = m\boldsymbol{a} \tag{2.1}$$

where $\boldsymbol{F}$ is the total force on an object of mass $m$, and $\boldsymbol{a} = \ddot{\boldsymbol{x}}$ is its acceleration (second time derivative of position). Both $\boldsymbol{F}$ and $\boldsymbol{a}$ are vectors, which have magnitude and direction; hence the acceleration points in the same direction as the force. For a circular orbit with angular velocity $\omega = v/r$, the acceleration $a = \omega^2 r$ and points radially inward (see Appendix B.1). That is, the orbiting body (moon) must be attracted to the central body (Earth) by a force $F = m\omega^2 r$.

Newton's First Law is also encoded in $\boldsymbol{F} = m\boldsymbol{a}$, as the statement that if $\boldsymbol{F} = 0$ then $\boldsymbol{a} = 0$, and so $\boldsymbol{v}$ (velocity) is constant. Equivalently, momentum $\boldsymbol{p} \equiv m\boldsymbol{v}$ is conserved.

Newton's Third Law says that if A exerts a force $\boldsymbol{F}$ on B, then B exerts force $-\boldsymbol{F}$ on A ("equal and opposite"). That's particularly important because if you draw a line around both A and B, and call them a "system", then the internal forces (A on B and B on A) net out to zero. In the absence of external forces, the momentum of the system is conserved. A rubber ball may consist of bazillions of atoms with strong interatomic forces, yet we can treat the ball as a simple system and use $\boldsymbol{F} = m\boldsymbol{a}$ with only the external forces (like gravity) acting on the ball.

## 2.4 The Apple and the Moon

Until Sir **Isaac Newton**, falling apples (Galileo) and pendulum clocks (Huygens) bore no apparent relation to the motion of heavenly bodies (Tycho Brahe, Johannes Kepler). Newton showed that the same law of gravity governs motion both terrestrial and celestial, which was a philosophical leap that I'll call the **First Unification**.

Newton's Law of Gravity is:

$$\boldsymbol{F} = G\frac{m_1 m_2}{r^2}(-\hat{\boldsymbol{r}}) \tag{2.2}$$

The $(-\hat{\boldsymbol{r}})$ just describes the vector direction: mass 1 attracts mass 2 (and vice-versa). There is a physical constant $G$ (Appendix A) which wasn't known at first, but fortunately wasn't needed for this argument. The $1/r^2$ dependence is very important: since the Moon is 60x further from the center of the Earth as an object on Earth's surface, Earth's gravity is $(60)^2$ times weaker there.

The fact that distances should be measured from the center of the Earth is actually the subtlest point, and required Newton to invent Calculus first. It's a classic result (requiring integration over spherical coordinates) that any spherical shell of uniform density produces a gravitational field (outside the shell) equivalent to what you would get if all the mass were concentrated at the center. Then the same can be said of a planet, which is roughly made up of concentric uniform-density shells (it doesn't matter that the inner shells may be denser than the outer ones). We'll just assume this result, and so for any object at Earth's surface we use $r = R_E = 6,400\,\text{km}$.

We apply Newton's Second Law, Eq. 2.1, and Newton's Law of Gravity, Eq. 2.2, to a falling apple. On the left side, $m_1 = M_E$ (mass of the Earth), $m_2$ is the mass of the apple, and $r = R_E$ as discussed in the previous paragraph. On the right side, the mass of the apple appears again, and its acceleration is the measured value $g = 9.8\,\text{m/s}^2$. So:

$$G\frac{M_E m_2}{R_E^2} = m_2 g \tag{2.3}$$

The mass of the apple $m_2$ drops out! *Everything* at Earth's surface accelerates downward at the same rate $g$ (ignoring air friction), which is what Galileo was telling us with his Tower of Pisa demonstrations, and now we know $g = GM_E/R_E^2$.

You can measure $g$ directly by dropping things, but it's more accurate to use a pendulum, whose period $T = 2\pi\sqrt{L/g}$. In fact, Chritiaan Huygens originally proposed defining a "meter" as the length $L$ of a pendulum with period $T = 2\,\text{s}$, in which case we'd have $g = \pi^2$ by definition; it didn't quite work out that way but it's close.

Now we apply Newton's Law of Gravity to the Moon, and use our result for orbital acceleration:

$$\frac{GM_E M_M}{r_M^2} = M_M \omega^2 r_M \tag{2.4}$$

And the mass of the Moon drops out! In the 17th century we didn't know $G$ or $M_E$, but we have $GM_E = gR_E^2$ from the apple. So the Moon's orbital period $T = 2\pi/\omega$ is:

$$T = 2\pi\sqrt{\frac{r_M^3}{gR_E^2}} = 27.3\,\text{days} \tag{2.5}$$

which is exactly right! That is approximately the time between full moons, and (even more roughly) the length of a "month". Between Eratosthenes and Newton, we have derived the orbital period of the Moon!

Really you'd like to start with observations and derive laws, so let's turn that around, now using the Moon's orbital period $T = 2\pi/\omega = 27.3\,\text{days}$ as an input. Newton's Law of Gravity is constrained by linearity arguments[1] to look like Eq. 2.2 except for the $r$-dependence, so let's say it's $F = Gm_1m_2/r^n$, where $n$ is an unknown exponent. Then we'd find:

$$n = \ln\left(\frac{g}{\omega^2 r}\right) / \ln\left(\frac{r}{R_E}\right) = 2.00 \tag{2.6}$$

and we have derived the inverse square law from measurements of the Moon!

## 2.5 Planetary Orbits and Electron Orbitals

We treated the Moon's orbit as a perfect circle around a fixed Earth, and that was good enough for our purpose. But let's say a little more about planetary orbits, because we will later make comparisons to electron orbitals in an atom. This whole subsection is "expert-level" and can be skipped if you like.

First, since the Earth is not infinitely massive, really the Earth and Moon both orbit around their common center of mass. Earth is about 80x the mass of the Moon, so that center of mass is actually located a little below the Earth's surface. You might think that analyzing this two-body system ($\boldsymbol{r}_M$ and $\boldsymbol{r}_E$) is much harder than a one-body system ($\boldsymbol{r}_M$ in a central potential), but a simple change of coordinates helps. The new coordinates are $\boldsymbol{r}_{CM}$ (the system's center of mass) and $\boldsymbol{r} \equiv \boldsymbol{r}_M - \boldsymbol{r}_E$ (the vector from Earth to Moon). Then the dynamics of $\boldsymbol{r}$ are right back to the one-body problem, except that we replace the Moon's mass with the **reduced mass** $M_M M_E/(M_M + M_E)$, which is approximately 1% smaller than the Moon's mass. That's it! There's just a 1% correction to the numbers derived above, that comes from replacing the Moon's mass by the reduced mass. Similarly, a precise calculation of hydrogen energy levels requires replacing the electron mass by the reduced mass of the electron plus proton, which is only about a 0.05% correction.

Second, most orbits are not circles, but rather ellipses, with the center of mass at one focus. We won't do it, but a neat way to derive this is by working in the rotating frame of the perfect circular orbit, where the gravitational force towards the Earth is offset by the "fictitious" centrifugal force that appears in the rotating frame. The net potential of these two forces has a minimum at the radius of the perfect circular orbit. Now perturb the Moon radially a bit, and you'll find it oscillates back and forth in this potential at exactly twice the frequency of its orbit, which turns the circle into an ellipse. The eccentricity of planetary orbits corresponds in quantum theory to the difference between s, p, d, *etc.* electron orbitals, with the highest angular momentum orbitals being the most "circular", and the lowest (s) orbitals the most eccentric.

An ellipse is characterized by the direction of its major axis, and that remains stable in classical Newtonian gravity. More formally, the **Laplace-Runge-Lenz vector** defined as $\boldsymbol{A} = \boldsymbol{p}\times\boldsymbol{L} - mk\hat{\boldsymbol{r}}$ is conserved for a central $1/r$ potential – we say it is "accidentally" conserved, because it depends on the particular form of the potential, and does not derive from a symmetry under Noether's Theorem (Appendix C). The Laplace-Runge-Lenz vector points along the direction of the ellipse's major axis, and since it is conserved it does not rotate (precess). Conservation of the Laplace-Runge-Lenz vector in the hydrogen atom system is responsible for the high degree of degeneracy in electron orbital energies, such as 2s and 2p having the same energy.

[1] Cleave mass 1 so that $m_1 = m_{1A} + m_{1B}$; the forces from the two pieces add linearly, so force must be linear in $m_1$. By Newton's Third Law, the same must be true of $m_2$.

Conservation of the Laplace-Runge-Lenz vector is broken by anything that alters the $1/r$ potential, such as the presence of other orbiting bodies. Some of Mercury's orbital precession is due to Jupiter. 2s and 2p orbitals are not quite degenerate in multi-electron atoms. These are the same phenomenon!

In the early 20th century, the precession of Mercury's orbit due to Jupiter (and other planets) was calculated, and it still did not quite match the measurement. The remaining **anomalous precession of Mercury** ($43''$/century) was successfully explained by Einstein's Theory of General Relativity, which also slightly alters the gravitational potential away from $1/r$.

# Chapter 3

# Light is Color: Trichromatism

In physics, "color" means the frequency of an electromagnetic wave. But our perception of color relates more to the physiology of the eye, which has three kinds of cone receptors.

## 3.1 Intensity and Wavelength

Light is an electromagnetic wave, as we'll explain in a later section. For now, if you took a snapshot of the fields in a monochromatic (single-color) beam of light, it would look like Fig. 3.1, and the two numbers we'll need to describe it are the **intensity** $I = |E_{\max}|^2$ (the square of the maximum electric field), and the **wavelength** $\lambda$ (the physical distance between peaks). Intensity is brightness, and wavelength is color.

Furthermore, *any* beam of light can be written as the sum of those monochromatic sinusoids. Since intensity is a measure of energy, the total intensity of the beam is the sum of the intensities of its components.

The process of decomposing a complicated periodic function into a sum of sinusoids is called **Fourier analysis**. A plot of intensity vs wavelength is the square of the Fourier transform of the original wave, also summed over polarizations. The fact that electromagnetic fields add linearly arises from the Abelian nature of the U(1) symmetry group.

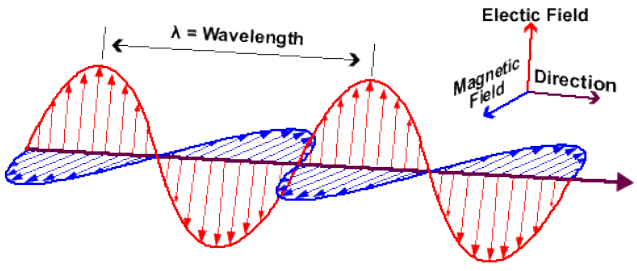
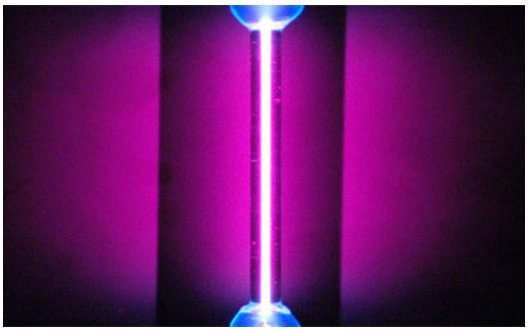


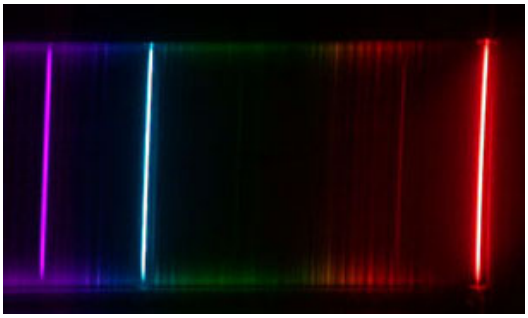

Figure 3.1: A monochromatic electromagnetic wave

The visible spectrum ranges in wavelength $\lambda$ from 380 nm (violet) to 740 nm (red). A nanometer (nm) is $10^{-9}$ m, making the wavelengths of visible light roughly 100x smaller than the width of a human hair.

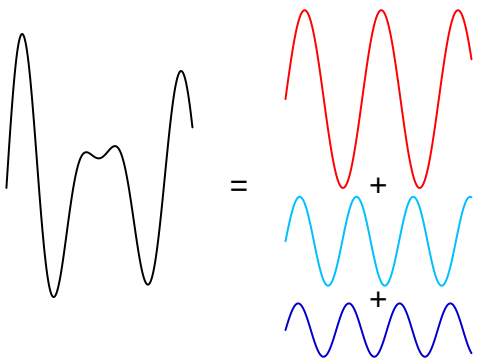
Fig. 3.2 shows the lavender glow emitted by a tube of hot hydrogen plasma. If we send that light through a **spectrometer**, it splits into several distinct lines at 656 nm, 486 nm, and 434 nm (there are more that I haven't shown). These are the first 3 lines of the **Balmer series**, which we will derive later from the quantum structure of the hydrogen atom.



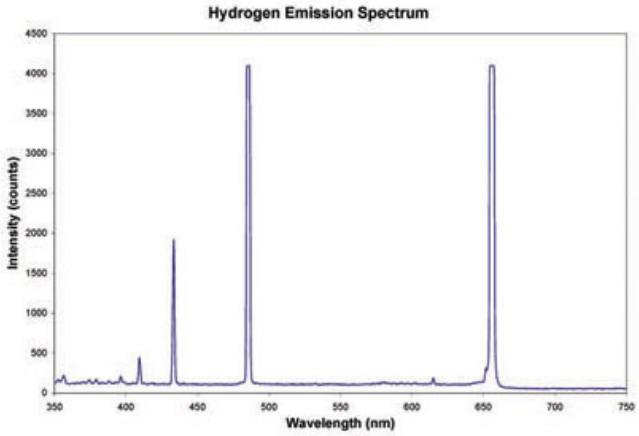
(a) Lavender glow of hydrogen plasma



(b) As seen through a spectrometer



(c) Waveform is a sum of sinusoids




(d) Intensity vs wavelength

Figure 3.2: Hydrogen's lavender glow arises from a few monochromatic components

## 3.2 Cones in the Human Eye

The **emission spectrum** of Fig. 3.2b, or equivalently the plot of intensity vs wavelength in Fig. 3.2d, is a full description of the light that reaches our eyes. A lot of chemistry and astrophysics involves running light through a spectrometer and analyzing spectra like this. But unfortunately the human eye is not a spectrometer – if it were, we'd all be experts at chemical analysis! Instead, the eye contains 3 types of **cones**, each of which is sensitive to a different range of wavelengths, and our brains register a single output level $x$ from each type. For simplicity, call them the "Red", "Green", and "Blue" cones, as that's where the sensitivity of each one peaks, but remember each one covers a broad range of wavelengths (and their more proper names are "L", "M", and "S", for long, medium, and short wavelengths). Fig. 3.3 shows the response curves of the three types of cones. Mathematically, the response $x_B$ of (say) a blue cone is the **convolution** of its response curve $R_B(\lambda)$ with the incoming light spectrum $S(\lambda)$: $x_B = R_B * S \equiv \int R_B(\lambda)S(\lambda)\mathrm{d}\lambda$. Convolution is an example of a **functional**, an operation that takes a function $S$ as input and returns a single number $x$.

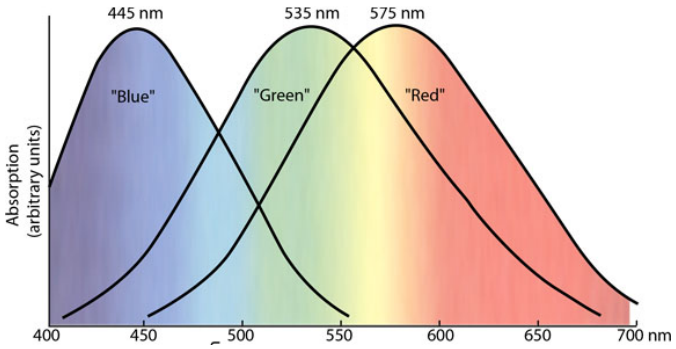



Figure 3.3: Cone Cell Responses

So the brain gets three numbers: $(x_R, x_G, x_B)$, and must interpret the result as a "color". Physiologically "color" is a 3-dimensional phenomenon; humans are **trichromats**. Note that's nothing to do with the physical nature of light, it's just that we have three types of cones. (The eye also contains cells called rods that take over in dim light, and since there's only one type of rod, we become monochromats.) Colorblind people may be dichromats. Many birds and pollinating insects are known to be tetrachromats, their range extending into the ultraviolet, experiencing a 4-D color space we can't even imagine, see **tetrachromacy**. There are some rare women (because it needs two X chromosome variants) who have a weak form of tetrachromacy, and people lacking a UV-blocking lens ("aphakia") can see deeper into the ultraviolet.

Philosophers ask whether our subjective experiences of color, or **qualia**, are the same. Is my "blue" the same as yours? Certainly they are not for people with color-blindness, tetrachromacy, or aphakia. In The Odyssey, Homer never uses the word "blue" ($\mu\pi\lambda\epsilon$), even though it takes place on the sea – which he calls "wine-dark". Jules Davidoff's 2006 study of the Himba tribe of Namibia, who have no word for blue, showed they had difficulty distinguishing green from blue, but could perceive shades of green that were very subtle for English speakers. How much does language influence our perception of color?

Note that the human eye has a pretty weird design flaw: the light-sensitive cells lie *below* the "wiring" (nerves) and blood vessels. The nerves and blood vessels have to pass through the **optic disk**, a hole in the retina, to reach the brain, and so we have a blind spot in each eye. Also, shadows of the nerves and blood vessels corrupt the image. In "post-processing", the brain has learned to ignore parts of the image that don't move, so we are not conscious of these defects. But now we'd have trouble seeing still objects, so the eye also jitters around several times a second in order to keep the real image moving. It's kind of a crazy system!

Let's distinguish **spectral colors**, or colors of the rainbow, from **perceived colors**. Spectral colors form a 2-dimensional space (intensity and wavelength), while perceived colors form a 3-dimensional space (due to our 3 types of cones). Spectral colors are single-frequency electromagnetic waves, while perceived colors may be generated by mixes of frequencies. The number of spectral colors (red, orange, yellow, green, blue, indigo, violet) is purely conventional, and is due to Isaac Newton wanting to make some analogy to the 7-note musical scale; these days you don't hear much mention of "indigo" which is just dark blue. And note that "violet" is a spectral color (the shortest visible wavelengths), but "purple" is a perceived color mix of red and blue. "Brown", "pink", and "lavender" are other examples of perceived colors not in the rainbow.

I want to emphasize that for any given perceived color, there are (infinitely) many ways to generate it. "Yellow", for example, could come from a monochromatic beam of wavelength 570 nm, or a combination of red and green pixels, or the broad thermal spectrum of a 5000 K hot

object (like a flame or the sun), or any other spectrum that generates the same three ($x_R$, $x_G$, $x_B$) responses. Trichromatism is great, but it's still no match for a full spectrographic analysis! Conversely, there are hypothetical cone responses that cannot be generated by any real spectrum. Imagine artificially stimulating the Green cones, for example, without exciting the Red or Blue ones. What would "hypergreen" look like?

People have several ways of mapping that 3-dimensional (trichromatic) perceived color space:

- **RGB** gives a value for red, green, and blue components that roughly correspond to the peak sensitivities of our three kinds of cones (Fig. 3.3), and also to the three kinds of pixels in a color television or monitor. Lavender, for example, is represented in RGB hex format as #E6E6FA. The first two hex digits E6 correspond to 230 in decimal, so the red value is 230/255; green is 230/255, and blue is 250/255. Red, green, and blue are also known as the **additive primaries**.

- **CMY** uses the **subtractive primaries** Cyan, Magenta, and Yellow, which are the typical color inks (or toners) used in printing. Starting with white paper, roughly speaking cyan ink subtracts red from it, magenta ink subtracts green, and yellow ink subtracts blue. Applying all three gives black, but that's inefficient for basic black-on-white printing, so printers also have black ink "K". Thus the "CMYK" color model is the practical standard (and you need to buy 4 ink cartridges, not 3).

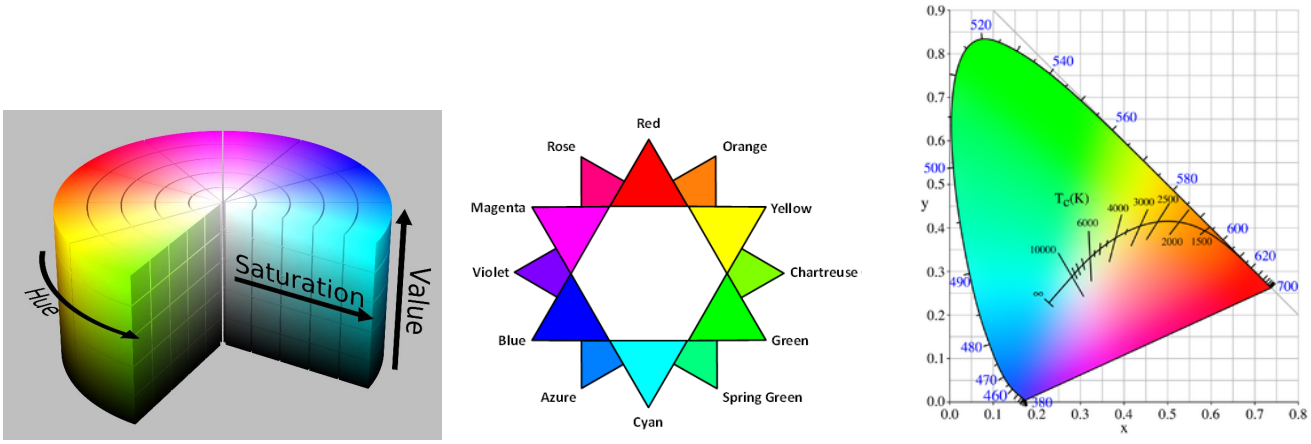
- **HSV** stands for Hue/Saturation/Value, see Fig. 3.4a. In cylindrical coordinates, hue (the color wheel) is the angular coordinate $\theta$, saturation (gray vs vibrant) is the radial coordinate $r$, and value (intensity) is the azimuthal coordinate $z$. Some variations on the idea are HSL (Hue, Saturation, Lightness) and YIQ (Luminance and Chromaticity); see **HSL and HSV**. Color television signals use models like these (YIQ is the NTSC standard) because the old black-and-white televisions can just read the one intensity channel and ignore the other two.

- The rim of the HSV cylinder ($S = V = 1$) roughly matches the classic **color wheel**, Fig. 3.4b. The additive primaries form a triangle, and the subtractive primaries are opposite to them. From red to violet (the long way) we see the **spectral colors** (wavelengths 740 nm to 380 nm), but then we get magenta and rose (and purple, not shown), which stimulate the red and blue cones but not the green cones, something no pure spectral color can do. This is more explicit in the **CIE 1931 color space** of Fig. 3.4c, where the wavelengths of spectral colors are labelled along the upper curve, but not on the **line of purples** (bottom straight edge).

- The **opponent process** theory says that the brain is wired to detect *differences* between cone outputs (and maybe rods?), specifically red-vs-green, blue-vs-yellow, and black-vs-white. This seems to correspond better to the deficiencies typically seen in color blindness, and also explains afterimages (*e.g.* stare at yellow then look away, and you'll briefly see blue) and many optical illusions.

## 3.3 Thermal Radiation and Blackbody Spectra

A hot low-pressure gas (or "plasma", if it's ionized), like the hydrogen in Fig. 3.2a, usually emits distinct spectral lines that correspond to specific electron level transitions. One example is sodium




(a) The HSV cylinder (b) The Color Wheel (c) CIE and Planckian Locus

Figure 3.4: The HSV Cylinder and Subsets

vapor, which has two closely-spaced yellow lines, and is common in (pre-LED) street lamps. Another is mercury vapor, which has a more bluish-white color, and is also common in street lamps. A **fluorescent lamp** is a mercury-vapor lamp with a phosphor coating to alter its spectrum. If you view a fluorescent lamp through a spectrometer, you can still detect the distinct mercury lines from light that bypassed the phosphor.

Solids, on the other hand, generally emit a broad band of radiation when they get hot. The tungsten filament of an incandescent light bulb is an example. A **blackbody** is the limiting case of an object that can absorb and emit all wavelengths, and in that limit the emission spectrum is completely determined by one parameter: its temperature $T$ (measured in Kelvin "K"). The spectrum of **black-body radiation** is given by **Planck's Law**, Eq. 11.13 and Fig. 11.1. Its peak moves to shorter wavelengths (higher frequencies) as $T$ increases. A blackbody at $T = 1500\,\mathrm{K}$ is "red-hot"; it literally glows red. The sun's surface is approximately a blackbody at $T = 5780\,\mathrm{K}$; it is "white-hot" (we think of the sun as yellow, and the sky blue, only because the atmosphere scatters some of its bluer light). The Earth emits infrared radiation corresponding roughly to $T = 254\,\mathrm{K}$. As the temperature increases, the blackbody peak goes through red, orange, yellow, white, and blue, as shown in the CIE **Planckian locus** of Fig. 3.4c.

Fire, such as a candle flame, is actually a little of both. The blue part of a flame is usually generated by particular transitions in molecular radicals. The orange-yellow part is usually blackbody radiation from hot soot, solid though microscopic, at around $T = 2000\,\mathrm{K}$. Varying the oxygen supply can determine how much soot is produced, and therefore which color dominates.

When you buy a light bulb, it's characterized by a "color temperature" marking the Planck spectrum that most closely matches the bulb's output, Fig. 3.5. Ironically, the low temperatures (red, orange, yellow) are called "warm" colors, while the high temperatures (white, blue) are called "cool", which seems backwards, doesn't it?

In real life, we rarely see anything "blue-hot" (blackbody with $T \geq 10,000\,\mathrm{K}$), except maybe lightning bolts. (Note that a blue methane or butane flame is not emitting blackbody radiation but rather specific hydrocarbon spectral lines.) Instead, most blue things we encounter are cold: the sky, the ocean, ice; so I guess that's why we associate blue with "cold". The sky is blue due to **Rayleigh scattering**, in which tiny particles in the air (much smaller than the wavelengths of visible light) scatter blue light from the sun; it's the same reason the sunset appears orange-red, as those are the rays from the sun that are not scattered. The sea appears blue both because water

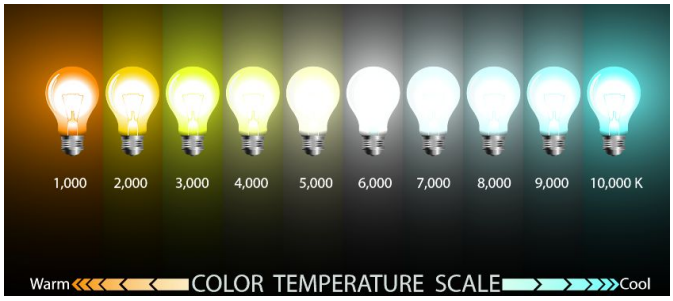



Figure 3.5: Color Temperature Seems Backwards

absorbs some red light, and because the sea reflects the blue sky.

### 3.3.1 The Electromagnetic Spectrum

"Visible light" has wavelengths between 380 nm (violet) and 740 nm (red), but often I will use "light" as a synonym for **electromagnetic radiation** of any wavelength. Fig. 3.6 shows a much broader spectrum, and the names given to various ranges. Note that you can parametrize the spectrum by wavelength $\lambda$ in meters (m), or by frequency $\nu$ in Hz, or even by photon energy $E$ in eV, as they are related by simple formulas $\lambda = c/\nu$ and $E = h\nu$, where $c$ is the speed of light and $h$ is the Planck constant, see Appendix A. Violet is 380 nm, or 790 THz, or 3.3 eV. Red is 740 nm, or 400 THz, or 1.7 eV. Just remember that **shorter** wavelengths mean **higher** frequencies or energies.

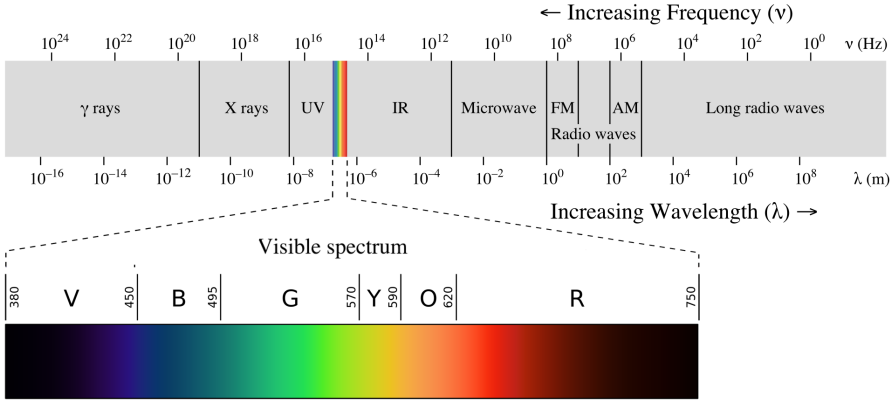



Figure 3.6: Electromagnetic spectrum

The visible range is noteworthy for a few reasons. First, it corresponds to the peak output of the sun, whose surface is at roughly $T = 5780\,\mathrm{K}$. Second, these photon energies $E = h\nu$ correspond to typical electron transitions, whose scale is set by the **Rydberg energy** $Ry = \alpha^2 m_e c^2/2 = 13.6\,\mathrm{eV}$. Third, both the universe and our atmosphere are nearly transparent to it.

Shorter wavelengths (higher energies) than violet are called "ultraviolet", and then "X-rays", and then "gamma rays". As you go up in energy, this radiation generally penetrates deeper and/or causes more damage to biomatter. In the other direction, longer wavelengths (lower energies) than

red are called “infrared”, then “microwaves”, then “radio waves”. Radio waves have macroscopic wavelengths and interact with macroscopic antennas, and surround us all the time harmlessly.

# Chapter 4

# Light is a Wave

Light is an electromagnetic wave, characterized by its direction, polarization, amplitude, frequency, and phase. Its properties give us an early glimpse of quantum behavior. We compare light with sound, which is another wave phenomenon.

## 4.1 Evidence for the Wave Nature of Light

The wave nature of light is clearly evident in the **light double-slit experiment**, Fig. 4.1. Monochromatic light from a single source (or maybe a beam-splitter) passes through two slits, and then hits a screen. At some points on the screen, the two paths to get there differ by a whole number of wavelengths, and so the two beams add (constructively interfere) to create a bright spot. At other points on the screen, the two paths differ by a half-integer number of wavelengths, so the two beams cancel (destructively interefere). The fringe pattern that results, along with an analysis of the geometry, tells you the wavelength of the light.

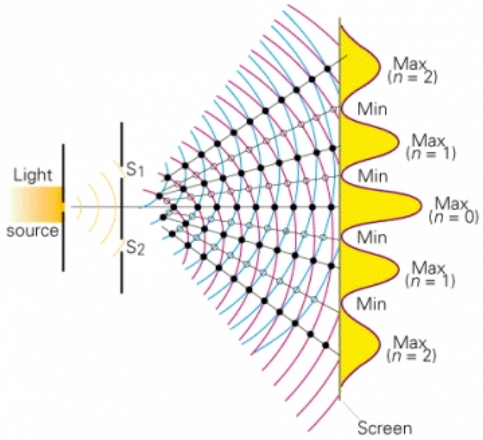



Figure 4.1: Light double-slit experiment

## 4.2 Polarization

Let's return to Fig. 3.1. The light is travelling to the right, which we'll call $\hat{\boldsymbol{z}}$, at speed $c = 3 \times 10^8$ m/s. The **electric field** (red arrows) is a vector $\boldsymbol{E}(\boldsymbol{x}, t)$ at every point in space and time,

representing the force that would be felt by a unit charge if one were placed there. Like any vector, $\boldsymbol{E}$ has a magnitude and a direction. In light, the electric field is always perpendicular to the direction of motion – it is a **transverse wave** – so $\boldsymbol{E}$ must lie in the x-y plane. In the figure, this electric field points up and down, call that $\hat{\boldsymbol{x}}$, and we say it is **linearly polarized** in the $\hat{\boldsymbol{x}}$ direction. We still won't say much about the magnetic field, except that it is perpendicular to both the electric field and the direction of motion (so $\hat{\boldsymbol{y}}$ here), and it is completely determined by the electric field. When we get to Maxwell's Equations, we'll see that changing electric fields generate magnetic fields, and those changing magnetic fields generate electric fields, and that (roughly speaking) is how the combination propagates.

Now you could also have light propagating along $\hat{\boldsymbol{z}}$ but with the electric field along $\hat{\boldsymbol{y}}$ instead of $\hat{\boldsymbol{x}}$. It would look like Fig. 3.1 but with the red and blue arrows switched. That would be linearly polarized in the $\hat{\boldsymbol{y}}$ direction.

So there are two independent **polarizations**, namely $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$, which form a **basis** for any other polarization you might imagine. For example, if your electric field points 45° between $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$, that's just the sum of an x-polarized wave and a y-polarized wave: $\boldsymbol{E} = E(\hat{\boldsymbol{x}} + \hat{\boldsymbol{y}})/\sqrt{2}$.

We're ready to write down a full mathematical description for a pure, coherent laser beam of monochromatic light traveling through vacuum in the $\hat{\boldsymbol{z}}$ direction and polarized in the $\hat{\boldsymbol{x}}$ direction. The electric field is:

$$\boldsymbol{E} = E_0\,\hat{\boldsymbol{x}}\ \sin(\omega(z/c - t) + \phi) \tag{4.1}$$

In addition to its direction and polarization, this beam is described by its **amplitude** $E_0$, **angular frequency** $\omega$, and **phase** $\phi$. The angular frequency $\omega$ is related to the **frequency** $\nu$ simply by $\omega = 2\pi\nu$, because there are $2\pi$ radians in a cycle. Note that the unit "Hertz" (Hz) is cycles per second, so is always used for $\nu$ not $\omega$.

It is also convenient to define **intensity** $I \sim E_0^2$, **wavelength** $\lambda = c/\nu$, and **wavenumber** $k = \omega/c$. Intensity measures how much energy is in the beam, while frequency (or wavelength or wavenumber) describes its color. As long as we are talking about light in a vacuum, the frequency and wavelength are always related by $\lambda = c/\nu$, so higher frequency means shorter wavelength ("bluer").

Eq. 4.1 describes both an oscillation in time (for fixed $z$) and a wave in space (at given time $t$). In time it repeats with period $T = 1/\nu$, and in space it has wavelength $\lambda = c/\nu$. And any given peak (*i.e.* where the argument of the sin is $\frac{\pi}{2}$) travels at $c$ along $\hat{\boldsymbol{z}}$ as claimed, because $(z/c - t) =$ constant implies $\frac{\mathrm{d}z}{\mathrm{d}t} = c$.

Light carries energy and momentum (see SubSec. 6.3.2). That may seem a little surprising, since in the absence of any electrical charges, no Coulomb forces are in play. Not every electric field (or magnetic field) carries energy, but the combination of fields in light does.

### 4.2.1 Projection

Shine natural light through a horizontal ($\hat{\boldsymbol{x}}$) polarization filter, and what gets through is horizontally polarized. Place a vertical ($\hat{\boldsymbol{y}}$) filter after that, and all the light is blocked. Now add a third, *diagonal* ($(\hat{\boldsymbol{x}} + \hat{\boldsymbol{y}})/2$) filter between them (it must go between), and some light gets through, Fig. 4.2. Please appreciate how weird that is! By way of contrast, there's nothing analogous you could do with color filters.

OK, it's easy enough to explain classically: the middle (diagonal) filter doesn't just block some light, it *interacts* with it, rotating the electric field into the diagonal. The diagonal filter **projects** the polarization onto the diagonal direction, and then the third (vertical) filter projects that onto

the vertical direction. But when you think in terms of light particles (photons), it shows that a filter does not simply block some photons, but actually alters the state of the ones it lets through. It is a hint of the quantum behavior that a measurement inevitably alters the system, projecting it onto an eigenstate. We'll revisit this when we discuss Bell's Inequality, because *it's even weirder than that.*

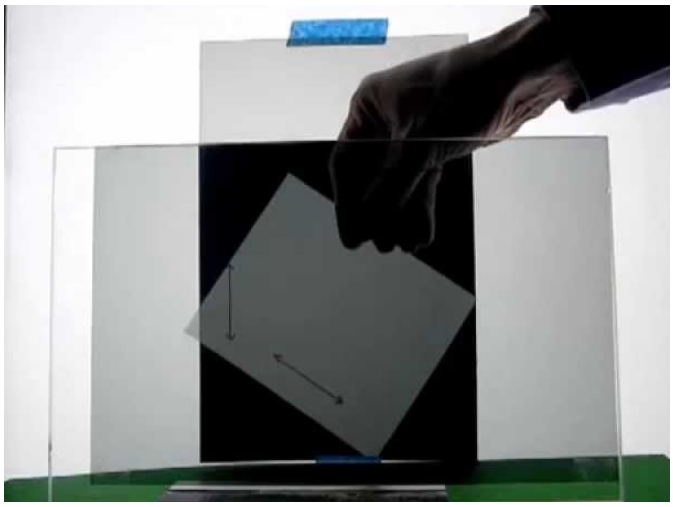


Figure 4.2: Three polarization filters allow more light through than two.

## 4.3 Linear Combinations

Electric fields add linearly. If two sources produce electric fields $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$, the resulting electric field is just $\boldsymbol{E}_1 + \boldsymbol{E}_2$. Two beams traveling in different directions simply cross paths without interacting. Here are a few interesting examples of linear sums of fields like Eq. 4.1:

1. $E_0\,\hat{\boldsymbol{x}}\,\sin(\omega(z/c-t))+E_0\,\hat{\boldsymbol{x}}\,\sin(\omega(z/c-t))$. Two identical beams add to double the amplitude ($2E_0$), which means quadrupling the intensity. This is essentially why laser beams can be so powerful: because all the light comes out in the same frequency and phase.

2. $E_0\,\hat{\boldsymbol{x}}\,\sin(\omega(z/c-t))+E_0\,\hat{\boldsymbol{y}}\,\sin(\omega(z/c-t))$. Here we've added x-polarized light and y-polarized light (of the same phase), to get light polarized in the $(\hat{\boldsymbol{x}}+\hat{\boldsymbol{y}})/\sqrt{2}$ direction. That's just a restatement of the claim above that any polarization in the x-y plane can be described in terms of the 2-element basis $(\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}})$. The new amplitude is $\sqrt{2}E_0$, so we've only doubled the intensity.

3. $E_0\,\hat{\boldsymbol{x}}\,\sin(\omega(z/c-t)) + E_0\,\hat{\boldsymbol{y}}\,\sin(\omega(z/c-t)-\pi/2)$. Here the x- and y-polarized components are $90° = \pi/2$ out of phase. Now the electric field rotates clockwise[1] rather than wiggling back and forth. This is called **circularly polarized light**. We could have chosen for our polarization basis clockwise and counterclockwise, aka right- and left-handed, aka negative- and positive-helicity.

4. $E_0\,\hat{\boldsymbol{x}}\,\sin(\omega(z/c-t))+E_0\,\hat{\boldsymbol{x}}\,\sin(\omega(z/c+t))$. Two waves traveling in opposite directions form a standing wave. This is how we describe light trapped in a box (or "cavity").

[1] as you face the oncoming beam, which is the optics convention, but is not universal

Linearity of the electric field is equivalent to saying there is no tree-level photon-photon interaction, which derives from the fact that the gauge group U(1) is Abelian. In fact QED introduces a 1-loop 4-photon interaction at $O(\alpha^4)$, so at very high energies there actually is some self-interaction, and linearity breaks down. We'll ignore that for now.

## 4.4 Natural Light is a Mix

Most light seen in nature, particularly any light from a hot source like the sun, is a mix of different frequencies, phases, and polarizations, each with its own amplitude. You can always Fourier-decompose the resulting field into a sum of terms like Eq. 4.1. In general, like cases (2) and (3) above, the energy flux is just the sum of the intensities of the components. Let's discuss how these mixes are perceived by the human eye:

- **Intensities** add, as intensity represents the energy density in any component. Between the rods and the cones, the human retina can detect light intensity over 10 orders of magnitude, from almost single-photon levels to the midday sun. As with other human senses, the perception of "brightness" is roughly $\log(I)$, the logarithm of total intensity.
- **Frequency** is seen as (spectral) "color", and our perception of a mix of frequencies was discussed in the prior chapter.
- **Phase** is not detected by the eye. In natural light it is unimportant. Mostly phase matters when it leads to interference as we saw in Fig. 4.1.
- **Polarization** is mostly undetectable to the human eye (but see **Haidinger's brush** for a subtle effect), so the intensities of different polarizations just add. Sunlight reflected off a horizontal surface, like the sea or a car hood, becomes partially polarized, making polarized sunglasses useful to filter out "glare" (a polarized filter only allows one polarization to pass through). Polarized glasses are also used in some 3D theaters; conveniently there are two independent polarizations, and we have two eyes.

## 4.5 Light and Sound: Compare and Contrast

**Compare** Sound is a compression wave in a medium (*e.g.* air, or a crystal), and has many similarities to light. Intensity is "volume" or "loudness", and frequency (or wavelength) is "pitch". Here is a small table of analogs:

| | Light | Sound |
|---|---|---|
| Intensity | Brightness (candles) | Loudness (dB) |
| Frequency | Color | Pitch |
| Speed | $c$ | Mach 1 |
| Quantum | Photon | Phonon |

Sound intensity (energy) is the square of pressure amplitude, and our perception of intensity ("loudness") is logarithmic. A **bel** is a factor of 10 increase in intensity, and the more commonly-known **decibel** (dB) is 1/10 of a bel, *i.e.* a factor of $\sqrt[10]{10}$ in intensity. Human hearing covers about 10 bels (a factor of $10^{10}$ in intensity), with 0 dB about the quietest thing you can hear, and

100 dB painfully loud. Human vision covers a similarly remarkable 9 orders of magnitude ($10^9$), from the faintest star on a dark night to the brightest sunlight you can see without damage.

Sound frequency is called **pitch**, and humans can hear in the range of 20 Hz to 20,000 Hz (the upper limit decreases with age). That's a much broader range than visible light, which covers only about a factor of two! With both light and sound, observers moving relative to the source will experience **Doppler shift**, discussed in SubSec. 4.5.1.

**Contrast** Sound in air is a longitudinal wave (while light is transverse), so there's no polarization. Sound requires a medium, and its speed depends on the medium, and is relative to the medium; for example in air at room temperature and pressure, sound travels 343 m/s (767 mph), called "Mach 1". The human ear can hear a much greater range of frequencies (1000x) than we can see (2x). But we have only limited ability for **sound localization**, *e.g.* using phase or arrival time differences between our two ears, nothing like the megapixel resolution of the retina.

**Music Theory** The human cochlea can distinguish many different frequencies of sound at once, unlike the mere 3 color channels of the human retina. This is what makes music such a rich experience. We are much more tuned in (see what I did there?) to **harmonics** of sound, frequencies that are in simple ratios to one another. An **octave** is a factor of 2 in frequency, *e.g.* $A_4$ (440 Hz by definition) and $A_5$ (880 Hz) are an octave apart.

In the equal-temperament **chromatic scale**, an octave is divided into 12 **semitones**, each $\sqrt[12]{2}$ apart in frequency, called C, C#, D, D#, E, F, F#, G, G#, A, A#, B. On a piano, the sharped notes are black keys, while the un-sharped (C, D, E, F, G, A, B) are white keys. Those white keys form the heptatonic major scale in the key of C, and the intervals (C to D, C to E, *etc.* ) are called major second, major third, perfect fourth, perfect fifth, *etc.* . These intervals are close to simple harmonics; for example, G/C is $2^{7/12} = 1.4983$ which is very close to 3/2. With some tweaking ("just tuning") of the semitone frequencies, you can make an octave $C'/C = 2$, a perfect fifth $G/C \approx 3/2$, a perfect fourth $F/C \approx 4/3$, and a major third $E/C \approx 5/4$, all nice simple harmonics that sound consonant (pleasant, complete). The notes of a major chord C:E:G have frequencies with ratios 4:5:6, which the ear finds appealing. Adding other semitones that are not simple harmonics introduces dissonance, a feeling of tension or yearning or incompleteness. See **music theory**, **musical tuning**, and **chromatic scale**.

### 4.5.1 Doppler Shift

A train approaches you with its whistle blowing. The sound waves "bunch up" and the wavelength appears to shorten (the pitch rises): $\lambda = \lambda_0 * (1 - v/v_s)$, where $v_s$ is the speed of sound, approximately 800 mph, also called Mach 1. If the train approaches at $v_s$, $\lambda \to 0$, and all the sound hits you at once as a sonic boom.

If, instead, the train is at rest (with respect to the air) and you move towards it, then $\lambda = \lambda_0/(1 + v/v_s)$; notice the singularity at $v = -v_s$, where you are moving away from the source so fast the sound never catches you. Both versions are called "Doppler shift", but the formulas are a little different depending on who's moving.

Light also exhibits Doppler shift, but since there's no medium, there's no distinction between you moving and the train moving. We need to do Lorentz transforms on the wave peaks. Suppose that Alice marks the spacetime coordinates $(ct, z)$ of two wavecrests as $(0, 0)$ and $(0, \lambda)$. Bob moves along $\hat{\boldsymbol{z}}$ with speed $v$; define $\beta = v/c$ and $\gamma = 1/\sqrt{1 - \beta^2}$. The marked points in Bob's frame are $(0, 0)$ and $(-\beta\gamma\lambda, \gamma\lambda)$. Bob knows the second crest will then move $c\beta\gamma\lambda$ in the $\hat{\boldsymbol{z}}$ direction

before his clock hits zero, which puts it at $x' = \gamma(1+\beta)\lambda$, thus $\lambda' = \sqrt{\frac{1+\beta}{1-\beta}}\lambda$, which is just Bob's Doppler-shifted wavelength.

I find it very amusing that the relativistic Doppler shift is kind of "half way" between the two results for sound derived above.

Since red is at the long-wavelength end of the visible spectrum, a Doppler shift to longer wavelengths (*e.g.* receding stars) is called a **red-shift**, and a Doppler shift to shorter wavelengths (*e.g.* approaching stars) is called a **blue-shift**. Most galaxies we observe are red-shifted, because the universe is expanding.

For either observer $\lambda = h/p$. Light waves Lorentz-transform properly.

# Chapter 5

# Light in Matter

The dominant effect light has on passing through matter is to wiggle the electrons, in the direction of the electric field polarization.[1] For longer wavelengths like radio waves, that produces classical oscillating electric currents. For shorter wavelengths, quantum effects (such as discrete electron energy levels) become important. Loosely-bound electrons like those in a metal reflect light, while the more tightly constrained electrons of a dielectric just slow it down.

## 5.1 Across the Electromagnetic Spectrum

Starting from the right side of Fig. 3.6, radio waves have long wavelengths, low frequencies, and low photon energies. They induce bulk movements of electrons in a metal (*e.g.* an antenna), over distances comparable to the wavelength. Radio waves are generally biologically harmless, as they don't carry enough energy to cause chemical changes. Microwaves are more energetic, and can cause **dielectric heating**. Note the metal screen on your microwave oven suffices to block the microwaves from exiting, because the holes are smaller than the microwave wavelengths, even as they let visible light through just fine. Infrared radiation, which can excite rotational and vibrational modes of molecules, is best known for being emitted from warm bodies; most of the radiation emitted from the Earth into space is infrared. Visible light contains enough energy to induce electron transitions, so the "wiggling" is now firmly in the quantum realm. As we pass to even shorter wavelengths of ultraviolet and X-rays, the ionizing radiation is increasingly able to penetrate matter and cause damage, particularly to living cells. Gamma rays can even disrupt nuclei.

## 5.2 Radio Antenna Theory

Large-scale "wiggling" is most obvious with radio waves (long wavelengths, low frequencies) impinging on metal. Metal consists of a fixed matrix of positive ions (nuclei plus closed shells of electrons) surrounded by a sea of nearly-free electrons. An $\hat{\boldsymbol{x}}$-polarized radio wave wiggles the electrons in the $\hat{\boldsymbol{x}}$ direction. Attach the metal **antenna** to a circuit (tuner) that resonates at a particular frequency, and what you have is a radio **receiver**. By time-reversal symmetry, a circuit that drives electrons back and forth in the antenna is then a **transmitter**, which emits radio waves. Wiggling electrons in the transmitter "here" cause wiggling of electrons in the receiver

[1] In fancier language, electric dipole (E1) interactions are stronger than either electric quadrupole (E2) or magnetic dipole (M1) by $1/\alpha^2$.

"there" via radio waves. There is a whole science of **antenna theory** about how best to detect radio waves when you may not know their direction or polarization.

## 5.3 Visible Light in Matter

### 5.3.1 Reflection

Metals, like the aluminum in your bathroom mirror, tend to reflect visible light. When light impinges on aluminum, the nearly-free electrons respond in a way that negates electric fields inside the metal, and in so doing generates a new beam of light emerging from the surface. A smooth surface results in **specular reflection**.

### 5.3.2 Transmission

Glass (like most transparent materials) is a **dielectric**, meaning that its electrons respond to an electric field by slightly polarizing and thereby creating a partially offsetting field; that is, the material dampens the amplitude of the electric field. This has the effect of slowing the propagation to an effective speed $c/n$, where $n$ is the **index of refraction**. A typical glass may have $n \approx 1.5$; water has $n = 1.333$, and air has $n = 1.0003$.

A simple model, a baby version of the **Clausius-Mossotti relation**, gives $n^2 = 1 + N\alpha$, where $N$ is number density of molecules and $\alpha$ is the atomic polarizability. Magnetic effects also contribute to $n$, but these are usually small.

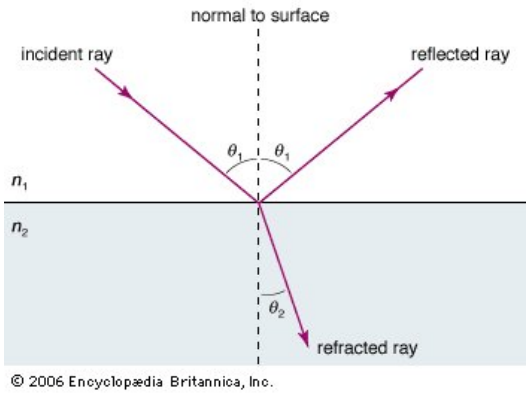
### 5.3.3 Refraction

The behaviour of light as it passes from one medium like air ($n_1$) to an optically denser one like glass ($n_2 > n_1$) is particularly interesting; see Fig. 5.1. It partly reflects, bouncing off the surface at the same incident angle $\theta_1$, and partly refracts, at an refracted angle $\theta_2$ given by **Snell's Law**: $\frac{\sin\theta_2}{\sin\theta_1} = \frac{n_1}{n_2}$. Note the light bends towards the vertical.

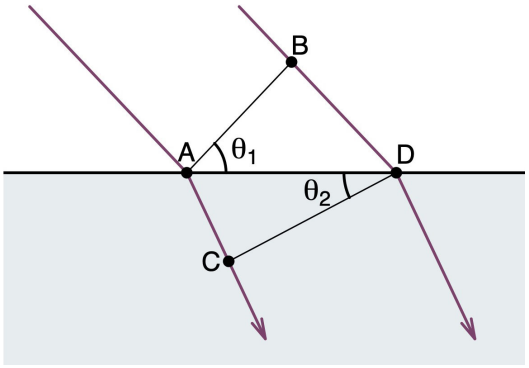
We can intuit the bending behaviour by imagining a line of soldiers marching on solid ground, encountering a marsh. The soldiers who hit the marsh first slow down first, bending the line towards the marsh.

With just a little geometry we can turn that into a proof, Fig. 5.1b. Here we have zoomed in on the light beam, and drawn wavefronts AB and CD. The angles marked $\theta_1$ (incident) and $\theta_2$ (refracted) are equal to those in Fig. 5.1a. Let $x = AD$ be the distance between the leftmost point of contact A and the rightmost point D, and let $\delta t$ be the time between when the wavefront hits A on the left, and when it hits D on the right. In that time, the left edge of the wavefront travels $AC = x\sin\theta_2 = \delta t/(c/n_2)$, while the right edge travels $BD = x\sin\theta_1 = \delta t/(c/n_1)$. Eliminating $\delta t$ gives $\frac{\sin\theta_2}{\sin\theta_1} = \frac{n_1}{n_2}$ as promised.

We can also derive Snell's Law from **Fermat's Principle** of least time, which we'll see again in Chapter 10. Fixing the starting point (in air) and the ending point (in glass), we find the point along the interface that minimizes the total travel time, and we get the same result. In the marsh analogy, if the soldiers know where they will start and end, they will wisely choose the bent path to get there fastest. This approach seems **teleological**, as though nature is following a plan, but that's a common feature of wave (and quantum) mechanics.




(a) Reflection and Refraction




(b) Proof of Snell's Law

Figure 5.1: Snell's Law

**The Brewster Angle and Polarized Sunglasses** In Fig. 5.1a, the reflected ray will be somewhat polarized horizontally, *i.e.* along the plane of the interface. When the reflected and refracted waves are perpendicular, this effect is complete: the reflected ray is fully polarized. The incident angle $\theta_B = \tan^{-1}(n_2/n_1)$ that results in this situation is called the **Brewster angle**. An important real-life example is sunlight reflecting off of water. That's why polarized sunglasses are always polarized vertically: to block some of the glare of sunlight reflected off horizontal surfaces.

### 5.3.4 Dispersion

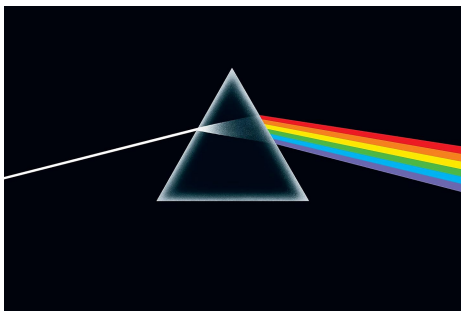
A system with resonance frequency $\omega_0$ encountering a periodic driving force of frequency $\omega$ eventually oscillates at the driving frequency $\omega$, with an amplitude that increases as $\omega$ approaches $\omega_0$. Both glass and water have resonances in the ultraviolet, so the index of refraction is larger for blue light than for red. Blue light goes slower inside the medium, and therefore bends more at the interface by Snell's Law.

In camera optics this is a nuisance called **chromatic aberration**. A **prism** uses this effect to split up white light into a rainbow of colors (Fig. 5.2a). And an actual **rainbow** occurs as sunlight passes through drops of water ($n = 1.333$) suspended in the air, see Fig. 5.2b. Sunlight refracts on its way into the droplet, reflects off the back surface, and refracts again on the way out, forming a virtual image in a characteristic 42° arc around the point opposite the sun. Dispersion puts the red arc slightly outside the blue arc. An even fancier path, involving 3 internal reflections, causes the fainter double-rainbow, with the color order reversed.

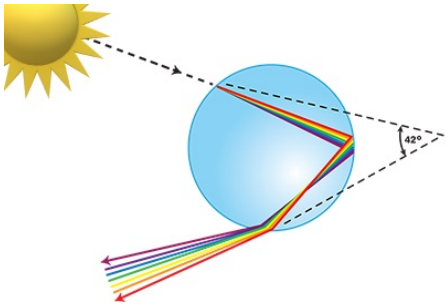
**Group velocity** $v_g$ is the speed at which information propagates in a light wave, *e.g.* information contained in a change of amplitude, frequency, or phase. Without dispersion, group velocity $v_g$ would be the same as phase velocity $v = c/n$. With dispersion, $v_g = v/(1 + \frac{dv}{d\lambda})$. Scientists working with longer-wavelength electromagnetic radiation, such as microwaves, have produced materials with very funny dispersion relations, giving them $v_g$'s close to zero or even negative.

### 5.3.5 Diffraction

When light passes through a slit of width comparable to its wavelength, it broadens out. If you pass light through a set of evenly-spaced slits, then let it fall on a screen, the beams from each slit will combine and interfere, similar to the double-slit of Fig. 4.1. All frequencies will constructively



(a) Prism (Pink Floyd version)




(b) Raindrop Refraction Forms a Rainbow

Figure 5.2: Rainbow from Prism and Raindrops

interfere at the center of the screen. Light of a given frequency, say "blue", will also constructively interfere a certain distance from the center, while "red" light will constructively interfere at a point further out; see Fig. 5.3. Thus a **diffraction grating** separately white light into its colors, similar to the dispersion effect of a prism (describe above). A **spectrometer** is essentially just a diffraction grating. I have described a transmissive diffraction grating, but you see a similar effect in a reflective diffraction grating, such as the rainbow colors reflected in a CD or DVD.

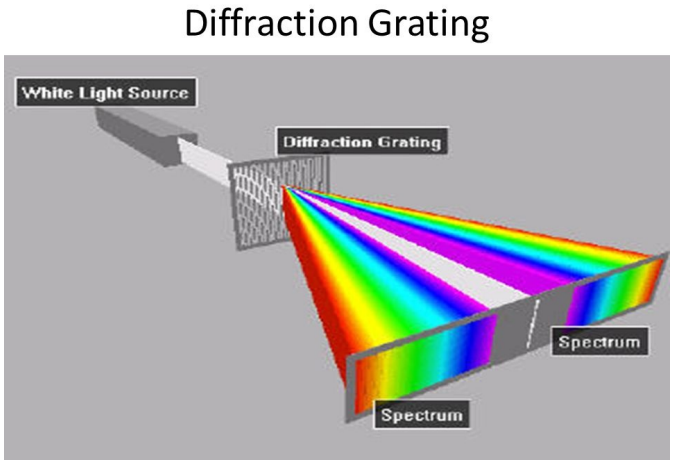



Figure 5.3: Diffraction Grating

A key tool for **solid-state physics** is **X-ray crystallography**, which is basically using a crystal as a diffraction grating. X-rays are appropriate because their wavelength is comparable to typical interatomic spacing of $\mathcal{O}(10^{-9}\,\mathrm{m})$ (or 10 Å if you prefer). Simple crystal structures, *e.g.* hexagonal or body-centered-cubic, give simple diffraction patterns. Complex organic molecules have also been examined this way, famously including Rosalind Franklin's crystallograph (Fig. 5.4) that hinted at the double-helix structure of DNA.

### 5.3.6 Emission

Inject energy into matter (*e.g.* with heat or electricity), exciting it into higher-energy states, then it will emit light as it drops back to lower-energy states. Those excited states might be electron orbitals above the ground state, or vibrational or rotational modes of molecules. Quantum

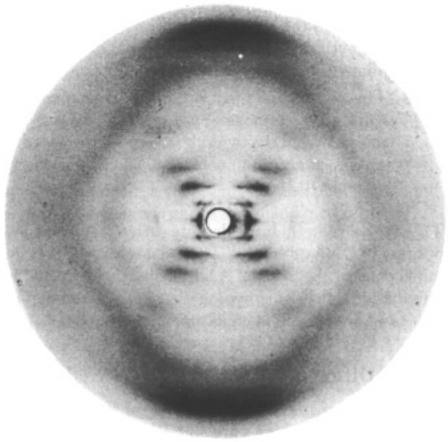


Figure 5.4: Franklin's DNA Crystallograph

mechanics only allows these states to have particular energies, so the jumps between them can only produce photons of certain discrete wavelengths. In the limit of a low-density gas of simple atoms or molecules, then, we see an **emission spectrum** consisting of narrow spectral lines, as in the Balmer series of Fig. 3.2.

Crowd those molecules together, though, and the lines broaden. Specific effects are called **pressure broadening**, **Doppler broadening**, and others. Think of a solid as an extreme case of pressure broadening, where constant interactions with neighbors smears out the narrow lines. In the limit of a **blackbody** solid at a temperature $T$, the emission spectrum is smoothed out into a Planck distribution, Eq. 11.13, determined solely by thermodynamics and depending only on $T$.

Fig. 5.5 shows a heuristic for understanding pressure broadening, from Rybicki and Lightman[7]. At top is the sine wave of electromagnetic radiation that would be produced by an isolated atom. Every time that atom gets bumped by a neighbor, imagine the phase of the wave gets randomly reset (bottom). The Fourier transform of the result is a broad distribution. So pressure broadening is significant when the frequency of atomic collisions is comparable to the frequency of light emitted.

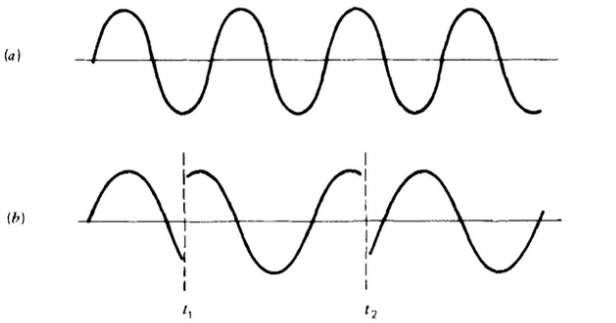



***Figure 10.3** Time-dependence of the electric field of emitted radiation which is (a) purely sinusoidal and (b) subject to random phase interruptions by atomic collisions.*

Figure 5.5: Pressure broadening heuristic

Most real-life thermal radiation lies between these extremes: a Planck-like spectrum punctuated by distinct emission lines. Examples include the sun, a burning log, or a tungsten filament.

**Hydrocarbon Flame** We often encounter burning hydrocarbons, particularly those of chemical structure $C_nH_{2n+2}$ such as methane ($n$=1), propane ($n$=3), butane ($n$=4), and paraffin (candle wax, $n \geq 16$). The "natural gas" in our stoves and heaters consists of methane, ethane, butane, and propane. In many cases, the flame is bluish at the base and orange-yellow further up. The bluish part is from **Swan band**s, which are transitions between vibrational modes of simple hydrocarbon radicals such as $CH$ and $C_2$, Fig. 5.6 (and see **vibronic spectroscopy**). By contrast, the orange or yellow parts are approximately blackbody radiation from glowing hot soot. Soot consists of microscopic carbon particles, but they still act as a solid and emit broad spectra. Hydrocarbon flame temperatures range from around 1500 K (candle) to 3600 K (propane), which is why they're orange or yellow. Often if you increase the oxygen supply to improve combustion and reduce soot (producing only $CO_2$ and $H_2O$), the orange/yellow part goes away.

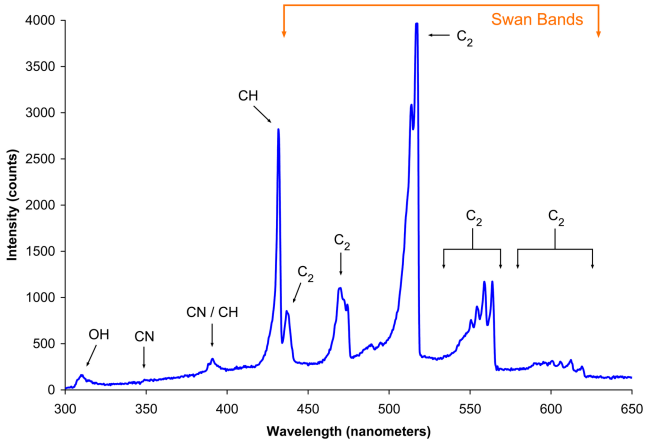



Figure 5.6: Swan bands producing blue hydrocarbon flames

**Fluorescence and Phosphorescence** **Fluorescence** is the process of absorbing light and then emitting light. Sometimes the de-excitation path can be different than the excitation, *e.g.* "black-light paint" may absorb ultraviolet light (causing electrons to jump from A to C), then emit visible light in the transitions from C to B and then B to A. **Phosphorescence** is fluorescence with a significant time delay between absorption and emission, like those glow-in-the-dark stickers.

Another way to excite electrons is with an electric current. A low-pressure sodium vapor street-lamp, for example, then de-excites with a very distinct double yellow line spectrum, see Fig. 5.7. A mercury vapor lamp also has a characteristic spectrum that is more blue. A **fluorescent lamp** consists of a tube of electrically-excited plasma, surrounded by a fluorescent material that converts the plasma light to more useful frequencies.

### 5.3.7 Absorption

Absorption is the time-reversal of emission. Matter with a given emission spectrum will have a corresponding absorption spectrum (Fig. 5.8)) when light passes through it. Low-pressure simple gases will absorb only discrete lines, while a **blackbody** will absorb all light impinging on it.

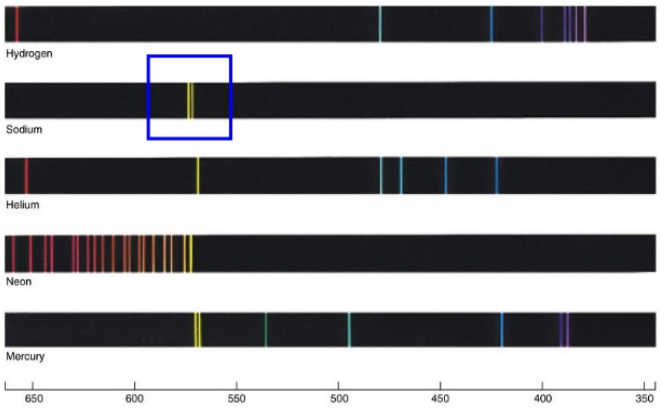



Figure 5.7: Sodium D-Lines, and Other Spectra

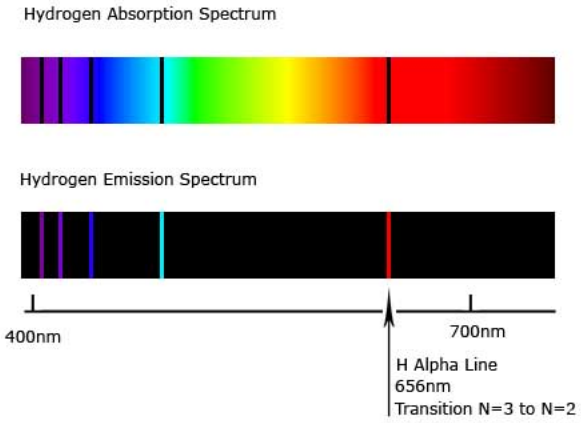



Figure 5.8: Hydrogen Absorption and Emission Spectra

## 5.4 Optics

A converging lens uses refraction to collect light emitted from one point, and re-focus it at another point (Fig. 5.9). Light from an extended object, passing through the converging lens, forms a **real image** on a focal plane. The human eye contains a converging lens, and the focal plane is the retina, where the image is detected by light-sensitive rods and cones. A camera has either film or a digital sensor at the focal plane.

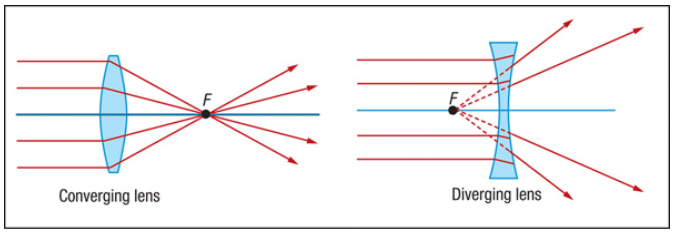



Figure 5.9: Converging and Diverging Lenses

If you are far-sighted or have presbyopia (have trouble seeing things up close), your glasses (reading glasses) are converging lenses, which add to the converging power of the eye.

The strength of the lens is measured in **diopters**, which have units of inverse meters. Light from an infinitely distant source will focus a distance $f$ meters from the lens, and we say the lens's strength is $1/f$ diopters. An object at distance $r_1$ from the lens will form a real image at $r_2$, where $\frac{1}{r_1} + \frac{1}{r_2} = \frac{1}{f}$. Remarkably, the strength of several lenses in series simply add: $\frac{1}{f_T} = \frac{1}{f_1} + \frac{1}{f_2}$, sometimes called the "lens equation".

There are also diverging lenses, which will take light from an infinitely far source and cause it to diverge, as if it came from a point $|f|$ meters from the lens (Fig. 5.9). Here we call the focal distance $f$ and the lens strength $1/f$ negative (and the lens equation still works!). When the eye sees light apparently diverging from this ghostly point left of the lens, it interprets that as a **virtual image**. If you are near-sighted (have trouble seeing things far away), your glasses are diverging lenses.

A **refracting telescope** combines a converging and a diverging lens to create a magnified virtual image of a distant object. (A microscope works similarly.) In practice, most large telescopes these days are **reflecting telescopes**, which use curved mirrors to achieve similar results.

### 5.4.1 Holography

A completely different way to create images with film is **holography**. In its simplest form, laser light is split into two beams. One beam reflects off the subject, then the reflected light meets up with the second beam at the film, which captures the complex interference pattern. Amazingly, if you pass laser light through the developed film, it re-creates a 3D image of the subject! The interference pattern on the film acts like Fresnel lenses, which create virtual images of the original subject with just the correct perspective, wherever you view it from. If you happen to visit San Francisco, you can see some great examples at the Holography Museum.

One remarkable feature of holography is that it encodes 3D information on a 2D surface (film). From an information theory perspective, you might worry that the 2D film can't contain that much information, and indeed there is a limit to the resolution you can achieve, related to the wavelength of the laser light.

# Chapter 6

# Light is Electromagnetic Radiation

Light is a wave solution to Maxwell's Equations.

## 6.1 Electric Fields

### 6.1.1 Coulomb's Law / Gauss's Law

A particle of charge $q_1$ repels a second particle of charge $q_2$ with a force given by **Coulomb's Law**:

$$\boldsymbol{F} = k_c \frac{q_1 q_2}{r^2} \hat{\boldsymbol{r}} \tag{6.1}$$

which is similar to Eq. 2.2 (Newton's Law of Gravity) except for the sign. See Appendix A for a discussion of units, but we will immediately adopt CGS Gaussian units where $k_c = 1$. The $\hat{\boldsymbol{r}}$ indicates that when the charges have the same sign, the force on the second particle points directly away from the first: same-sign charges repel; opposites attract.

Instead of saying charge 1 acts on charge 2, we can say charge 1 creates an electric field $\boldsymbol{E} = (q_1/r^2)\,\hat{\boldsymbol{r}}$, and charge 2 feels a force $\boldsymbol{F} = q_2\boldsymbol{E}$. The electric field points radially outward from a positive charge (or inward towards a negative charge), see Fig. 6.1. As long as we are doing **electrostatics**, the electric field is a mere convenience, it carries no energy or momentum, doesn't do anything except convey Coulomb's Law. Note that in determining the force on charge 2 (as in Fig. 6.1), we do not include the electric field generated by charge 2. This issue is revisited much later in renormalization theory. Sometimes we refer to charge 2 as a "test charge" in this context.

The net force on a test charge due to several charges $q_i$ is the vector sum of the Coulomb forces $\boldsymbol{F} = \sum_i \boldsymbol{F}_i$, and so the net electric field $\boldsymbol{E} = \sum_i \boldsymbol{E}_i$. We say electric fields add linearly.

Divergence and the **divergence theorem** are defined in Appendix B.3.1. Coulomb's Law can be restated as **Gauss's Law**, the first of Maxwell's Equations, which in differential form is: $\nabla \cdot \boldsymbol{E} = 4\pi\rho$. Using the divergence theorem, the integral form of Gauss's Law states that the flux of $\boldsymbol{E}$ over a closed surface equals $4\pi Q$ where $Q$ is the enclosed charge. Imagine a sphere of radius $r$ around a point charge $q$, then the flux is $4\pi r^2 E$, and from $4\pi r^2 E = 4\pi q$ we recover Coulomb's Law $E = q/r^2$. These are all the same law stated different ways. Using well-chosen Gaussian surfaces, you can similarly show that the electric field around a line charge $E = 2\lambda/r$ (where $\lambda$ is charge per unit length), and inside a capacitor $E = 4\pi\sigma$ (where the charge per unit area is $+\sigma$ on one sheet and $-\sigma$ on the other).

In electrostatics, where charges don't move and there are no changing magnetic fields, we can also say the **curl** vanishes: $\nabla \times \boldsymbol{E} = 0$. Electric fields don't go around in a circle. When your

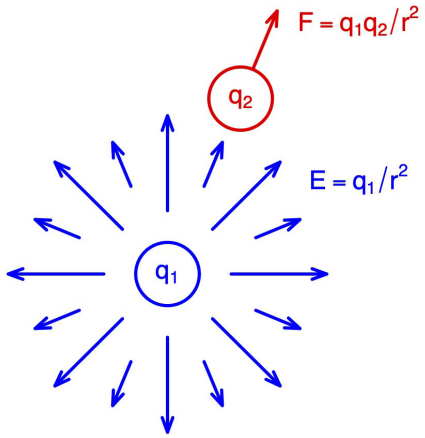
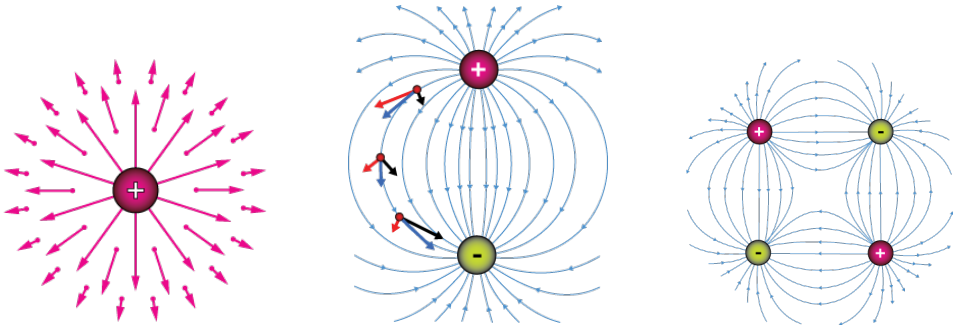



Figure 6.1: Electric field and Coulomb force from a point charge

system consists of discrete point charges, and everywhere else in space $\nabla \cdot \boldsymbol{E} = 0$ and $\nabla \times \boldsymbol{E} = 0$, the electric field vectors are highly constrained. In fact, you can avoid the awkwardness of drawing little $\boldsymbol{E}$ vectors entirely, just draw **field lines** that are parallel to $\boldsymbol{E}$ everywhere, and the strength of the field $|\boldsymbol{E}|$ is automatically proportional to the density of field lines.

### 6.1.2 Multipole Expansion

Fig 6.2 shows the electric field lines from a point charge (aka electric monopole), a dipole, and a quadrupole. If you have a localized static system of charges, and you look at the electric field sufficiently far away, it is to first order the Coulomb field $Q/r^2$ from the net charge $Q$ of the system. To next order, add a dipole field which goes like $1/r^3$, and then a quadrupole field like $1/r^4$, and so on. The charge distribution can always be expanded this way, called a **multipole expansion**. Note that a dipole consists of two opposite charges separated by some distance; a quadrupole consists of two opposite dipoles separated by some distance; and that pattern continues to octopoles and beyond.



(a) Monopole (b) Dipole (c) Quadrupole

Figure 6.2: First 3 Elements of Multipole Expansion

If you start moving those charges slowly ("quasi-statically"), remembering that total $Q$ is conserved, the quasi-static dipole field dominates with dependence $1/r^3$. Comparing that to a radiation field whose strength drops off as $1/r$, we see that electromagnetic radiation ("far-field")

always dominates over static and quasi-static electric fields ("near-field") sufficiently far from the source. Loosely speaking, the (quasi-)static fields stay "attached" to the source, while radiation "detaches" and propagates.

### 6.1.3 Charge Conjugation

If all charges in the universe flipped sign, it wouldn't alter Coulomb's Law or any other physical predictions of electrodynamics. The $\boldsymbol{E}$ field would change sign, but forces would not. We say electromagnetism is invariant under **charge conjugation** "**C**".

Ben Franklin actually set the convention that became positive protons and negative electrons. It's a little unfortunate, because electrons do the actual moving in a current-carrying wire, so we sometimes have to remember that current density $\boldsymbol{J}$ to the right comes from electrons moving left.

Applying charge conjugation to matter turns it into antimatter, which is a real thing, with positively charged anti-electrons (also called positrons) orbiting around negatively charged antiprotons. Small amounts of antimatter are created in accelerator labs and cosmic rays all the time, but the universe contains far more matter than antimatter. So while electrodynamics and gravity are invariant under **C**, the actual content of the universe is not **C**-symmetric, and we don't completely know why.

## 6.2 Magnetic Fields

### 6.2.1 Ampère and the Magnetic Force

Magnetism was first discovered in lodestones, a form of iron ore. It was not at all clear that it had anything to do with electricity. We now know that natural magnets arise from the electric current of orbiting electrons in atoms. So more fundamentally, a magnetic field $\boldsymbol{B}$ is generated from a current $I$, or rather the current density $\boldsymbol{J}$. If you run a current through a straight wire, and measure $\boldsymbol{B}$ with, say, a magnetic compass, you'll see that $\boldsymbol{B}$ circles around the wire. This describes a vector field with **curl**, see Appendix B.3.1, and we find $\nabla \times \boldsymbol{B} = \frac{4\pi}{c}\boldsymbol{J}$, which in various forms is known as Ampère's Law or the Law of Biot-Savart. This will become our fourth Maxwell Equation, though there is still a piece missing.

There are (to our knowledge) no magnetic charges (or "monopoles"), so $\nabla \cdot \boldsymbol{B} = 0$, which will be our second Maxwell Equation, sometimes called Gauss's Law of Magnetism.

Recall that the divergence theorem relates the differential and integral forms of Gauss's Law. Similarly, Stokes's theorem relates the differential and integral forms of Ampère's Law. Draw a closed loop around a current-carrying wire, and any surface bounded by that loop. The integral form of Ampère's Law says that loop integral of $\boldsymbol{B}$ equals $\frac{4\pi}{c}$ times the total current flux through the surface. Thus around a wire carrying current $I$, the magnetic field at distance $r$ has magnitude $B = 2I/(rc)$.

Another classic problem using the integral form of Ampère's Law is the magnetic field inside a **solenoid**, which is a coil of wire carrying current $I$, having $n$ turns per centimeter. In the limit that the solenoid is very long, the magnetic field inside is uniform, points along the axis, and has magnitude $B = 4\pi In/c$.

I alluded to measuring $\boldsymbol{B}$ with a compass, but again that results from the interaction with orbiting electrons in the compass's iron atoms. The more fundamental statement is that a magnetic field creates a force on moving charges, $\boldsymbol{F} = q\frac{\boldsymbol{v}}{c} \times \boldsymbol{B}$. That's weird, right? The force is perpendicular

to $\boldsymbol{B}$, and also to $\boldsymbol{v}$, and you have to use the "right-hand rule" to get the sign right: point your right fingers along $\boldsymbol{v}$, curl them towards $\boldsymbol{B}$, then your thumb points in the direction of the force (if $q > 0$). In the presence of both electric and magnetic fields, the Lorentz force law for a charge $q$ is:

$$\boldsymbol{F} = q\left(\boldsymbol{E} + \frac{\boldsymbol{v}}{c} \times \boldsymbol{B}\right) \tag{6.2}$$

### 6.2.2 Parity

If you observe the world through a mirror, it still looks very sensible; the same laws of physics work for everyday phenomena. We say that electromagnetism and gravity are invariant under **parity** "**P**". Sure, people's hearts are now apparently on their right sides, and DNA twists the other way, but those are just accidents of history anyway. Magnetism still works the same way, *e.g.* two wires carrying parallel currents still attract. But through the mirror you see scientists calculating the $\boldsymbol{B}$ field using a *left*-hand rule, and then the magnetic force again with another *left*-hand rule, corresponding to the cross-products of Ampère's Law and the Lorentz Force Law (Eq. 6.2). Even though they got the right final answer, they had the sign of $\boldsymbol{B}$ wrong. In the mirror world, for us to use the *right*-hand rule, $\boldsymbol{B}$ has to change sign. We say $\boldsymbol{B}$ is a **pseudovector**.

To be precise, mirror reflection is parity inversion ($\boldsymbol{x} \to -\boldsymbol{x}$) plus a 180° rotation. Under parity alone, all true vectors ($\boldsymbol{x}$, $\nabla$, $\boldsymbol{E}$, $\boldsymbol{J}$) change sign, but pseudovectors ($\boldsymbol{B}$) do not. Pseudovectors can be formed *e.g.* from the cross product of two true vectors: $\boldsymbol{B} = \nabla \times \boldsymbol{A}$, since under parity both of those true vectors change sign. For completeness, the combination of three true vectors $\boldsymbol{U} \cdot (\boldsymbol{V} \times \boldsymbol{W})$ would be a pseudoscalar.

### 6.2.3 Ferromagnets, How Do They Work?

Let's get back to those lodestones, or in modern culture, refrigerator magnets. What do they have to do with magnetic fields from currents? Any charged particle with angular momentum (orbital or spin) is effectively a tiny little loop current, and therefore a magnetic dipole. Electrons have charge $-e$ and spin $\hbar/2$, so every electron is a tiny magnet. While magnetic moments in a filled electron shell must cancel out, the 6 electrons in an iron atom's unfilled 3d shell do not cancel, so each iron atom is a dipole magnet.

Dipoles do **not** naturally align with one another; in fact it takes some energy to do so. But aligned electron spins, due to Pauli exclusion, have a different spatial distribution from anti-aligned spins, and in ferromagnets the lowest-energy crystal structure favors the aligned distribution. Even so, absent external influence, regions of alignment (called "domains") are microscopically small, and different domains point in different directions. It takes either a strong magnetic field, or a weak field plus heat, to make a domain macroscopic and permanent.

Those lodestones were made from iron born in the hearts of ancient supernovae, collected and heated in Earth's gravitational well. They formed ferromagnetic iron compound crystals, and then cooled under the influence of Earth's planetary magnetic field, to become permanent magnets.

Permanent magnets interact with each other, of course, but also with unmagnetized ferromagnetic material such as your refrigerator door. There, the permanent magnet induces temporary dipole alignment in the door, sufficient to cause attraction, but the alignment disappears when the permanent magnet is removed.

## 6.3 Changing Fields and Maxwell's Equations

If you move a permanent magnet around, the changing magnetic field induces an electric field. In fact it causes $\boldsymbol{E}$ to have curl, which is useful for, say, producing a current in a closed loop. This is **Faraday's Law of Induction**: $\nabla \times \boldsymbol{E} = -\frac{1}{c}\frac{\partial \boldsymbol{B}}{\partial t}$, which will be our third Maxwell Equation. It is of tremendous practical value! You could say that Faraday's Law is the basis for our entire electrical infrastructure, allowing mechanical work to be converted into electrical energy. The turbines of a hydroelectric dam, for example, turn a **magneto** which creates an electric current, the current flows over wires to our homes, and then through a **motor** which converts the energy back into mechanical. The promise of a new form of energy made electromagnetism a primary focus of 19th century physics.

At last we get to James Clerk Maxwell himself, who realized that the equations we have collected so far were inconsistent. To repair the problem, it had to also be true that a changing electric field produces curl in the magnetic field, just as current does; in fact the $\partial \boldsymbol{E}/\partial t$ term is sometimes called the "displacement current", and modifies Ampère's Law.

Putting these insights together, in CGS units, **Maxwell's Equations** are:

$$\nabla \cdot \boldsymbol{E} = 4\pi\rho \qquad \text{Coulomb / Gauss} \tag{6.3a}$$

$$\nabla \cdot \boldsymbol{B} = 0 \qquad \text{No magnetic monopoles} \tag{6.3b}$$

$$\nabla \times \boldsymbol{E} = -\frac{1}{c}\frac{\partial \boldsymbol{B}}{\partial t} \qquad \text{Faraday} \tag{6.3c}$$

$$\nabla \times \boldsymbol{B} = \frac{4\pi}{c}\boldsymbol{J} + \frac{1}{c}\frac{\partial \boldsymbol{E}}{\partial t} \qquad \text{Ampère-Maxwell} \tag{6.3d}$$

### 6.3.1 Propagation

Changing magnetic fields induce electric fields, and changing electric fields generate magnetic fields. Sources ($\rho$, $\boldsymbol{J}$) may initiate these fields, but even far from those sources that back-and-forth allows the fields to propagate on their own, a phenomenon known as electromagnetic radiation, or light. To be concrete, here is a solution to Maxwell's Equations in the absence of sources:

$$\boldsymbol{E} = E_0\,\hat{\boldsymbol{x}}\,\sin(k(z-ct)) \qquad \boldsymbol{B} = E_0\,\hat{\boldsymbol{y}}\,\sin(k(z-ct)) \tag{6.4}$$

This is an electromagnetic wave propagating in the $\hat{\boldsymbol{z}}$ direction at speed $c$. $\boldsymbol{E}$ points along the x-axis, and $\boldsymbol{B}$ points along the y-axis. The amplitude $E_0$ and the wavenumber $k$ are unspecified, but we note the wavelength $\lambda = 2\pi/k$, and the angular frequency $\omega = ck$.

### 6.3.2 Poynting Vector

The **Poynting vector** $\boldsymbol{S} = \frac{c}{4\pi}\boldsymbol{E} \times \boldsymbol{B}$ gives the energy flux (or power flow) of an electromagnetic field. For light, $\boldsymbol{S}$ points as expected in the direction that the light travels. It also shows how energy flows in a coaxial cable or electrical circuit.

### 6.3.3 Light in Matter

Place a chunk of matter in an electric field, and the matter tends to **polarize** in a way that reduces the field inside the chunk. It's useful to define $D = E + 4\pi P$, where $P$ is the average polarization, $E$ is the actual (reduced) electric field, and $D$ is sort of what the field there would have been if

not for the polarized matter. Matter in a magnetic field tends to **magnetize**, but usually in such a way to enhance the field, so we define $H = B - 4\pi M$ where $M$ is the average magnetization. Remarkably, these fields $D$ and $H$ obey equations like Eq. 6.3, except that the speed $c$ is replaced by $c/n$, where $n$ is the **index of refraction**. In other words, light passes through matter slower than $c$ but otherwise in a very similar fashion.

In practice, polarization is usually the dominant effect, so $n$ is mostly a measure of the matter's polarizability. If $\varepsilon$ is the dielectric constant at the appropriate frequency, then $n \approx \sqrt{\varepsilon}$. $n$ is frequency-dependent, and that's the reason for dispersion (rainbows).

## 6.4 Potentials

### 6.4.1 The Electric Potential

Let's return to electrostatics (or even magnetostatics, as long as $\dot{\boldsymbol{B}} = 0$), where $\nabla \cdot \boldsymbol{E} = 4\pi\rho$ and $\nabla \times \boldsymbol{E} = 0$. In Appendix B.3.1 we state that the curl of a gradient always vanishes $\nabla \times (\nabla\Phi) = 0$, suggesting we try the substitution $\boldsymbol{E} = -\nabla\Phi$, where $\Phi$ is the **electric potential**.

The minus sign is the same as the sign in $\boldsymbol{F} = -\nabla V$ where $V$ is potential energy associated with any conservative force; in fact for the Coulomb force we just have $V = q\Phi$. A positive test charge experiences a force pushing it from high potential to low.

If by chance you are feeling lost at this point, you may be happy to learn you already know this part! Electric potential is also called **voltage**. $\Phi$ differs by 9 volts between the terminals of a 9-volt battery. $\Delta\Phi = 9\,\mathrm{V}$ is what pushes electrons through a circuit.

Only potential differences matter. Since the absolute level of $\Phi$ is arbitrary, in practice we often choose to set $\Phi = 0$ for the Earth itself, and anything tied to that is "grounded".

Gauss's Law turns into $\nabla^2\Phi = -4\pi\rho$, which is Poisson's Equation. In practice we often have only a handful of sources (point-, line-, or sheet-charges), so everywhere else it's just Laplace's Equation ($\nabla^2\Phi = 0$) with boundary conditions. Mathematicians have studied Laplace's and Poisson's Equations for hundreds of years. We will have to modify that a bit when we move away from magnetostatics.

### 6.4.2 The Magnetic Potential

Let's try a similar trick with $\boldsymbol{B}$. Its divergence vanishes, and we know (Appendix B.3.1) that the divergence of a curl vanishes, so we write $\boldsymbol{B} = \nabla \times \boldsymbol{A}$ for some **vector potential** $\boldsymbol{A}$. When $\boldsymbol{A}$ is time-dependent, Faraday's Law tells us we now must write $\boldsymbol{E} = -\nabla\Phi - \dot{\boldsymbol{A}}$.

Maxwell's Equations in potential form become just these two:

$$\nabla^2\Phi + \frac{\partial}{\partial t}(\nabla \cdot \boldsymbol{A}) = -4\pi\rho \tag{6.5a}$$

$$\left(\nabla^2\boldsymbol{A} - \frac{1}{c^2}\frac{\partial^2\boldsymbol{A}}{\partial t^2}\right) - \nabla\left(\nabla \cdot \boldsymbol{A} + \frac{1}{c^2}\frac{\partial\Phi}{\partial t}\right) = -\frac{4\pi}{c}\boldsymbol{J} \tag{6.5b}$$

plus the field definitions

$$\boldsymbol{E} = -\nabla\Phi - \dot{\boldsymbol{A}} \tag{6.6a}$$

$$\boldsymbol{B} = \nabla \times \boldsymbol{A} \tag{6.6b}$$

### 6.4.3 Gauge Invariance

In magnetostatics ($\dot{\boldsymbol{B}} = 0$) we saw that we could shift $\Phi$ by a constant everywhere in space (even a time-dependent one), because only $\nabla\Phi$ is physical. More generally, we can perform a **gauge transformation** without affecting the physical fields:

$$\Phi' = \Phi - \frac{\partial \Lambda}{\partial t} \qquad \boldsymbol{A}' = \boldsymbol{A} + \nabla\Lambda \tag{6.7}$$

A clever choice of **gauge** may clean up Eq. 6.5. The **Coulomb gauge** is a choice of $\Lambda$ that restores Eq. 6.5a to Poisson's Equation, though it makes Eq. 6.5b even uglier. The **Lorenz gauge** is a choice that turns Eq. 6.5 into inhomogeneous wave equations that are Lorentz invariant.[1]

## 6.5 A Dialogue on the Weirdness of Magnetism

DAVE: Electrostatics is so elegant, it's just the mathematics of a scalar field that obeys Poisson's Equation. What's up with magnetism, with its crazy right-hand rules, curls and cross products, pseudovectors, and vector potential? $\boldsymbol{B}$ doesn't even point anywhere sensible. Who ordered that?

DR. B: Magnetism is a relativistic consequence of the Coulomb theory you love so much. No avoiding it in a relativistic world.

DAVE: OK I see all those $c$'s in the denominators of Eq. 6.3 and the Force Law. Our everyday experience with mechanics is that we can take the $c \to \infty$ limit of relativity to recover Newtonian physics. Why can't I take $c \to \infty$ and ignore magnetism?

DR. B: $c$ just relates units of time and space; in the next section we're going to set $c = 1$. How can you have $c = 1$ and $c \to \infty$ at the same time?

DAVE: Well I mean that velocities we normally encounter are small: $v/c \ll 1$.

DR. B: Electrons in atomic orbitals are moving somewhat relativistically, with root-mean-square velocity $v \sim Z\alpha c$, where $Z$ is the atomic number and $\alpha = 1/137$. That's a quantum result, as is the magnetic dipole moment arising from spin angular momentum. You can't ignore relativistic effects in atomic physics.

DAVE: I heard that the electrons in a current-carrying wire move pretty slowly, so why is the magnetic force between two wires detectable?

DR. B: If the wires held only electrons, no protons, the Coulomb force between them would indeed be tremendous. But wires, like everything else, are very nearly electrically neutral. Magnetism (and gravity!) can compete because neutrality hides the strength of electrostatic forces.

DAVE: Where else do we need to understand magnetism?

DR. B: Light is a purely relativistic phenomenon where $\boldsymbol{E}$ and $\boldsymbol{B}$ play similar roles. Also the Poynting vector that traces energy through circuits. And magnetos for generating electric power. In all these cases the relativistic phenomenon of magnetism is critical.

DAVE: So why all those curls and cross products? Why is $\boldsymbol{B}$ a pseudovector?

DR. B: These are hints that $\boldsymbol{B}$ is not really a vector, but rather the three space-space components of an antisymmetric 4-tensor $F^{\mu\nu}$. It happens to work in $D$=4 spacetime that the number of these components $(D-1)(D-2)/2$ is the same as for a vector $D-1$. Curls and cross products would not work the same in any other number of dimensions. The equations are just crying out to be re-written in 4-tensors.

[1] It's a quirk of history that Ludvig Lorenz and Hendrik Lorentz both show up in that sentence.

DAVE: I guess we'd better start learning Special Relativity.

# Chapter 7

# Light on a Geodesic: Special Relativity

The speed of light $c$ is embedded in Eq. 6.3, and its wave solution Eq. 6.4, as a constant of nature (also known as the **Einstein constant**). If you are more familiar with SI units, there are two constants $\epsilon_0$ and $\mu_0$. One combination is $c = 1/\sqrt{\epsilon_0\mu_0}$. The other combination $\sqrt{\epsilon_0/\mu_0}$ just relates units of $\boldsymbol{E}$ and units of $\boldsymbol{B}$.

At first physicists imagined that light travelled at $c$ with respect to some invisible background medium, the **aether**. After all, that's what sound does, the medium being air. But the Michelson-Morley experiment proved there is no aether! (Specifically it showed the speed of light is unaffected by Earth's rotation around the sun; we presume the aether does not move with our planet.) The physical constant had to be the same for all obervers, but that seemed impossible. How can light recede from you at $c$, and then when you start running to catch it, it still recedes at $c$? The resolution required a fundamental re-thinking of space and time.

## 7.1 Lorentz Transformations

### 7.1.1 Galilean Relativity

**Galilean relativity** is a name for our common-sense notion of relative motion (which turns out to be wrong). Imagine two observers, named "unprimed" and "primed", with the primed observer moving at speed $v$ to the right ($+\hat{\boldsymbol{x}}$) with respect to the unprimed. There's some event, say a firecracker going off, which both observers set as their origin: $(t, x, y, z) = (t', x', y', z') = (0, 0, 0, 0)$. For any other event, the coordinates are related by $t' = t$, $x' = x - vt$, $y' = y$, $z' = z$. Both observers' clocks always agree. An object moving right with velocity $V$ relative to the unprimed observer appears to the primed observer to have velocity $V - v$. If the "object" is light, with $V = c$, then the primed observer sees the light travel slower than $c$ under Galilean Relativity, *i.e.* Maxwell's Equations do not hold.

### 7.1.2 Special Relativity

Let's set up some thought experiments for these two observers, and insist that both see light travel at $c$; what are the implications for the coordinate transformations? For convenience we define $\beta \equiv v/c$ and $\gamma \equiv 1/\sqrt{1-\beta^2}$.

Suppose the primed observer tracks a beam of light from the firecracker at $(0, 0, 0, 0)$ in the $\hat{\boldsymbol{y}}'$ direction, Fig. 7.1. It bounces off a mirror at $y' = L$ and returns to him at $ct' = 2L$. To the

unprimed observer, it travels a distance $2\sqrt{L^2 + (vt/2)^2}$, so it takes time $ct = 2\sqrt{L^2 + (vt/2)^2}$. Thus $t^2 = t'^2/(1-\beta^2)$, or $t = \gamma t'$.

We have discovered **time dilation**. To the primed observer, the light bouncing off the mirror is a form of clock, *e.g.* he could calibrate it so that $2L/c$ is 1 second. To the unprimed observer, that clock runs slow; it takes $\gamma$ seconds to tick. Note $\gamma \geq 1$, so the moving clock always runs slow. In the famous **twin paradox**, the twin who flies through space returns home younger.

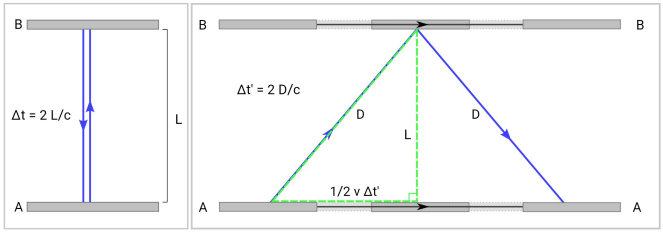



Figure 7.1: Time Dilation Experiment: Light Bouncing in $\hat{\boldsymbol{y}}$ Direction

Now let's send the light beam in the $\hat{\boldsymbol{x}}'$ direction instead, Fig. 7.2. Say the primed observer's spaceship has length $L'$ (as he sees it). The light travels from the back end at $ct' = 0$, hits a mirror at the front at $ct' = L'$, and returns to the back at $ct' = 2L'$. The unprimed observer thinks the spaceship has a different length $L$. To reach the front mirror, light takes time $ct_1 = L/(1-\beta)$, because it must travel a distance $(L + vt_1)$ at speed $c$. Then the return trip takes $ct_2 = L/(1+\beta)$, for a total time $ct = 2L/(1-\beta^2)$. But we already know from time dilation that $t = \gamma t'$, so $L = L'/\gamma$. We have discovered **length contraction**: the moving spaceship appears shorter to the stationary observer.

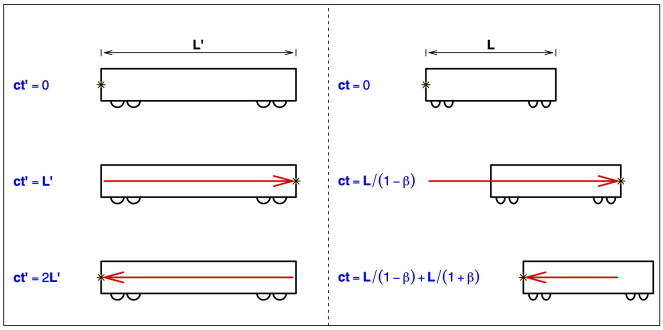



Figure 7.2: Length Contraction Experiment: Light Bouncing in $\hat{\boldsymbol{x}}$ Direction

We assumed (correctly) that lengths in the $\hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}}$ directions are unaffected. So, suppressing those and setting $c = 1$, the **Lorentz transformation** is:

$$\begin{pmatrix} t' \\ x' \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} t \\ x \end{pmatrix} \tag{7.1}$$

The inverse transformation, which is just the same but with $\beta \to -\beta$, is:

$$\begin{pmatrix} t \\ x \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} t' \\ x' \end{pmatrix} \tag{7.2}$$

You can even write this in a form highly suggestive of ordinary rotations, but with hyperbolic trig functions replacing the ordinary ones. Defining **rapidity** (aka **boost**) $\zeta = \tanh^{-1}\beta$,

$$\begin{pmatrix} t \\ x \end{pmatrix} = \begin{pmatrix} \cosh\zeta & \sinh\zeta \\ \sinh\zeta & \cosh\zeta \end{pmatrix} \begin{pmatrix} t' \\ x' \end{pmatrix} \tag{7.3}$$

If we restore $\hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}}$, time and space coordinates form a **4-vector** $x^\mu$ (with $x^0 \equiv t$), and the Lorentz transformation is $x'^\mu = \Lambda^\mu_\nu x^\nu$. Here $\mu$ and $\nu$ are indices that take values 0-3. We use the **Einstein implied summation** convention: when an index appears twice on the same side of an equation (as $\nu$ does here), one upper (contravariant) and one lower (covariant), there's an implied sum over it.

### 7.1.3 The Metric Tensor and Contravariant/Covariant Notation

You might have noticed that indices may appear as superscripts (contravariant) or subscripts (covariant). 4-vectors with units of length, like $x^\mu$, are naturally contravariant.[1] The gradient operator $\partial_\mu$, with units of inverse length, is naturally covariant.

You can raise or lower indices, however, with the **metric tensor** $g^{\mu\nu}$. In flat Minkowski space $g_{\mu\nu} = \text{diag}[+1, -1, -1, -1]$, so effectively you lower the index of a contravariant 4-vector by changing the sign of its spatial elements. $g^{\mu\nu}$ is numerically the same as $g_{\mu\nu}$, and $g^\mu_\nu = \delta^\mu_\nu$ is the identity matrix.

It is tempting, and sometimes useful, to perform a **Wick rotation** to imaginary time: $t = -i\tau$, then the metric for $(\tau, x, y, z)$ is the identity matrix and you can forget all about metric tensors and co-/contravariant indices. Our world, absent gravity, is sort of a simple Euclidean 4-space, with one coordinate always imaginary. But in General Relativity, the metric tensor itself becomes a dynamic quantity, rendering the Wick rotation view less useful, and requiring a much more careful tracking of co-/contravariant indices.

### 7.1.4 The Invariant Interval

The **invariant interval** between two events, one at the origin and one "target" event at $x^\mu = (t, x, y, z)$, is (in Minkowski space)

$$s^2 = |x|^2 = x^\mu x_\mu = g_{\mu\nu} x^\mu x^\nu = t^2 - x^2 - y^2 - z^2 \tag{7.4}$$

Two inertial observers (like "unprimed" and "primed" above) may disagree about times and lengths, but they always agree on the invariant interval between events. Lorentz transformations, including (but not limited to) the boost of Eq. 7.1, preserve the invariant interval, analogous to how in 3-space rotations and translations preserve distance.

The **Lorentz transformation** matrices $\Lambda^\mu_\nu$ are $4 \times 4$ symmetric matrices that preserve the invariant interval $|\Lambda x|^2 = |x|^2$. They form the (restricted homogeneous) **Lorentz group** which is $SO(1,3)$, analogous to how spatial rotations form the rotation group $SO(3)$. They represent six independent transformations: 3 rotations and 3 boosts. If you add translations in space and time, you get the **Poincaré group**.

The signs of $s^2$ and $t$ categorize all possible target events $x^\mu$ by where they lie on the **light cone**, Fig. 7.3. If $s^2 > 0$ and $t > 0$, the target event lies within the **future light cone** of the origin

[1] If you make the length units larger, *e.g.* from cm to m, the numerical values in $x^\mu$ get smaller, thus the "contra".

event, and may be causally influenced by it. If $s^2 > 0$ and $t < 0$, then the target event lies within the **past light cone** and may causally influence it. If $s^2 < 0$, the target is causally disconnected from the origin. If $s^2 = 0$ (on the light cone), the two events may be connected by a light beam. That is, even in General Relativity, light follows a **geodesic** along which $\mathrm{d}s^2 = 0$ everywhere.

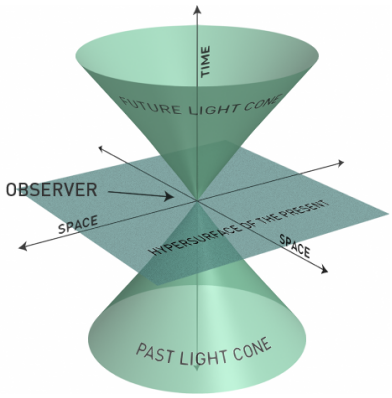



Figure 7.3: Light Cone

Along the path of a massive particle, $\mathrm{d}s^2 = \mathrm{d}t^2(1-\beta^2)$, where now $\beta$ is the particle's speed $\mathrm{d}|\boldsymbol{x}|/\mathrm{d}t$. In the particle's rest frame, where $\beta = 0$, a clock measures the particle's "proper time" $\mathrm{d}\tau$, and we see $\mathrm{d}s = \mathrm{d}\tau$; the interval is the same thing as the proper time. In any other frame, then,

$$\mathrm{d}t = \gamma\,\mathrm{d}\tau \tag{7.5}$$

and we have recovered the formula for time dilation. In the limit $\beta \to 1$ no proper time passes at all; one could say that a photon's clock never ticks.

### 7.1.5 Four-Vector Frenzy

Meaningful relativistic 4-vectors are often formed by pairing scalars with vectors, for example:

- $x^\mu = (t, \boldsymbol{x})$ location in time and space,
- $\partial_\mu = (\partial_t, \nabla) = \partial/\partial x^\mu$ rate of change and gradient,
- $p^\mu = (E, \boldsymbol{p})$ energy and momentum,
- $k^\mu = (\omega, \boldsymbol{k})$ frequency and wavenumber,
- $A^\mu = (\Phi, \boldsymbol{A})$ scalar and vector potentials,
- $J^\mu = (\rho, \boldsymbol{J})$ charge density and current density.

Each of these can be transformed between frames with a Lorentz transformation, *e.g.* $J'^\mu = \Lambda^\mu_\nu J^\nu$.

Physical laws should be writeable in Lorentz-invariant form. For example, local charge conservation is expressed elegantly by

$$\partial_\mu J^\mu = 0 \qquad \text{Charge conservation} \tag{7.6}$$

Written out, that's $\partial_t \rho = \nabla \boldsymbol{J}$, meaning[2] that in any defined volume of space, the change in the charge enclosed equals the flow of current out the surface. Take a moment to convince yourself that this is what conservation means!

## 7.2 Classical Interlude II: 4-Force

Velocity, acceleration, and force $\boldsymbol{F} = \mathrm{d}\boldsymbol{p}/\mathrm{d}t$ are **not** the spatial parts of 4-vectors, because they involve derivatives with respect to observer time, which we now know is not absolute. We can fix this by defining new quantities which are derivatives with respect the the particle's proper time $\tau$ and using Eq. 7.5, so these new quantities relate to the old ones by factors of $\gamma$. Thus we have 4-velocity $u^\mu = \mathrm{d}x^\mu/\mathrm{d}\tau = (\gamma, \gamma\boldsymbol{\beta})$, and note that 4-momentum is just $p^\mu = mu^\mu$. The 4-force $f^\mu = (\gamma\boldsymbol{\beta}\cdot\boldsymbol{F}, \gamma\boldsymbol{F})$. The new form of Newton's Second Law is

$$f^\mu = \frac{\mathrm{d}p^\mu}{\mathrm{d}\tau} = m\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2} \qquad \text{Newton's Second Law} \tag{7.7}$$

The old form $\boldsymbol{F} = \mathrm{d}\boldsymbol{p}/\mathrm{d}t$ remains valid, as long as you remember that $\boldsymbol{p} = \gamma m v$ now. And so we must abandon Eq. 2.1 the mantra of high school physics $\boldsymbol{F} = m\boldsymbol{a}$; that's no longer true. To recap, while the spatial part of 4-current $J^\mu$ is the 3-current $\boldsymbol{J}$, the spatial part of 4-force $f^\mu$ is **not** the 3-force $\boldsymbol{F}$, but rather $\gamma\boldsymbol{F}$, and this sometimes leads to confusion.

## 7.3 Transformations of the Electric and Magnetic Fields

### 7.3.1 The Electromagnetic Tensor $F^{\mu\nu}$

The electric and magnetic fields $\boldsymbol{E}$ and $\boldsymbol{B}$ are **not** spatial components of 4-vectors, but rather together they form the 6 components of an antisymmetric rank-2 tensor, the **electromagnetic tensor** $F_{\mu\nu}$. A rank-2 tensor looks like a $4\times4$ matrix in any given frame.

We asserted above that $A^\mu = (\Phi, \boldsymbol{A})$ is a contravariant 4-vector which combines the electric (scalar) potential with the magnetic (vector) potential. Its covariant form is $A_\mu = (\Phi, -\boldsymbol{A})$. $F_{\mu\nu}$ is defined by derivatives of $A_\mu$:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{pmatrix} \tag{7.8}$$

We see, for example, that $F_{21} = B_z = \partial_x A_y - \partial_y A_x$, which just recovers the curl formula $\boldsymbol{B} = \nabla\times\boldsymbol{A}$. And $F_{01} = E_x = -\partial_x\Phi - \partial_t A_x$, recovering the formula $\boldsymbol{E} = -\nabla\Phi - \dot{\boldsymbol{A}}$ from Eq. 6.6.

$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is what mathematicians called an "exterior derivative" ($F = \mathrm{d}A$ in differential geometry notation), and it looks like a 4-dimensional version of the curl. The spatial part is just $\boldsymbol{B} = \nabla\times\boldsymbol{A}$; in 3 dimensions an antisymmetric rank-2 tensor maps to a pseudovector. $\boldsymbol{E}$ is kind of curl-ish too but with one direction being time. In electrostatics where the $\partial_t$ piece vanishes, the remaining term looks like a 3-gradient of the component $A_0 = \Phi$.

[2] Integrate both sides over a volume $V$, then apply the **divergence theorem** from Appendix B.3.1, just as we did earlier to relate Coulomb's and Gauss's Laws.

Under a Lorentz transformation, a rank-2 tensor needs two copies of the Lorentz transformation matrix shown in abbreviated form in Eq. 7.1. In linear algebra notation, $F' = \Lambda^T F \Lambda$. Showing the indices, that's $F'_{\alpha\beta} = \Lambda^\mu_\alpha F_{\mu\nu} \Lambda^\nu_\beta$.

The Lorentz force law of Eq. 6.2 becomes

$$f^\mu = \frac{\mathrm{d}p^\mu}{\mathrm{d}\tau} = m\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = qF^{\mu\nu}u_\nu \tag{7.9}$$

using the 4-force $f^\mu$, 4-velocity $u^\mu$, and proper time $\tau$ introduced in the previous subsection.

Now we will consider a particular example to show how an electric field in one frame becomes an electric and a magnetic field in another, cementing the notion that electricity and magnetism are two aspects of the same phenomenon. This was the **Second Unification**.

### 7.3.2 Parallel Rods of Charge

Consider two parallel infinite rods of linear charge density $\lambda$ and mass density $\mu$, at rest in the primed frame. The rods extend in the $\hat{\boldsymbol{x}}$ direction, and are separated by a distance $r$ in the $\hat{\boldsymbol{y}}$ direction. We want to calculate the repulsive force per unit length $\mathcal{F}$ exerted by one rod on the other. Transverse force per unit length is Lorentz-invariant,[3] so $\mathcal{F}' = \mathcal{F}$.

In the primed frame there is only an electric field. Using Gauss's Law we have shown the field is $E'_y = 2\lambda/r$, producing a repulsive force per unit length on the second rod $\mathcal{F}' = 2\lambda^2/r$.

In the unprimed frame, the rods are moving at $+v\hat{\boldsymbol{x}}$. The net force per unit length is a combination of electric repulsion and magnetic attraction. Length contraction increases the charge density on each rod by $\gamma$, so the unprimed $E_y = 2\gamma\lambda/r$ and the electric force per unit length is $\mathcal{F}_E = 2\gamma^2\lambda^2/r$. The current in each rod is $I = \gamma\lambda v$, the magnetic field is $B_z = 2I/(rc) = 2\beta\gamma\lambda/r$, and the resulting magnetic force per unit length is $\mathcal{F}_B = -2\gamma^2\lambda^2\beta^2/r$. The sum is $\mathcal{F} = \mathcal{F}_E + \mathcal{F}_B = 2\lambda^2/r$, in agreement with the primed frame. Pure Coulomb repulsion in the primed frame becomes a combination of electric and magnetic forces in the unprimed frame.

In the primed frame, the only non-zero component of $F'_{\mu\nu}$ is $E'_y = 2\lambda/r$. Let's now write out the inverse Lorentz transformation (Eq. 7.2, the one with all plus signs) to find $F_{\mu\nu}$ in the unprimed frame:

$$\begin{pmatrix} 0 & 0 & \gamma E'_y & 0 \\ 0 & 0 & -\beta\gamma E'_y & 0 \\ -\gamma E'_y & \beta\gamma E'_y & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & E'_y & 0 \\ 0 & 0 & 0 & 0 \\ -E'_y & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{7.10}$$

from which we recover the correct unprimed fields $E_y = \gamma E'_y = 2\gamma\lambda/r$ and $B_z = \beta\gamma E'_y = 2\beta\gamma\lambda/r$. Our success should convince you of our original assertion that $A^\mu = (\Phi, \boldsymbol{A})$ is a true contravariant 4-vector.

[3] To show this, fix the first rod and let the second drift in the $\hat{\boldsymbol{y}}$ direction for a time $\mathrm{d}t'$ in the primed frame. Using Newton's Second Law $\mathcal{F}' = \mu a'$, we get $\mathrm{d}y = \frac{1}{2}(\mathcal{F}'/\mu)\mathrm{d}t'^2$ (non-relativistic approximation is okay for small $\mathrm{d}t'$). In the unprimed frame, due to time dilation, the allotted time is $\mathrm{d}t = \gamma\mathrm{d}t'$. Due to length contraction, the mass per unit length is $\gamma\mu$. Since relativistic momentum $p = \gamma m v$, Newton's Second Law is $\mathcal{F} = \gamma(\gamma\mu)a$, and $\mathrm{d}y = \frac{1}{2}(\mathcal{F}/(\gamma^2\mu))(\gamma\mathrm{d}t)^2$. The two frames must agree on the transverse displacement $\mathrm{d}y$, all the $\gamma$'s cancel out, so $\mathcal{F}' = \mathcal{F}$, QED. Note that if we give the rods finite rest length $L \gg r$, the total transverse force $F_\perp = F'_\perp/\gamma$, in agreement with Purcell and Morin (2nd ed) chapter 5 eq. 31.

### 7.3.3 The Fields of a Moving Charge

Now imagine a single point charge $Q$ moving to the right at speed $\beta$ and crossing the origin at $t = 0$: $\boldsymbol{x} = \beta t\,\hat{\boldsymbol{x}}$. Does Coulomb's Law Eq. 6.1 still apply? Place a test charge $q$ at $P = (0, r, 0)$, *i.e.* $r$ light-seconds above the origin in the y-direction. At $t = 0$ does it feel a force pointing straight up, directly away from where the source $Q$ is *right now*, and does that violate causality?

We can re-use the Lorentz transform of Eq. 7.10. In the primed frame, $Q$ is just a charge at rest with a static Coulomb field, and at point $P$ (which is also $(0, r, 0)$ in the primed frame) the field points up with $E'_y = Q/r^2$. Eq. 7.10 tells us that in the unprimed frame, $E_y = \gamma Q/r^2$, while $E_x = E_z = 0$, and also there is a magnetic field (but our test charge doesn't care because it's not moving). The source's full electric field looks like Fig. 7.4: stronger than Coulomb (by $\gamma$) perpendicular to the direction of motion, and weaker (by $1/\gamma^2$) along it. Indeed, if the source $Q$ were a small sphere, length contraction would make it look like an oblate spheroid and $\boldsymbol{E}$ would still be normal everywhere to that surface. So the answer to our first question is no, Coulomb's Law does not work here.

Also Newton's Third Law fails, since the force on the test charge is $\gamma qQ/r^2$, while the force on the source is $-qQ/r^2$; the "system" appears to gain net momentum in the $\hat{\boldsymbol{y}}$ direction. This is a hint that momentum (and energy) are also carried by the fields themselves, and we must include them as part of the "system".

The test charge at $t = 0$ cannot have causal contact with the source $Q$ after $t = -r\gamma$, when the source is at $x = -r\beta\gamma$, and yet the force on it emanates from the source's position at $t = 0$. Causality is not violated because that position was completely predictable. In fact, if something deflects the source during the window $-r\gamma < t < 0$, the test charge will not know that. In this case the apparent origin of the force is not where the source is, was, or will be, but only where it *would have been* if not for the deflection. Deflection means acceleration, accelerating charges radiate, and that radiation carries information about the change outward at speed $c$.

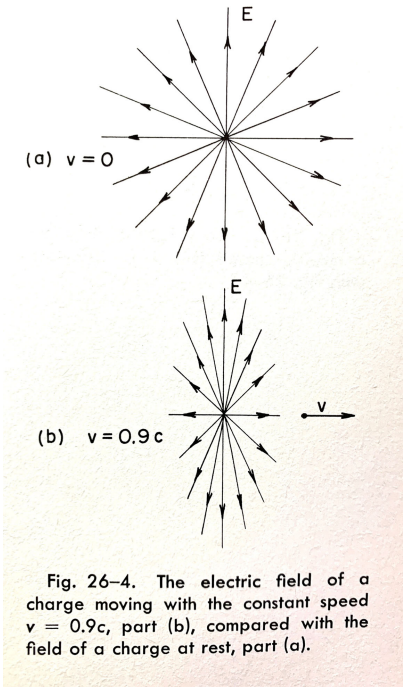



Fig. 26–4. The electric field of a charge moving with the constant speed v = 0.9c, part (b), compared with the field of a charge at rest, part (a).

Figure 7.4: The Electric Field of a Moving Charge

We now quote the complete electric field from a moving point source, Jackson[3] Eq. (14.14):

$$\boldsymbol{E}(\boldsymbol{x},t) = e\left[\frac{\boldsymbol{n}-\boldsymbol{\beta}}{\gamma^2(1-\boldsymbol{\beta}\cdot\boldsymbol{n})^3R^2} + \frac{\boldsymbol{n}\times\{(\boldsymbol{n}-\boldsymbol{\beta})\times\dot{\boldsymbol{\beta}}\}}{c(1-\boldsymbol{\beta}\cdot\boldsymbol{n})^3R}\right]_{\text{ret}} \tag{7.11}$$

It's a lot to digest but we can now recognize a few features. The first thing to note is that the entire expression is evaluated at the "retarded time" when the source passes through the past light cone of the test charge, so information travels at light-speed and causality is preserved. Second, $R\boldsymbol{n}$ is a vector from the "projected" position of the source to the test charge at $(\boldsymbol{x},t)$; the projected position is where the source would be at time $t$ if it had continued at constant velocity $\boldsymbol{\beta}$. Third, the expression has two terms: a squished Coulomb field like Fig. 7.4 that goes like $1/R^2$, and a radiation field that goes like $\dot{\boldsymbol{\beta}}/R$ and is transverse to $\boldsymbol{n}$. Here $c\dot{\boldsymbol{\beta}}$ is the acceleration at the retarded time. If the acceleration is periodic, then the radiation field will be periodic (light). The $\boldsymbol{B}$ field is just $\boldsymbol{B} = \boldsymbol{n}\times\boldsymbol{E}$, also of course evaluated at the retarded time, and splits into a Biot-Savart term and a radiation term.

## 7.4 Relativistic Electrodynamics

Maxwell's Equations become:

$$\partial_\mu F^{\mu\nu} = 4\pi J^\nu \qquad \text{Gauss + Faraday} \tag{7.12a}$$

$$\partial_{[\alpha}F_{\beta\gamma]} = 0 \qquad \text{No monopoles + Ampère-Maxwell} \tag{7.12b}$$

The square brackets around indices in the second equation indicate a fully antisymmetric sum. The first ("inhomogeneous") equation fully describes how fields arise from sources, while the second ("homogeneous") equation follows from the definition of $F_{\mu\nu}$.

If you like compact notation, you'll love differential geometry, where $F = \mathrm{d}A$, from which it follows that $\mathrm{d}F = 0$, and the inhomogeneous Maxwell's Equation is $\mathrm{d}^*F = 4\pi J$.

Recalling Eq. 6.7, $F_{\mu\nu}$ is invariant under the gauge transformation

$$A'_\mu = A_\mu - \partial_\mu\Lambda \tag{7.13}$$

where $\Lambda$ is any scalar function of $x^\mu$. That's now obvious from its definition, Eq. 7.8, since $F_{\mu\nu}$ changes by $-\partial_\mu\partial_\nu\Lambda + \partial_\nu\partial_\mu\Lambda = 0$.

A particular gauge choice is given (implicitly!) by the Lorenz gauge-fixing condition $\partial_\mu A^\mu = 0$. Then the inhomogeneous Maxwell Equation is $\Box A^\nu = 4\pi J^\nu$ (where $\Box \equiv \partial_\mu\partial^\mu$ is the **d'Alembertian** operator). Away from sources, $J^\nu = 0$, and $\Box A^\nu = 0$ is a wave equation for light.

### 7.4.1 Duality

I have presented the Coulomb force as primary, with magnetism a relativistic by-product, and that's reasonable given the existence of electric charges but no magnetic monopoles. Yet, some Grand Unified Theories predict the existence of monopoles, in which case there's more symmetry between $\boldsymbol{E}$ and $\boldsymbol{B}$. You can write down the "Hodge dual" of $F_{\mu\nu}$:

$$\tilde{F}^{\mu\nu} \equiv \frac{1}{2}\varepsilon^{\mu\nu\alpha\beta}F_{\alpha\beta} = \begin{pmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & E_z & -E_y \\ B_y & -E_z & 0 & E_x \\ B_z & E_y & -E_x & 0 \end{pmatrix} \tag{7.14}$$

where $\varepsilon^{\mu\nu\alpha\beta}$ is the **Levi-Civita symbol**. It should be clear that if you had **only** magnetic monopoles, you'd end up with Maxwell's Equations but with $\boldsymbol{E}$ and $\boldsymbol{B}$ switched. And in the absence of sources, *e.g.* light traveling through empty space, $\boldsymbol{E}$ and $\boldsymbol{B}$ are equal partners.

### 7.4.2 Invariants

$\frac{1}{2}F^{\mu\nu}F_{\mu\nu} = B^2 - E^2$ and $-\frac{1}{4}\tilde{F}^{\mu\nu}F_{\mu\nu} = \boldsymbol{B}\cdot\boldsymbol{E}$ are both Lorentz invariant.

### 7.4.3 Electromagnetic Stress-Energy Tensor

The **electromagnetic stress-energy tensor**, which is what gravity couples to, is defined as $T^{\mu\nu} = \frac{1}{4\pi}\left(F^{\mu\alpha}F^{\nu}_{\alpha} - \frac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right)$. The 00 component is energy density $\rho = (E^2 + B^2)/(8\pi)$. The time-space components are the Poynting vector $\boldsymbol{S} = \frac{1}{4\pi}\boldsymbol{E}\times\boldsymbol{B}$, which gives the energy flux (or equivalently the momentum density) of the electromagnetic field. Energy conservation (compare to Eq. 7.6) requires that

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\boldsymbol{S} + \boldsymbol{E}\cdot\boldsymbol{J} = 0 \tag{7.15}$$

which in Lorentz-invariant notation is $\partial_\nu T^{\mu\nu} + f^\mu = 0$, with $f^\mu$ the Lorentz 4-force of Eq. 7.9. If you work hard, or follow Feynman[8] vol 2 chapter 27-3, you'll find this follows from Maxwell's Equations.

# Chapter 8

# Light is a Particle: The Photon

Classically (and historically), light is a wave and the electron is a particle. But in the quantum world, both can be seen traveling as waves and interacting as particles. The photoelectric effect shows that light interacts as a particle (with a discrete amount of energy and momentum), and the **electron double-slit experiment** shows that electrons travel as waves (exhibiting interference). In this chapter we discuss **photons**, the particles of light, while in the next chapter we dive into the quantum theory of electrons.

## 8.1 The Photoelectric Effect

The **photoelectric effect**, first observed in 1887, provided early evidence for the quantum nature of light (and also got Albert Einstein a Nobel Prize). If you shine light on a metal, sometimes electrons pop off the surface. Under the wave model of light, the oscillating electric field wiggles the electrons until they have enough energy to escape the binding energy of the metal matrix. You'd think the energy transferred would depend on the light intensity, frequency, and time applied. But actually it only depends on the frequency!

Fig. 8.1 plots the kinetic energy of the liberated electrons against light frequency $\nu$, and you can see it's a straight line with slope $h$ (the **Planck constant**). Einstein's interpretation was that each electron had absorbed a single particle of light (photon) with energy $E = h\nu$. A fixed amount of that energy went into escaping the metal, and the rest became kinetic energy. More intense beams of light just produced a larger number of liberated electrons.

It follows from $E = pc$ that the momentum transferred is $p = h/\lambda$. This was later confirmed experimentally in **Compton scattering** of X-rays off of free electrons.

The Planck constant $h$ has units of angular momentum and value $h = 6.626 \times 10^{-34}\,\text{Joule} \cdot \text{s}$. We also define the **reduced Planck constant** $\hbar = h/2\pi$. Recall that $\omega = 2\pi\nu$ and $k = 2\pi/\lambda$, so we commonly write

$$E = \hbar\omega \quad , \qquad p = \hbar k \tag{8.1}$$

We really showed that *transferred* energy is quantized, but we tend to say light *consists* of particles (**photons**) with energy $E = \hbar\omega$ and momentum $p = \hbar\omega/c$. Since observing light involve energy and momentum transfer, the distinction is subtle. Still, one simple answer to the age-old question "wave or particle?" is that light travels as a wave but interacts as a particle. That may violate our intuition, but we need to get used to it, since it turns out that everything works that way.

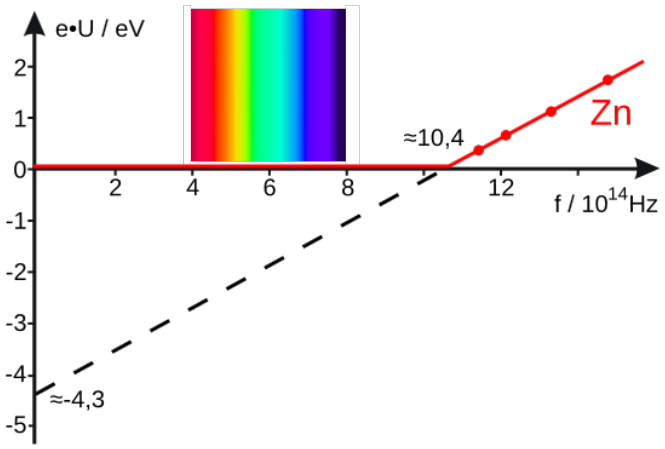



Figure 8.1: The Photoelectric Effect

## 8.2 Phonons

The phonon is to sound as the photon is to light. Here "sound" means vibrations in a crystal lattice, such as salt (NaCl). Unlike sound in air, which only supports longitudinal waves, **acoustic phonons** in crystal can be longitudinal or transverse. Modes that vibrate ions (like the $Na^+$ and $Cl^-$) with respect to one another are called **optical phonons**, as they interact with the electromagnetic field. The phonon is an example of a quasiparticle, in that it describes a collective behavior, but has particle-like properties such as quantized energy and momentum.

Just as blackbody radiation can be described as a thermal bath of photons, heat energy in a crystal can be described as a thermal bath of phonons; this view can be used to predict a crystal's heat capacity and thermal conductivity.

## 8.3 Polarization as a Two-State System

There are two common approaches to describing a quantum state: the wavefunction $\psi(x)$ or the state vector $|\psi\rangle$. The relation is that the wavefunction $\psi(x)$ is the projection of state $|\psi\rangle$ onto the infinite-dimensional basis of position eigenstates $|x\rangle$, that is, $\psi(x) = \langle x|\psi\rangle$. Usually students learn about wavefunctions first, then generalize to the concepts of state vector and basis. However, we defer discussion of wavefunctions until the next chapter. Here we discuss a photon's **polarization** as a two-state system. Now is a good time to review Appendix B.6 on Linear Algebra.

Quantum mechanics is linear algebra on a complex Hilbert space.

### Basis

A photon traveling in the $\hat{z}$ direction can be polarized in either the $\hat{x}$ or the $\hat{y}$ direction, so these two states $|x\rangle$ and $|y\rangle$ form a basis for the possible states: $|\psi\rangle = a|x\rangle + b|y\rangle$, where $a$ and $b$ are complex, and $|a|^2 + |b|^2 = 1$ to describe one photon. We can alternatively use the notation $\binom{a}{b}$, as long as the basis has already been specified.

You are free to choose a different basis. One example is a pair of linear polarizations rotated by angle $\theta$ from our initial basis: $|x'\rangle = \cos\theta\,|x\rangle + \sin\theta\,|y\rangle$ and $|y'\rangle = \cos\theta\,|y\rangle - \sin\theta\,|x\rangle$. Another choice

of basis consists of the circularly polarized[1] states $|L\rangle = \frac{1}{\sqrt{2}}(|x\rangle + i|y\rangle)$ and $|R\rangle = \frac{1}{\sqrt{2}}(|x\rangle - i|y\rangle)$.

To be pedantic, a "two-state system" can be in any one of an infinite number of possible states, parametrized by $a$ and $b$. But it takes exactly two orthonormal states (like $|x\rangle$ and $|y\rangle$) to span the space. The state space is a two-dimensional complex Hilbert space.

### Classical vs Quantum Description

We know that for light polarized in the $\hat{x}' = \cos\theta\,\hat{x} + \sin\theta\,\hat{y}$ direction and passed through an x-polarized filter, the final intensity is $\cos^2\theta$ times the initial intensity. Classically, the $\boldsymbol{E}$ vector is projected onto the $\hat{x}$ direction (the $E_y$ component is eliminated), and intensity is proportional to $\boldsymbol{E}^2$. In a quantum description, the state $|x'\rangle$ is collapsed into the state $\cos\theta\,|x\rangle$, and therefore the probability of a photon passing is $\cos^2\theta$. Importantly, each photon that does make it through is now x-polarized, and is still "full-strength" ($E = \hbar\omega$).

Our classical and quantum descriptions are really quite similar, with the direction of $\boldsymbol{E}$ replaced by the state vector, and its squared magnitude by the number of photons $N$. Even the use of complex numbers for circularly-polarized states has a classical precedent. But as $N \to 1$ the idea of projecting (and therefore weakening) the field is replaced by a probability for passing (or blocking) each photon. Single-photon experiments are all about statistics.

### Other 2-State Systems

There are other interesting 2-state systems. A spin-1/2 particle, such as an electron, can be described as $|\psi\rangle = a|\uparrow\rangle + b|\downarrow\rangle$. An ammonia molecule[8] can have its nitrogen atom above or below the plane of the hydrogens, so $|\psi\rangle = a|\text{Above}\rangle + b|\text{Below}\rangle$. Schrödinger's cat is in the state $|\psi\rangle = a|\text{Alive}\rangle + b|\text{Dead}\rangle$. A qubit in a quantum computer is $|\psi\rangle = a|0\rangle + b|1\rangle$. Mathematically these systems all look the same.

## 8.4 Lasers

Consider a mirrored cavity filled with atoms in a particular excited state $|b\rangle$. You might think that, as the atoms relax to state $|a\rangle$, they would emit photons of frequency $\omega = (E_b - E_a)/\hbar$ but with different random polarizations and directions. A surprising feature of bosons (such as photons) is that they prefer to enter the same state as others in their environment. So if this cavity is filled with x-polarized photons, each atom will tend to also emit an x-polarized photon. This bunching effect, called **stimulated emission**, is responsible for the **laser** ("Light Amplification by Stimulated Emission of Radiation"). Laser light consists of many photons all with the same energy (frequency), polarization, and direction $\boldsymbol{k}$, which is called **coherent** light. That is, coherent light is a highly excited state of a single mode $(a^\dagger)^n|0\rangle$, rather than singly-excited states of many different modes.

Einstein taught us to think of the emission rate as the sum of **spontaneous emission** $AN_b$ and **stimulated emission** $BU(\omega)N_b$, where $N_b$ is the density of atoms in the excited state, $U(\omega)$ is the blackbody photon density (Eq. 11.13), and $(A, B)$ are the **Einstein coefficients**. In thermal equilibrium, detailed balance sets constraints on the coefficients, notably that the same coefficient $B$ applies to stimulated emission and (stimulated) absorption. This makes some sense classically, as they both represent coupling of the atomic dipole moment to the electric field.

[1] In an unfortunate clash of conventions, the positive helicity state is called "left circularly polarized" because, as the light approaches you, the $\boldsymbol{E}$ vector appears to rotate counterclockwise.

A laser is not in thermal equilibrium. It must have a **population inversion**, $N_b > N_a$, which is achieved by other means (such as exciting a higher state $c$ which then quickly decays to $b$). But the stimulated emission rate is still $BU(\omega)N_b$, proportional to the photon density in the cavity. In a continuous laser, a relatively small fraction of the photon bath is allowed to escape, while a high $U(\omega)$ is maintained inside the cavity.

## 8.5 Classical Limits

If the rules of quantum mechanics (or quantum field theory) apply equally to photons (light) and electrons, why are their classical limit so different? Away from the quantum regime, why is light a wave but electrons are particles? Peierls[9] sec. 1.3 shows how these are the necessary classical limits for a massless boson and a massive fermion respectively. In a nutshell: a classical wave needs both amplitude $A \sim \sqrt{N}$ and phase $\phi$ specified accurately, but $\Delta N \Delta\phi \geq 1$ by the Uncertainty Principle, so you need large $N$ (bosons) and the ability to alter $N$ at low energy cost (massless or nearly so). On the other hand, a classical particle implies density and current operators that form a 4-vector $J^\mu$ from the square of field operators, which you can do for fermions ($J^\mu = i\bar{\Psi}\partial^\mu\Psi$) but not bosons.

## 8.6 Quantum Philosophy

We saw that a photon that passes an x-polarizer collapses into the $|x\rangle$ state. In fancier terms, the system collapses to an eigenstate of the Hermitian operator that represents the observation. Here's the problem: The natural evolution of a quantum state (*e.g.* Schrödinger's Equation) is unitary, but wavefunction collapse is not unitary. "Unitary" has a precise mathematical meaning ($\psi(t) = U(t)\psi(0) = e^{-iHt}\psi(0)$ where $U^\dagger U = 1$), but part of that is the implication that $|\psi|^2$ remains unchanged. When $a|x\rangle + b|y\rangle$ collapses to $a|x\rangle$ its norm is not preserved. Is observation a supernatural phenomenon? Do we need one set of rules for evolution of states, and another for observations?

Of course we have to mention **Schrödinger's cat**. A box contains a cat whose life depends on the result of a radioactive decay, which is a quantum event. Until we open the box to observe, is the cat in a superposition of alive and dead? Does the cat count as an observer? Is human consciousness required? (But then consider **Wigner's friend**.) Mystics have had a field day with this idea.

There are several schools of thought here:

1. The **Copenhagen interpretation** says that observation collapses the wavefunction.

2. The "shut up and calculate" approach says that if you can predict the outcome of experiments, you don't need to worry about philosophy.[2]

3. The statistical approach says that $|\psi\rangle$ describes the state of an ensemble of identical systems; an individual system does not have a state.

4. The **hidden variables** hypothesis suggests that there really is an underlying classical state, we just don't know it. This is demonstrably incorrect, as shown by Bell's Inequality in the next section.

[2]Really that's just a cynical restatement of the Copenhagen Interpretation, probably due to N. David Mermin.

5. The **Many-Worlds Interpretation** says that collapse is the illusion; what happens is that the quantum system **entangles** with the observer.

### 8.6.1 Many Worlds

If the Schrödinger's Cat box is opened after one half-life, Many-Worlds says its state is

$$\psi = \frac{1}{\sqrt{2}}(|\uparrow\rangle|\text{Alive}\rangle|\text{Observe Alive}\rangle + |\downarrow\rangle|\text{Dead}\rangle|\text{Observe Dead}\rangle) \tag{8.2}$$

where the first ket ($|\uparrow\rangle$ or $|\downarrow\rangle$ refers to the radioactive decay, the second to the cat's health, and the third to the observer's brain. But the complex, macroscopic nature of the observer's brain disallows any practical possibility of interference terms in the future evolution of the state, so its two parts ("worlds") thereafter evolve independently. This process is called **quantum decoherence**.

The Many Worlds Interpretation (MWI) states that every quantum event since the beginning of time has split the wavefunction of the universe into an unimaginably large sum of terms, each of which represents a different history. First proposed by Hugh Everett III in his 1957 Ph.D. thesis, MWI was initially ridiculed, but has since gained acceptance. It has also diffused into popular culture (*e.g.* the 2022 movies "Everything Everywhere All at Once" and "Doctor Strange and the Multiverse of Madness"). One could speculate that the wavefunction of the universe at the Big Bang is simple and low-entropy (*e.g.* see the **Hartle-Hawking state**), but its evolution can be described as a sum of very complex decoherent histories, including of course our own, something roughly like

$$|\Psi(t)\rangle = e^{-iHt}|0\rangle = \Sigma_i a_i|\text{World}_i\rangle \tag{8.3}$$

There used to be a website called "Universe Splitter", where you could pose a question like, "Should I ask her out?" On the back end, a single photon was passed through a beam splitter. If it went one way the website responded "yes", otherwise "no". By entangling you with a clearly quantum system, it evaded the more difficult question of whether ordinary human decisions are deterministic or quantum (and whether that has any relation to "free will"). Did the "Universe Splitter" really bifurcate the wavefunction of the universe at every use?

One more step down the rabbit hole: If time travel were possible (and I'm not saying it is), one objection is the **grandfather paradox**. If you go back in time and kill your grandfather, how were you ever born to commit the deed? MWI says your grandfather was in a superposition of $|\text{Alive}\rangle$ and $|\text{Dead}\rangle$ (like Schrödinger's Cat, Eq. 8.2), and your temporal travel just moved you from one branch to the other. "Many worlds" then becomes equivalent to "alternate timelines". Serious papers have been written[10] about consistent quantum states in the presence of closed time-like curves.

### 8.6.2 The Anthropic Principle

Earth is a beautiful rocky planet with plenty of water and carbon for life, in a stable nearly-circular orbit around a stable star that keeps Earth's temperature suitable for liquid water. How many of the other roughly $10^{23}$ planets in the observable universe are suitable for life? Our one observation (Earth itself) tells us little, because we have to be on such a planet to ask the question. This is a weak (and somewhat vacuous) form of the **anthropic principle**.

It took a few billion years for complex, multi-cellular life to develop, with deoxyribonucleic acid (DNA) encoding instructions to form proteins from amino acids, and epigenetic modification

leading to cell differentiation. Is that a reasonable amount of time for evolution to do the job, or were we insanely lucky? Are most of the MWI branches sterile at 13.8 Gy? Maybe we find ourselves on the rare MWI branch where all the right evolutionary events occurred because we are the ones collapsing the wavefunction. That's a stronger anthropic principle.

Young stars fuse hydrogen to helium, while older stars produce heavier elements like carbon by fusing helium. The bottleneck **triple-alpha process**[11] is two heliums combining to form an unstable beryllium-8 (not the stable form beryllium-9), which in its brief existence can fuse with another helium to form the **Hoyle state** of carbon-12, which then relaxes to regular carbon-12. The Hoyle state of carbon, by a fantastic coincidence of physical constants, is an almost perfect energy match (resonance) to beryllium-8 plus helium. Without that resonance, almost none of the heavier elements that make up Earth and life would exist. Is there some very strong anthropic principle that makes the physical constants themselves take values that allow their own observation?

We once hoped that string theory would explain the physical constants. So far that has not worked, but string theory does seem to predict a huge number of "string vacua" as the 10-dimensional theory compactifies to 4D, and these may have different values for the constants. So string theory compactification provides a mechanism for the strong anthropic principle to select a world with observers. That's not a very satisfying explanation, but it's the best we've got.

## 8.7 Entangled Photons and Bell's Inequality

Let's show[12] that quantum mechanics is **inconsistent** with a hidden-variables theory. We set up an experiment in which an atom emits two photons, in opposite directions $-\hat{\boldsymbol{z}}$ and $\hat{\boldsymbol{z}}$, in a state with positive parity and zero angular momentum. On each side, a light-minute away, there is a linear polarization filter followed by a photon detector. As the photon approaches, experimenter 1 rotates polarizer 1 to direction $\hat{a}$, so it will pass an $\hat{a}$-polarized photon on to the detector but block[3] the orthogonal **polarization**. Similarly, experimenter 2 rotates polarizer 2 to direction $\hat{b}$, which is at angle $\theta_{ab}$ with respect to $\hat{a}$, that is $\hat{a}\cdot\hat{b}=\cos(\theta_{ab})$.

We define a correlation function $C_{ab}=(P_{++}+P_{--})-(P_{+-}+P_{-+})$, where $P_{++}$ is the fraction of events where both detectors fire, etc. If the two detectors always get the same result, they are 100% correlated: $C_{ab}=+1$. If the two detectors always get opposite results, they are 100% anticorrelated: $C_{ab}=-1$.

Quantum says that the system is in state

$$|\psi\rangle=\frac{1}{\sqrt{2}}(|\hat{x}_1\hat{x}_2\rangle+|\hat{y}_1\hat{y}_2\rangle) \tag{8.4}$$

where we could use any orthogonal basis $(\hat{x},\hat{y})$, and we choose $\hat{x}=\hat{a}$. When a photon passes polarizer 1, the state collapses to $\frac{1}{\sqrt{2}}|\hat{a}_1\hat{a}_2\rangle$, and the probability photon 2 also passes is $P_{++}=\frac{1}{2}\cos^2\theta_{ab}$. Similarly, $P_{--}=\frac{1}{2}\cos^2\theta_{ab}$, and $P_{+-}=P_{-+}=\frac{1}{2}\sin^2\theta_{ab}$, so (using a double-angle identity)

$$C_{ab}=\cos(2\theta_{ab}) \tag{8.5}$$

This result is confirmed by experiment, for example the calcium cascade decay $4p^2\ {}^1S_0\to 4s4p\ {}^1P_1\to 4s^2\ {}^1S_0$ which starts and ends in $J=0$ states.

[3] To avoid concerns about counting, buy fancier polarizers that redirect "blocked" photons to a different counter.

Using sloppy language, we might say that when photon 1 is detected or blocked, photon 2 is (superluminally!) put into the same polarization state. In particular if $\hat{a} = \hat{b}$ (by luck or prearrangement) then $C_{ab} = 1$; when photon 1 passes we know that photon 2 must pass, and when photon 1 is blocked then photon 2 must be blocked. Unfortunately, you can't build a superluminal communicator this way, because the correlations aren't evident until the experimenters meet later to share data. Still, Einstein called this "spooky action at a distance."

We try to build a model where the atom imparts some hidden variable $\lambda$ to the photons, whose behavior at the polarizers is then local and deterministic (if we knew $\lambda$). That is, there is some $A_a$, a binary function implicitly of $\lambda$ and explicitly of the unit vector $\hat{a}$, which is either $+1$ if photon 1 passes polarizer 1 aligned to $\hat{a}$, or $-1$ if it doesn't. Importantly, $A_a$ doesn't depend on what's happening at polarizer 2. Similarly there is a binary function $B_b$ for photon 2. The atom hands out values of $\lambda$ (to both photons) with some distribution $\rho(\lambda)$. The correlation function $C_{ab} = \int \mathrm{d}\lambda\, \rho(\lambda) A_a B_b$. I'll now suppress the $\mathrm{d}\lambda\, \rho(\lambda)$ under every integral sign and just write $C_{ab} = \int\{A_a B_b\}$. We introduce two new directions $\hat{c}$ and $\hat{d}$, and write

$$\begin{aligned} |C_{ab} - C_{ad}| &= |\int\{A_a B_b(1 - A_c B_d)\} - \int\{A_a B_d(1 - A_c B_b)\}| \\ &\le \int\{|A_a B_b||1 - A_c B_d|\} + \int\{|A_a B_d||1 - A_c B_b|\} \\ &\le \int\{2 - A_c B_d - A_c B_b\} \end{aligned} \tag{8.6}$$

In the first line we subtracted and added $\int\{A_a B_b A_c B_d\}$; in the second we used the triangle inequality, and in the third we used $|A_a B_b| = 1$. We end up with this (piece of) **Bell's Inequality**:

$$C_{ab} - C_{ad} + C_{cd} + C_{cb} \le 2 \tag{8.7}$$

Our quantum (and experimentally verified) result Eq. 8.5 sometimes violates Bell's Inequality. Let's take the particular example where $(\hat{a}, \hat{b}, \hat{c}, \hat{d})$ are unit vectors at angles $(0, 1, 2, 3)\epsilon$ from the x-axis, and $\epsilon$ is small enough that we can approximate Eq. 8.5 with $C_{ab} = 1 - 2\theta_{ab}^2$ for all of the pairings. Then Eq. 8.7 becomes $2(-1 + 9 - 1 - 1)\epsilon^2 \le 0$ which is blatantly false, thus disproving the hidden variable hypothesis.

The idea of hidden variables is also called "realism" or "counterfactual determinism"; the latter means that there are definite (though unknown) results for experiments that were not performed. Here we assumed $A_c$ and $B_d$ have definite values, based on $\lambda$, even when the experiment performed has the polarizers at $\hat{a}$ and $\hat{b}$. People sometimes think that quantum theory violates locality or relativity, but actually it only violates counterfactual determinism.

# Chapter 9

# Light and Quantized Electron Orbitals

Light is emitted or absorbed by atoms when electrons jump between **orbitals** of different energy. Like planets orbiting around the sun (Sec. 2.5), negatively-charged electrons orbit the positively-charged nucleus, in a $1/r$ attractive potential. But quantum mechanics dominates at the atomic scale, resulting in a set of discrete energy levels. A typical "life-cycle" for a photon, particularly of visible or infrared frequency, is that it is created (emitted) in one atom when an electron drops to a lower energy level, travels through space, and then is destroyed (absorbed) in another atom as it raises an electron to a higher energy level. This chapter is devoted to the quantum mechanics of electron orbitals, with our goal to find these discrete energy levels; that is, we will "solve the hydrogen atom".

## 9.1 De Broglie Generalizes the Quantum

The **electron double-slit experiment** (Fig. 9.1) shows that electrons travel and self-interfere like waves. A beam of electrons is blocked by a barrier which has two open slits, beyond which is a phosphor screen. Either slit by itself would allow electrons through to form a Gaussian pattern on the screen, so you'd think that with both slits open the pattern would just be the sum of two Gaussians. But instead, you get an interference pattern! Interference is a sure indication of a wave phenomenon. Just like light, electrons travel as waves and interact as particles.

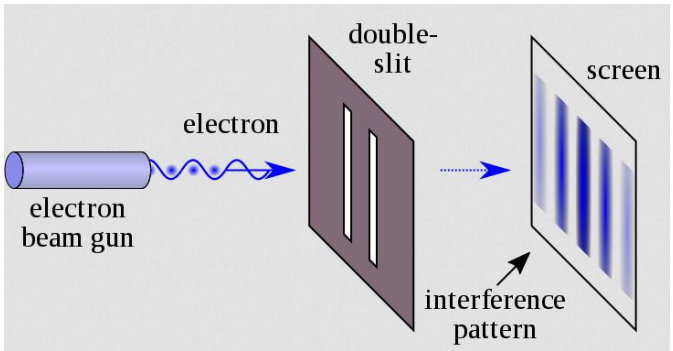



Figure 9.1: The Electron Double Slit Experiment.

Louis de Broglie defined the **de Broglie wavelength** $\lambda_{dB} = h/p$ for electrons in analogy to light, and that correctly predicted the double-slit results. (Note we haven't said anything yet about what it's a wave *of*.)

Rather than just two slits, solid crystals provide a regularly-spaced lattice of scatterers. In 1927, Davisson and Germer shone electrons onta a nickel crystal. The electrons that backscattered off the small, hard nuclei displayed an interference pattern, again demonstrating the wave-like nature of electrons; see Fig. 9.2.

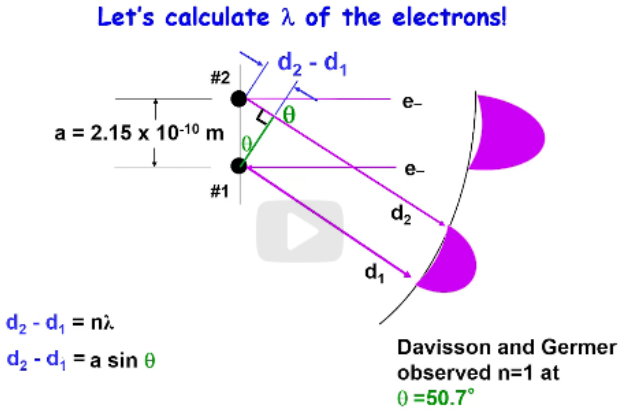



Figure 9.2: Davisson and Germer Electron Backscattering

## 9.2 Hydrogen Energy Levels and Emission Spectra

Point a spectrometer at a sodium vapor lamp, and you will see the light it emits is primarily in two distinct yellow lines, Fig. 5.7. If you excite hydrogen (a more dangerous experiment!) and view it with an instrument that sees into the ultraviolet, you'll see the top row of Fig. 5.7.

Lyman showed that the hydrogen lines are all at energies $\frac{1}{2}\mathcal{E}\left(\frac{1}{n^2}-\frac{1}{m^2}\right)$ where $n$ and $m$ are integers, and correctly concluded a photon was being emitted as an electron dropped from energy level $m$ to energy level $n$. Why are electrons found only in discrete energy levels, with energy $-\frac{1}{2}\mathcal{E}/n^2$? What is the constant $\mathcal{E}$? In fact, why are atoms stable at all? Classically, an orbiting electron is accelerating and therefore should be radiating (losing energy) all the time, spinning down into the nucleus in a death spiral.

The **Bohr model** suggests that electrons have classical circular orbits, but a whole number of wavelengths must fit around the circumference, and the wavelength is the de Broglie wavelength corresponding to the classical momentum. Do the math, and it works! The Bohr model isn't quite right, but it gave early credence to the idea of electron waves responsible for quantization of energy.

Such a wave, however, utterly fails to obey basic rules of relativity (Einsteinian or Galilean)! For example, suppose a non-relativistic electron is moving with speed $v$ and you claim there is a wave of wavelength $\lambda_{dB}=h/(mv)$. An observer co-moving with the electron would disagree; in fact his observed de Broglie wavelength is infinite! So right off the bat we have to say these waves are not directly observable.

(We note, by contrast, that waves of light Lorentz-transform correctly, as we saw in Sub-Sec. 4.5.1. So light waves can be directly observable.)

The solution is to postulate a **wavefunction** $\psi$ which is complex, but complex numbers are not observable. Only $|\psi|^2$ is real and can be physical; we interpret it as a probability density for observing the electron. An electron with fixed momentum has (at a given time) $\psi=e^{ipz/\hbar}$, which has the desired (but unobservable) wavelength $\lambda=h/p$ and uniform probability density. That

wavefunction, depending as it does on $p$, is completely observer-dependent. Yet all observers agree that the electron could be anywhere along the $z$-axis with equal likelihood. (That's our first peek at the Uncertainty Principle: completely known momentum implies completely unknown position.)

Since momentum-energy is a 4-vector $p^\mu$ in relativity, just like space-time $x^\mu$, we generalize the wavefunction to be a function of time as well as space. The full **free-particle wavefunction** is:

$$\Psi = e^{-ip\cdot x/\hbar} = e^{-iEt/\hbar} e^{i\boldsymbol{p}\cdot\boldsymbol{x}/\hbar} \tag{9.1}$$

and thus see the electron wavefunction "rotates" (in complex phase) in time with frequency $\omega = E/\hbar$, as well as in space with wavenumber $k = p/\hbar$, very reminiscent of circularly polarized light.

## 9.3 Classical Interlude III: Harmonic Motion

Newton's Second Law says $\boldsymbol{F} = m\boldsymbol{a}$, where $\boldsymbol{F}$ is the net force on a particle, $m$ is its mass, and $\boldsymbol{a} = \ddot{\boldsymbol{x}}$ is its acceleration (second derivative of position with respect to time). For a **conservative force** $\boldsymbol{F} = -\nabla V$, where $V$ is the potential energy and $\nabla$ is the gradient (derivative with respect to $\boldsymbol{x}$). As a simple example, consider a mass on a spring in one dimension, so $F = -kx$ and $V = \frac{1}{2}kx^2$. Newton's Second Law gives $\ddot{x} = -\omega^2 x$ where $\omega^2 = k/m$. As with any 2nd-order differential equation, the solution requires two constants determined by initial conditions, such as initial position and speed, and can be written in these three different (but equivalent) ways:

$$\begin{aligned} x &= A\sin(\omega t) + B\cos(\omega t) \\ &= C\sin(\omega t + \phi) \\ &= \frac{1}{\sqrt{2m\omega}}(a^* e^{i\omega t} + a e^{-i\omega t}) \end{aligned} \tag{9.2}$$

Note the third expression is real even though it contains complex numbers; the two constants of motion are the real and imaginary parts of $a$; and the normalization $1/\sqrt{2m\omega}$ is chosen in anticipation of the quantum result Eq. 9.8. The particle oscillates back and forth like a sine wave, which is why this is called a **simple harmonic oscillator**. If you plot the particle's position $x$ and momentum $p = m\dot{x}$, you'll see it traces out a circle in $(x, p)$ **phase space**.

In classical mechanics, a particle's state is fully described by a point $(x, p)$ in phase space. The particle's motion under Newton's 2nd Law is determined by a pair of first-order differential equations $\dot{x} = p/m, \dot{p} = F = -V'$, or equivalently a single second-order equation $\ddot{x} = -V'/m$, also known as $F = ma$. Here $V(x, t)$ is the potential. For a system of $N$ particles, the evolution can be described as a path in 6N-dimensional phase space.

## 9.4 The Schrödinger Equation

In (1D non-relativistic) quantum mechanics, a particle's state is fully described by its wavefunction $\Psi$, a complex function which obeys a first-order differential equation called the **Schrödinger Equation** which we will derive. $\rho = |\Psi|^2$ provides a probability density for finding a particle at any given point, and $J = \frac{\hbar}{2im}(\Psi^*\Psi' - \Psi\Psi^{*\prime})$ is the corresponding probability current, related by the **continuity equation**

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} + \frac{\mathrm{d}J}{\mathrm{d}x} = 0 \tag{9.3}$$

This equation describes any conserved quantity, which here is probability (we'll see the same equation later for electric charge). It basically says that any increase in a region's charge is due to inflow into the region. For the free-particle wavefunction, $\rho = 1$ (you can't normalize this wavefunction because it has infinite range in $x$), and $J = p/m$.

The free-particle wavefunction also contains information about the particle's momentum, which we can extract by noting that $\hat{p}\Psi = p\Psi$ where

$$\hat{p} = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x} \tag{9.4}$$

is a **momentum operator** that acts on $\Psi$. Similarly we note that $i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\Psi = E\Psi$. And since $E = \frac{p^2}{2m}$, it follows that $i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\Psi = \frac{\hat{p}^2}{2m}\Psi$. This equation, in conjunction with the requirement of probability conservation, actually gives you back Eq. 9.1.

The 1D generalization of this operator equation is the **Schrödinger Equation**:

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}\Psi = \left(\frac{\hat{p}^2}{2m} + V\right)\Psi = \left(\frac{-\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V\right)\Psi \tag{9.5}$$

which is obeyed by a non-relativistic particle in potential $V$.

In Linear Algebra (Appendix B.6), we distinguish a vector (like an arrow drawn on the blackboard) from its representation in a particular coordinate system (like $(2, 1)$). Similarly we distinguish the quantum state $|\Psi\rangle$ from its representation in the position basis $\Psi(x) = \langle x|\Psi\rangle$. The state can be presented in other bases as well, such as the momentum basis $\tilde{\Psi}(p) = \langle p|\Psi\rangle$, which is related to $\Psi(x)$ by a Fourier transform.

We introduce the idea of **operators** and their **eigenvalues**. An observable $O$ (like $x$ or $p$) becomes an operator $\hat{O}$ (like $\hat{x}$ or $\hat{p}$) that acts on $\Psi$. When we are working in the position representation (describing $\Psi$ as a function of $x$), $\hat{x} = x$ (multiplication by $x$), and $\hat{p} = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}$. If $\hat{O}\Psi = v\Psi$ for a fixed number $v$, $\Psi$ is said to be an **eigenfunction** of $\hat{O}$ with **eigenvalue** $v$.

We want to find **energy eigenstates**, *i.e.* eigenenstates of the Hamiltonian. If you decomposed a particle's state into energy eigenstates at time $t = 0$, then you know the state at all times, because each energy eigenstate evolves as $e^{-iEt/\hbar}$. In position representation, $\Psi(x, t) = e^{-iEt/\hbar}\psi(x)$, where the trick is to find $\psi(x)$ for a given $E$. In first semester Quantum, you spend a lot of time finding the energy eigenstates $\psi(x)$ of simple static potentials. Here are a few examples.

### 9.4.1 Quantum Warmup I: The Square Well

The **infinite square well** is just a potential that is zero between $x = -a$ and $x = a$, and infinite everywhere else. The particle is trapped in the region $-a \le x \le a$, and in that region the solutions to the Schrödinger Equation are sinusoids. To match the left boundary condition, I'm going to write the solutions as $\sin(...(x+a))$. To match the right boundary condition, a half-integer number of wavelengths must fit the well. We'll index the solutions by a non-negative integer $n$, with $n = 0$ the ground state:

$$\psi = \frac{1}{\sqrt{a}}\sin\left(\frac{(n+1)\pi}{2a}(x+a)\right), \quad E_n = \frac{(n+1)^2\pi^2\hbar^2}{2m(2a)^2} \qquad (n \ge 0) \tag{9.6}$$

That's not the most common convention, but I think it fits better with the examples that follow. I have normalized the wavefunction so that $\int_{-a}^{a}|\psi|^2\mathrm{d}x = 1$, because the total probability of finding

the particle somewhere in the well is one. The energy $E_n$ of any level is calculated using $E = p^2/(2m)$ as an operator equation: $\hat{H}\psi = \hat{p}^2/(2m)\psi$, with $\hat{p} = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}$.

We see the allowed energy levels are discrete; with any other value of the energy you could not satisfy the boundary conditions. The **ground state** $\psi_0$ has an energy $E_0 > 0$, unlike the classical solution where a particle just sitting there would have $E = 0$. The ground state has no **nodes** (points where $\psi = 0$, excluding the edges), and state $\psi_n$ has $n$ nodes. Since the potential is symmetric about $x = 0$, even states have symmetric wavefunctions, and odd states have antisymmetric wavefunctions (with a node at $x = 0$).

The **finite square well** is shown in Fig. (fig). Now you have to match a sinusoid in $-a \le x \le a$ with exponentials for $|x| \ge a$, and the exponentials have to get small for large $|x|$ (see Appendix B.3). The solutions are shown in Fig. (fig). As with the infinite well, energy levels are discrete, but now there are only a finite number of them that are bound ($E < V_0$). A particle with higher energy may be influenced by the well but is "just passing through". And critically, even a bound state has some probability of finding the particle outside the walls, in the classically forbidden region. This is related to the quantum phenomenon of **tunneling**.

You can build more complicated potentials that are piecewise flat, and solve the 1D Schrödinger Equation by patching together sinusoids where $V < E$ and exponentials where $V > E$. Unless you have infinitely steep walls or delta-functions, both $\psi$ and $\psi'$ must be continuous across boundaries. Bound states always require discrete energies, found by requiring $\psi$ to be small far from the well. Where $V$ is far below the energy, the wavefunction oscillates rapidly (high momentum) and amplitude remains small; where $V$ is below but closer to $E$, the wavelength gets longer and amplitude higher – reproducing the classical notion that a particle is most often found in those places where it goes slowest!

### 9.4.2 Quantum Warmup II: The Harmonic Oscillator

The **harmonic oscillator** potential is a simple quadratic: $V = \frac{1}{2}kx^2$. Classically any particle in this potential oscillates with angular frequency $\omega = \sqrt{k/m}$. One reason it's important is that almost any particle near a potential minimum acts like it's in a harmonic oscillator for low energies, as you can see just by Taylor-expanding $V$ around $V_0$. The energy levels are discrete and evenly spaced: $E_n = \left(n + \frac{1}{2}\right)\hbar\omega$, again with the ground state $E_0 = \frac{1}{2}\hbar\omega > 0$. The wavefunctions are products of Hermite polynomials with a Gaussian, shown in Fig. (fig).

Rather than solve for the Hermite polynomials explicitly, let's set $\hbar = \omega = m = 1$ for now, and write the Hamiltonian

$$H = \frac{1}{2}(x^2 + p^2) = \frac{1}{2}(x - ip)(x + ip) + \frac{1}{2} \tag{9.7}$$

where that last term arose from the commutation relation $[x, p] = i$ (remember $\hbar = 1$ for now). Define the **annihilation operator** $a = \frac{1}{\sqrt{2}}(x + ip)$, and its conjugate **creation operator** $a^\dagger = \frac{1}{\sqrt{2}}(x - ip)$, which satisfy $[a, a^\dagger] = 1$; then (restoring constants) $H = \hbar\omega(a^\dagger a + \frac{1}{2})$. For any energy eigenstate $|n\rangle$ we have $\langle n|H|n\rangle = \hbar\omega(|a|n\rangle|^2 + \frac{1}{2})$, and the lowest value $E_0 = \frac{1}{2}\hbar\omega$ (ground state energy) occurs when $a|0\rangle = 0$. Turned back into a first-order differential equation, that shows that $\psi_0 = \langle x|0\rangle$ is a Gaussian.

We label the ground state $|0\rangle$, with $\langle x|0\rangle$ a simple Gaussian. The first excited energy eigenstate is $|1\rangle$, and so on. Then $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$; the raising operator transforms any energy eigenstate into the one above it. Similarly $a|n\rangle = \sqrt{n}|n-1\rangle$, and $a|0\rangle = 0$. The combination $a^\dagger a = N$ has the simple behavior $N|n\rangle = n|n\rangle$.

Now the energy in light can be written $H = E^2 + B^2$, and that looks a little like the harmonic oscillator. Indeed, for any given oscillation mode, the energy $E_n = \frac{n+1}{2}\hbar\omega$ can be interpreted as $n$ photons. Notably there is a zero-point energy $E_0$ even in the absence of photons. One case where that matters is a metal box, where bringing the walls together changes the available modes in such a way as to reduce the zero-point energy. Thus there is a small attractive force between any two conducting plates, called the **Casimir effect**.

### 9.4.3 Unitary Evolution and the Heisenberg Picture

In the **Schrödinger Picture** that we have used so far, and restricting ourselves for now to time-independent Hamiltonians, the state evolves as $\psi(x,t) = U(t)\psi(x,0)$ with the unitary operator $U(t) = e^{-iHt}$. Unitarity ensures that one particle remains one particle, *i.e.* the norm of the state ($\int |\psi|^2 = 1$) is preserved. Operators like $\hat{x}$ and $a^\dagger$ are time-independent, and observables $\langle\psi|\hat{O}|\psi\rangle$ evolve because the state does.

By contrast, in the **Heisenberg Picture**, states are static while the operators are time-dependent. In order to achieve the same predictions for observables, we define the Heisenberg version of any operator $\hat{O}_H(t) = U^\dagger(t)\hat{O}U(t)$. Heisenberg operators obey equations of motion, *e.g.* $\dot{O} = i[H, O]$, that look a lot like classical equations of motion, *e.g.* $\dot{O} = \{H, O\}$ where $\{,\}$ are **Poisson brackets**.

For the harmonic oscillator in the Heisenberg picture, the raising and lowering operators become $a^\dagger(t) = a^\dagger e^{i\omega t}$ and $a(t) = ae^{-i\omega t}$. The position operator is

$$\hat{x}_H(t) = \frac{1}{\sqrt{2m\omega}}\left(a^\dagger e^{i\omega t} + ae^{-i\omega t}\right) \tag{9.8}$$

which will be useful when we discuss field theory, where the dynamical variables $\Phi(\boldsymbol{x}, t)$ are treated as Heisenberg operators. You see we have effectively promoted the integration constants $a$ and $a^*$ of Eq. 9.2 to quantum operators. We can also write

$$\hat{x}_H(t) = \hat{x}\cos(\omega t) + \frac{1}{m\omega}\hat{p}\sin(\omega t) \tag{9.9}$$

which looks like the classical result; if you start in a **coherent state** with non-zero $\langle\hat{x}\rangle$ and/or $\langle\hat{p}\rangle$, then $\langle\hat{x}_H(t)\rangle$ sloshes back and forth as your classical intuition would expect.

## 9.5 The Hydrogen Atom

An electron bound to a nucleus of $Z$ protons obeys the 3D Schrödinger Equation with $V = -Ze^2/r$ ($Z$ is the atomic number, so $Z = 1$ for hydrogen). That is, the energy eigenstates $\psi$, which we call **electron orbitals**, obey:

$$\frac{-\hbar^2}{2m_e}\nabla^2\psi - \frac{Ze^2}{r}\psi = E\psi \tag{9.10}$$

or equivalently, using the angular momentum operator $\hat{\boldsymbol{L}}$,

$$\frac{-\hbar^2}{2m_e}\frac{1}{r}\frac{\partial^2}{\partial r^2}r\psi + \frac{\hat{\boldsymbol{L}}^2}{2m_e r^2}\psi - \frac{Ze^2}{r}\psi = E\psi \tag{9.11}$$

The latter version suggests a solution in spherical coordinates of the factorized form

$$\psi_{nlm} = r\,u(r)Y_l^m(\theta,\phi)\chi \tag{9.12}$$

Here $\chi$ is the spin state, which does not appear in the lowest-order Hamiltonian (left side of Eq. 9.10), and therefore only contributes a 2-fold degeneracy – in other words, each orbital can be occupied by one "spin up" electron and one "spin down".

The **spherical harmonics** $Y_l^m$ are characterized by two integers: $l \geq 0$ and $-l \leq m \leq l$. As a set, they form a complete basis for any function of $(\theta, \phi)$. If you've seen drawings of electron orbitals in chemistry class, then you have an idea how they look. $Y_0^0 = 1/\sqrt{4\pi}$ is a uniform spherical shell, like s-orbitals in chemistry; the three $Y_1^m$ (for $m$ in (-1,0,1)) make the p-orbitals; the five $Y_2^m$ (for $m$ in (-2,-1,0,1,2)) make the d-orbitals, and so on. $Y_l^m$ is an eigenfunction of $\hat{\boldsymbol{L}}^2$ with eigenvalue $\hbar^2 l(l+1)$, and also of $\hat{L}_z$ with eigenvalue $\hbar m$.

After replacing $\hat{\boldsymbol{L}}^2$ with $\hbar^2 l(l+1)$ in Eq. 9.11, the radial equation for $u(r)$ looks a lot like a 1-D Schrödinger equation:

$$\frac{-\hbar^2}{2m_e}u'' + \left(\frac{\hbar^2 l(l+1)}{2mr^2} - \frac{Ze^2}{r}\right)u = Eu \tag{9.13}$$

To account for the coordinate constraint $r \geq 0$, we impose $u(0) = 0$, as if there's an infinite wall there. For given $l$ (regardless of $m$), the radial solutions $u(r)$ are Hermite polynomials times a decaying exponential. Like our 1-D examples above, they are enumerated by the number of nodes $n_r$, starting with $n_r = 0$. Also note that (regardless of $n_r$) the behavior near $r = 0$ is always $u \sim r^{l+1}$; as a result, only s-orbitals ($l = 0$) can overlap with the nucleus.

We define the **principal quantum number** $n \equiv n_r + l + 1$, which is basically the sum of the radial excitation number and the angular excitation number (and it starts at 1 by convention).

Noting that the lowest Hermite polynomial (like the lowest spherical harmonic) is just a constant, the ground state of hydrogen ($Z = 1$, $n = 0$, $l = 0$) is $\psi = 1/\sqrt{\pi a_0^3}\, e^{-r/a_0}$, where the **Bohr radius**

$$a_0 = \frac{\hbar^2}{m_e e^2} = 5.29 \times 10^{-11}\,\mathrm{m} \tag{9.14}$$

effectively defines the size of an atom. There are no (suppressed or explicit) factors of $c$ because everything we did is non-relativistic, so in fact this is the only length scale available from the constants in Eq. 9.10. More generally the "decaying exponential" in $u(r)$ is always of the form $e^{-Zr/(na_0)}$.

Since Eq. 9.11 doesn't depend on $\hat{L}_z$, energy levels do not depend on $m$, but only on $n_r$ and $l$. That is, there is a degeneracy of $(2l+1)$ for each state $(n_r, l)$ from the possible values $m$ can take. This is true for *any* central potential.

But there is additional degeneracy specific to the $1/r$ potential, and it comes from the conservation of the **Laplace-Runge-Lenz vector** discussed in Sec. 2.5. The proof is beyond our scope, but as a result, the energy depends (to first approximation) only on $n$, not on $l$. For example, $n = 2$ states include both $(n_r = 1, l = 0)$ (one s-orbital) and $(n_r = 0, l = 1)$ (three p-orbitals), for a total degeneracy (ignoring spin) of four. In fact, the degeneracy (ignoring spin) for any given $n$ is $n^2$. Continuing the analogy, $n$ corresponds to the semi-major axis of an orbit (which solely determines its energy), while $l$ corresponds to its eccentricity, and $m$ its tilt. Note that s-orbitals ($l = 0$) are the **most** eccentric, corresponding to a planet that dives right through the sun with zero angular momentum; this matches our result that only s-orbitals overlap with the nucleus, and the spherical symmetry comes from the fact that the electron can dive through from any solid angle $(\theta, \phi)$.

The energy levels are

$$E_n = \frac{-Z^2}{2n^2}\mathcal{E} \qquad \text{where} \quad \mathcal{E} = \frac{e^4 m_e}{\hbar^2} = \frac{e^2}{a_0} = \alpha^2 m_e c^2 = 2\,\mathrm{Ry} = 27.2\,\mathrm{eV} \tag{9.15}$$

The ionization energy of ground-state hydrogen is (about) 1 Rydberg, or 13.6 eV. We have also introduced the **fine structure constant** $\alpha \equiv e^2/(\hbar c) = 1/137$, which does contain a $c$ because it is a relativistic concept, noting that $\mathcal{E}$ which sets the scale for orbital energies (chemistry) is $\alpha^2$ times the electron rest mass energy.

Remarkably, we get the same discrete energy levels $E_n$ as we got in the de Broglie model, but with better descriptions of their degeneracy, shape, and angular momentum (the ground state has angular momentum zero, not one). These electron orbitals (somewhat distorted by the presence of other electrons) form the basis for all chemistry. We have explained the Lyman alpha lines

$$\hbar\omega = \frac{1}{2}\mathcal{E}\left(\frac{1}{n^2} - \frac{1}{m^2}\right) \tag{9.16}$$

as the result of an electron jump from principal quantum number $n$ to $m$.

## 9.6 Hydrogen Atom Corrections

There are textbooks-full of interesting corrections to the derivation above, all of which can be seen in the spectra of various atoms. Here is a brief overview.

- The nucleus has a finite mass $m_N$, so the fixed central potential was an approximation. As we know from classical mechanics, though, the 2-body problem can be reduced to a 1-body problem by replacing $m_e$ with a reduced mass $\mu = m_e m_N/(m_e + m_N)$. In hydrogen $\mu = 0.99946\, m_e$, so this is pretty small and does not cause line splitting. You can, however, see the effect by comparing hydrogen and deuterium spectra.

- The electron spin, which generates a magnetic moment, interacts with any external magnetic field, and also (due to relativity) with the nuclear electric field ("spin-orbit coupling"), and also with the nuclear magnetic moment. So the "spin up" and "spin down" states are not exactly degenerate. When spin matters, we may change to the "coupled" basis labeled by eigenvalues of $\hat{\boldsymbol{J}}^2$ and $\hat{J}_z$, where $\boldsymbol{J} = L + S$. Notation like $3P_{1/2}$, meaning $(n = 3, l = 1, j = 1/2)$, indicates the coupled basis.

- The finite size of the nucleus produces a "Lamb shift" in the energy of s-orbitals (the only ones to overlap the nucleus).

- The position uncertainty of the electron produces a "Darwin correction" in the energy of s-orbitals (the only ones to overlap the nucleus).

- Our analysis was non-relativistic, so we must include relativistic corrections. To write the Hamiltonian operator, what do we do with the expression $E = \sqrt{p^2 + m_e^2}$ ? You could expand in $p/m_e$, but we will do better using the Dirac Equation in Subsection 9.8.

- There are several corrections, including the relativistic just mentioned, that adjust the energy levels by $\mathcal{O}(\alpha\mathcal{E})$, and are called **fine structure**.

- There are also corrections of $\mathcal{O}(\alpha^2\mathcal{E})$ called **hyperfine structure**.

- Our analysis has been for a single electron around a nucleus of charge $Ze$. For example when $Z = 6$ we have described the positive ion $C^{+++++}$. By the time you put six electrons in there,

their mutual interactions become pretty important. The **counting** of states remains valid ($1s^2 2s^2 2p^2$), but the orbitals are highly distorted (*e.g.* "s-p hybrid orbitals"). Mitigating this, though, is the fact that a "full shell" (a complete set of orbitals with a given $n$) is spherically symmetric, and so only affects higher shells by partially shielding the nuclear charge. Atoms with only full shells are chemically inert, and are called "noble gases". Atoms with similar valence (outermost) shells but different $n$'s have similar chemical properties, and lie in the same column of the **periodic table**.

- In chemistry we care about interactions between atoms, which is in general a very complicated multi-particle system. For example, put sodium (Na, $Z = 11$) together with chlorine (Cl, $Z = 17$), and if you solve the Schrödinger Equation for 28 electrons around two attractors with parameterized separation $R$, you will find a minimal-energy solution for some value of $R$. It is much easier to say that the sodium donates its valence electron to the chlorine, and then the charged ions form an "ionic bond" $Na^+Cl^-$.

## 9.7 Spin and Degeneracy

The **Stern-Gerlach experiment** (1922) sends a beam of silver atoms through a magnetic field whose gradient $\nabla B$ defines the z-axis. Silver atoms have a single valence electron, so the atoms are diverted by the interaction between that electron's magnetic moment and the magnetic field. An electron's magnetic moment is proportional to its spin, by $\boldsymbol{\mu} = \frac{1}{2} g_e (-e)/m_e \, \boldsymbol{S}$, where $g_e \approx 2$ is the **gyromagnetic ratio**. The key result is that the silver beam splits in two, indicating that the electron spin (projected onto the z-axis) takes two possible values. Measurement of the diversion shows that those values are $S_z = \pm\frac{1}{2}\hbar$. We say electrons are spin-1/2.

We can designate these two spin eigenstates $|\uparrow\rangle$ and $|\downarrow\rangle$, or by the Pauli spinor values $\binom{1}{0}$ and $\binom{0}{1}$. The most general state the electron can be in is $\chi = \binom{a}{b}$ with $a$ and $b$ complex, and $|a|^2 + |b|^2 = 1$ for normalization.

A current loop, and any system made up of current loops, has a magnetic moment $\mu = qL/(2m)$, and therefore a gyromagnetic ratio $g = 1$. Since the electron $g_e \approx 2$, it cannot be thought of (except maybe qualitatively) as a "spinning ball of charge". Also orbital angular momentum derived from $\boldsymbol{L} = r \times p$, when quantized, must have integer spin. So electron spin $\boldsymbol{S}$ (with $s = 1/2$) must be a different beast. It is still angular momentum, and obeys commutation relations like those for $\boldsymbol{L}$, and combines with $\boldsymbol{L}$ to give total angular momentum $\boldsymbol{J} = L + S$, and yet it is not rotation. Our challenge is to derive $g_e = 2$, and later to calculate the QED corrections that make the actual value about 2.002319304361. Theory and experiment actually agree to at least that many decimal places! Similarly for the muon, and precision measurements of the "muon anomalous magnetic moment" $(g_\mu - 2)$ are a powerful test of particle physics models.

We can represent the spin operator as $\boldsymbol{S} = \frac{1}{2}\hbar\boldsymbol{\sigma}$, where $\sigma_i$ are the **Pauli matrices**:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{9.17}$$

The Pauli matrices are Hermitian ($\sigma_i^\dagger \sigma_i = I$), involutory ($\sigma_i^2 = I$), and traceless ($\mathrm{Tr}(\sigma_i) = 0$). Along with the identity matrix $I$, they form a basis for the real vector space of $2 \times 2$ Hermitian matrices.[1] They provide the right commutation relation for spin: $[S_i, S_j] = i\hbar\epsilon_{ijk} S_k$, or equivalently

[1] They are infinitesimal generators of the Lie group SU(2), and the fact that they form a vector in 3-space shows

$\boldsymbol{S} \times \boldsymbol{S} = i\hbar \boldsymbol{S}$. They obey this identity:

$$(\boldsymbol{\sigma} \cdot \boldsymbol{a})(\boldsymbol{\sigma} \cdot \boldsymbol{b}) = (\boldsymbol{a} \cdot \boldsymbol{b})I + i(\boldsymbol{a} \times \boldsymbol{b}) \cdot \boldsymbol{\sigma} \tag{9.18}$$

Using Eq. 9.18, Pauli noticed that we can substitute $p^2 = (\boldsymbol{\sigma} \cdot \boldsymbol{p})(\boldsymbol{\sigma} \cdot \boldsymbol{p})$ in the Hamiltonian kinetic term. Then we turn $p$ into the operator $\hat{p} = -i\hbar\partial_x$ and convert derivatives to covariant derivatives with Eq. 12.2, giving

$$H_{\text{kinetic}} = \frac{1}{2m_e}(\boldsymbol{\sigma} \cdot (-i\hbar\nabla + e\boldsymbol{A}))(\boldsymbol{\sigma} \cdot (-i\hbar\nabla + e\boldsymbol{A})) \tag{9.19}$$

We now apply the identity Eq. 9.18 in reverse, but now[2] a $\nabla \times A$ appears, and that's the magnetic field $\boldsymbol{B}$. We end up with

$$H_{\text{kinetic}} = \frac{1}{2m_e}(\hat{p} + eA)^2 + \frac{e\hbar}{2m_e}(\sigma \cdot B) \tag{9.20}$$

That is exactly the magnetic moment interaction term, with $g_e = 2$!

## 9.8 Fine Structure and the Dirac Equation

The Schrödinger Equation of Eq. 9.5 describes **non**-relativistic electron quantum dynamics, both in using $E = p^2/2m$ and in the normalization $\int |\Psi|^2 = 1$. Unitary evolution ensures that one particle remains one particle $\int |\Psi(t)|^2 = 1$, disallowing the possibility that sufficient energy could create or annihilate electrons.

Let's try to write an analog of the Schrödinger Equation for $E = \sqrt{p^2 + m^2}$. Immediately we are stymied by the square root! You could expand $E = \sqrt{p^2 + m_e^2} = m_e + p^2/(2m_e) - p^4/(8m_e^3) + \cdots$, and indeed the $p^4$ term is correct. But, inspired by Pauli, Dirac tried looking for a matrix-based approach.

Let $E = \gamma^0 \boldsymbol{\gamma} \cdot \boldsymbol{p} + \gamma^0 m_e$ for a 3-vector of matrices $\boldsymbol{\gamma}$ and a fourth matrix $\gamma^0$. To recover the correct (scalar) energy relation, we want $(\gamma^0 \boldsymbol{\gamma} \cdot \boldsymbol{p} + \gamma^0 m_e)^2 = p^2 + m_e^2$. You might have hoped the Pauli matrices $\boldsymbol{\sigma}$ and the $2 \times 2$ identity $I_2$ would work here, but alas they do not. Instead Dirac found these $4 \times 4$ **gamma matrices** formed from them:

$$\boldsymbol{\gamma} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{pmatrix} \qquad \gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix} \tag{9.21}$$

While the Pauli matrices form a natural 3-vector, the gamma matrices form a natural 4-vector, and Feynman invented this useful "slash" convention $\not{p} \equiv p_\mu \gamma^\mu$.

Now we must invent 4-component Dirac spinors $\psi = \binom{\chi}{\eta}$ where each of $\chi$ and $\eta$ are (2-component) spinors. We interpret them roughly as representing particles (electrons) and antiparticles (positrons), so Dirac is credited with showing how a relativistic Hamiltonian requires the existence of antiparticles. This makes sense: if there's enough energy to be relativistic, then there's (nearly) enough energy for particle creation and annihilation.

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a deep relation between SU(2) and SO(3). Mathematicians say there is a "2-to-1 homomorphic mapping of SU(2) onto SO(3)".

[2]The key here is that $\boldsymbol{a} \times \boldsymbol{a}$ doesn't vanish if $\boldsymbol{a}$ is an operator that doesn't self-commute. Suppressing the prefactors $(-i\hbar, e)$, but restoring the implied state vector, $(\partial_i + A_i)(\partial_j + A_j)\psi = \partial_{ij}^2\psi + A_i(\partial_j\psi) + \partial_i(A_j\psi) + A_{ij}^2\psi$. Applying the chain rule and restoring the $\epsilon_{ijk}$, what remains is $\epsilon_{ijk}(\partial_i A_j)\psi$, where those bold parentheses imply that $\partial_i$ acts only on $A_j$, not $\psi$. But $\epsilon_{ijk}(\partial_i A_j) \equiv B_k$.

Back to the hydrogen atom, we treat $\eta$ as small, eliminate it from the dynamics, and end up with a Hamiltonian only in $\chi$, which is (in the absence of a magnetic field):

$$\hat{H} = \frac{\hat{\boldsymbol{p}}^2}{2m_e} + V - \frac{\hat{\boldsymbol{p}}^4}{8m_e^3} + \frac{1}{2m_e^2 r}\frac{\mathrm{d}V}{\mathrm{d}r}\hat{\boldsymbol{S}}\cdot\hat{\boldsymbol{L}} + \frac{\hbar^2}{8m_e^2}\nabla^2 V \tag{9.22}$$

Plugging in the Coloumb potential,

$$\hat{H} = \frac{\hat{\boldsymbol{p}}^2}{2m_e} - \frac{Ze^2}{r} - \frac{\hat{\boldsymbol{p}}^4}{8m_e^3} + \frac{Ze^2}{2m_e^2 r^3}\hat{\boldsymbol{S}}\cdot\hat{\boldsymbol{L}} + \frac{\pi Ze^2\hbar^2}{2m_e^2}\delta(r) \tag{9.23}$$

The third term is exactly the first-order relativistic correction we expected. The fourth term is the **spin-orbit interaction**, as the electron spin interacts with the magnetic field it "feels" as it orbits around in the Coulomb field of the nucleus. The fifth term is the **Darwin correction** that only affects s-orbitals ($l = 0$), as only they overlap the nucleus. Together, these explain the fine structure ($\mathcal{O}(\alpha^2)$ corrections) of the hydrogen atom.

We reiterate that all these corrections lead to line splittings, and therefore to predictions for atomic spectra. For example, electrons dropping from $3s$ to $2p$ in hydrogen produce a red line at about 656 nm, but fine splitting of $2P_{3/2}$ from $2P_{1/2}$ turns that into a tight pair of red lines, differing in wavelength by about 0.016 nm.

# Chapter 10

# Light and the Principle of Least Action

The Lagrangian and Hamiltonian formulations of mechanics lay the groundwork for our study of field theory.

## 10.1 Lagrangian and Hamiltonian

A force is **conservative** if $\boldsymbol{F} = -\nabla V$ for some potential $V$. It conserves energy as long as $V$ is included in the accounting. For example near Earth's surface $V = mgh$, and "falling" is just conversion of gravitational potential energy into kinetic energy. There are many useful and interesting non-conservative forces, such as friction, but the fundamental forces like gravity and electromagnetism are conservative. We now look for a reformulation of classical mechanics in terms of energy, momentum, and potentials.

We write the **Lagrangian** $L = T - V$, where $T$ is the expression for kinetic energy. $L$ depends on a set of generalized coordinates $q$ (or $q_i$ if you like, but we suppress the index), which need not be Cartesian, and their time derivatives $\dot{q}$. For example, a rigid body is described by 3 center-of-mass coordinates and 3 Euler angles (and $T$ is a rather complicated function of these 6 variables involving the inertia tensor). The generalized form of $F = ma$ is the **Euler-Lagrange equation**:

$$\frac{\mathrm{d}L}{\mathrm{d}q} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\mathrm{d}L}{\mathrm{d}\dot{q}} = 0 \tag{10.1}$$

Euler derived this using a lot of coordinate transformation math, but we will only show that it gives the right answer in a simple case: for a particle in a 1-dimensional potential, with a single coordinate $q = x$, the Lagrangian is $L = \frac{1}{2}m\dot{q}^2 - V(q)$, and the Euler-Lagrange equation gives $-V'(q) = m\ddot{q}$, which is just $F = ma$.

The **action** is $S = \int L\,\mathrm{d}t$ evaluated along a path. The Principle of Least Action states that the Euler-Lagrange equation Eq. 10.1 derives from extremizing the action ($\delta S = 0$) among all paths with the given endpoints. That is, the classical path is defined by the property that small deviations from that path negligibly change the action. This principle will be motivated later by the path integral formulation of quantum mechanics. For a single particle, if you specify starting point $(x_1, t_1)$ and endpoint $(x_2, t_2)$, the minimal-action path between them is the one that obeys $F = ma$. For light, action is proper time, *even inside matter*, and so you can use the Principle of Least Action to derive Snell's Law.

Generalized coordinates $q$ have corresponding generalized momenta $p = \partial L/\partial\dot{q}$. $q$ and $p$ together define the **phase space**. If you write the **Hamiltonian** $H = p\dot{q} - L = T + V$ in terms

of $q$ and $p$, then Eq. 10.1 is transformed into **Hamiltonian mechanics** with $\dot{q} = \partial H/\partial p$ and $\dot{p} = -\partial H/\partial q$. Returning again to the simplest case $q = x$, with $H = p^2/(2m) + V$, these equations give $p = mv$ and $F = ma$ respectively.

## 10.2 The Path Integral Approach to Quantum Mechanics

The Lagrangian and Hamiltonian are powerful tools for solving difficult problems in classical mechanics. But they also provide critical stepping stones to quantum mechanics and quantum field theory. Canonical coordinates as quantum operators obey canonical commutation relations: $[x, p] = i\hbar$, which leads for example to the uncertainty principle $\Delta x\, \Delta p \geq \hbar/2$.

We have already seen how the Hamiltonian leads to Schrödinger's Equation: $\hat{H}\Psi = E\Psi$, where $\hat{H}$ is an **operator** that looks like the classical expression for energy. Eigenstates of $\hat{H}$ ("energy eigenstates") have simple time evolution $\Psi(x,t) = e^{-iEt}\psi(x)$, where $\psi(x)$ is called a **stationary state**. Once you have described a system's state (at time $t = 0$) as a superposition of stationary states, its time evolution is straightforward. More generally, any state $|i\rangle$ evolves in time as $e^{-i\hat{H}t}|i\rangle$. The Hamiltonian is Hermitian, and therefore $U = e^{-i\hat{H}t}$ is unitary, meaning it preserves the norm of the state.

Let's recall the double slit experiment, where electrons start at a source at initial position $q_i$, pass through a screen with two slits, and end (at time $T$) at a wall with some distribution of final positions $q_f$. The amplitude $\langle q_f|e^{-i\hat{H}T}|q_i\rangle$ is the sum of the amplitude for the electron to pass through slit 1 and land at $q_f$, plus the amplitude for it to pass through slit 2 and land at $q_f$. The key feature is the interference pattern on the wall, because it is amplitudes that add, not probabilities.

The story goes that Feynman asked what happens if you add more slits, and keep doing so until the screen is gone. The amplitude $\langle q_f|e^{-i\hat{H}T}|q_i\rangle$ must be the sum (integral) of amplitudes over **all** intermediate positions $q_j$. Furthermore you place these empty screens at each of $N$ steps along the way from source to wall, or equivalently (for some given electron velocity) at equal time steps $\delta t$ (with $T = N\delta t$), so you integrate amplitudes over all possible paths. We are about to show that each path's amplitude is $e^{iS}$ where $S = \int L\, dt$ is the action evaluated along that path, that is,

$$\langle q_f|e^{-i\hat{H}T}|q_i\rangle = \int D[q]\, e^{iS[q]} \tag{10.2}$$

I won't discuss the measure over paths $D[q]$, so I'll drop constants (like $\frac{1}{2\pi}$), but rather focus on how the Hamiltonian on the left turns into an action on the right. See Zee[13] for a more complete derivation.

We break the left side into time slices $0 < j < N$, with $q_0 = q_i$ and $q_N = q_f$, so

$$\langle q_f|e^{-i\hat{H}T}|q_i\rangle = \prod_j \int dq_j\, \langle q_{j+1}|e^{-iH\delta t}|q_j\rangle \tag{10.3}$$

That product over $j$ (time) will eventually become a sum (integral) over time in the exponent, while the $\prod_j \int dq_j$ becomes the measure $D[q]$. Let us now concentrate on a particular time slice, and insert a complete set of momentum eigenstates $\sum |p\rangle\langle p|$, so that we can turn the Hamiltonian operator into ordinary $p$'s and $q$'s:

$$\langle q_{j+1}|e^{-iH\delta t}|q_j\rangle = \int dp\, \langle q_{j+1}| \exp\left[-i\left(\frac{p^2}{2m} + V(q)\right)\delta t\right] |p\rangle\langle p|q_j\rangle \tag{10.4}$$

Now $\langle q|p\rangle = e^{ipq}$ and $\langle p|q\rangle = e^{-ipq}$, so this term becomes

$$\int \mathrm{d}p \, \exp\left[-i\left(\frac{p^2}{2m} + V(q)\right)\delta t + ip(q_{j+1} - q_j)\right] \tag{10.5}$$

Divide by $\delta t$ (it gets absorbed into $D[q]$), and $(q_{j+1} - q_j)/\delta t$ just becomes $\dot{q}$. We have a Gaussian integral which we can do using Appendix B.4.1, eliminating $p$ to give for each time slice

$$\exp\left[i\left(\frac{1}{2}m\dot{q}^2 - V(q)\right)\right] \tag{10.6}$$

and now we recognize the Lagrangian in the exponent. The product over slices $j$ becomes an integral over time inside this exponent, and $\int L \, \mathrm{d}t = S$, giving $e^{iS}$ which is the right side of Eq. 10.2.

## 10.3 The Principle of Least Action

Restoring $\hbar$ for a moment, a path's amplitude is actually $e^{iS/\hbar}$, and a system acts "classically" when the action is much larger than $\hbar$. In that case the phase on the right side of Eq. 10.2 varies very rapidly as you modify paths, and the integral over phases therefore vanishes *except* near a stationary point where $\delta S = 0$. The usual situation is that there is only one such path, and it's a minimum (not maximum) of $S$, so this is the "path of least action". The **calculus of variations** then gives us the Euler-Lagrange equation of Eq. 10.1, which in turn gave us $F = ma$. Thus the **Principle of Least Action**, aka **stationary-action principle**, allows us to derive Newton's Second Law from quantum mechanics.

For light, action is proper time, *even inside matter*, and so you can use the Principle of Least Action to derive Snell's Law.

In his brilliant popular book *QED*, Feynman imagined each beam of light as carrying a little clock face at its head, representing phase. All the virtual paths were visualized as rapidly spinning clocks all out of sync and cancelling, whereas along the classical path only, the clocks were still. The breadth of paths over which the clocks don't significantly differ gives you an idea of the quantum width of an idealized classical path, which determines the diffraction width.

## 10.4 A First Look at Quantum Field Theory

In **field theory** the coordinates $q$ are replaced by fields $\Phi(\boldsymbol{x}, t)$ that depend on both space and time; think of a real scalar field $\Phi$ as an infinite number of dynamical variables $q(t)$, one at each point in space. Other types of fields (charged scalar, fermion, vector boson) may require a complex number, spinor, or 4-vector at each point. Because we've added spatial dependence, we now speak of the **Lagrangian density** $\mathcal{L}$, with $L = \int \mathcal{L} \, \mathrm{d}^3x$. (Physicists often sloppily drop the word "density".) In relativistic field theory, we require Lorentz invariance of the Lagrangian density, and write the relativistic Euler-Lagrange equation as

$$\frac{\partial \mathcal{L}}{\partial \Phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial\left(\partial_\mu \Phi\right)} = 0 \tag{10.7}$$

The action for a field theory is then $S = \int \mathcal{L} \, \mathrm{d}^4x$.

# Chapter 11

# Light is a Thermal Bath of Photons: Thermodynamics

Thermodynamics introduces the new and important concept of entropy. We can think of light from a hot object (ideally a blackbody) as a thermal bath of photons.

## 11.1 Statistical Mechanics

Consider two subsystems $A_1$ and $A_2$ in thermal contact, *i.e.* they can exchange energy, while the total system energy $E = E_1 + E_2$ remains fixed. For example, two boxes of gas, initially isolated and at different temperatures, joined together by a thermally conducting wall. You know intuitively what happens: heat flows from the hotter gas to the colder, until equilibrium is reached where they both have the same temperature $T$.

**Statistical mechanics**[14] (aka statistical thermodynamics) teaches us to analyze such systems by counting the number $\Omega$ of possible microstates (detailed descriptions of the position and momentum of each particle) in a given macrostate (bulk description). Macrostate variables can be **extensive** (energy, volume, number of particles) if they sum over subsystems (*e.g.* $E = E_1 + E_2$), or **intensive** (temperature, pressure) if a single measure describes every part of the system. Since $\Omega = \Omega_1 \Omega_2$, we can form an extensive variable $S$ called **entropy** using the logarithm:

$$S = k_B \, ln(\Omega) \tag{11.1}$$

so that $S = S_1 + S_2$. Here $k_B$ is the **Boltzmann constant**, see Appendix A. (Eq. 11.1 is enscribed on Ludwig Boltzmann's tombstone.) Within given constraints (like total energy), the likeliest macrostate is the one with the most microstates, *i.e.* the highest entropy. The **Second Law of Thermodynamics** states that the entropy of a closed system never decreases: $\mathrm{d}S \geq 0$. As a consequence, any **reversible** process must be **isentropic** ($\mathrm{d}S = 0$).

In normal systems, $\Omega$ increases rapidly with the amount of energy $E$ available to the system, and we define inverse temperature $\beta$ as:

$$\beta \equiv \frac{1}{k_B T} = \frac{\partial \ln(\Omega)}{\partial E} = \frac{1}{k_B} \frac{\partial S}{\partial E} \tag{11.2}$$

"Hey", you may say, "I thought temperature measured the average kinetic energy of molecules." The Equipartition Theorem, Eq. 11.6, makes that true of many systems, but not all. For a counterexample, consider the non-interacting Ising model: $N$ spin-1/2 dipoles in a magnetic field.

If $n$ of them are antialigned with $\boldsymbol{B}$, the energy is $E_n = 2n\mu B$, and the number of microstates is $\Omega = \binom{N}{n} = \frac{N!}{n!(N-n)!}$. Once the energy exceeds $E_{N/2} = N\mu B$, then $\Omega$ begins to decrease, and by Eq. 11.2 the temperature is negative!

You might worry about how to count classical microstates, or how precisely we specify the system energy $E$, or whether all microstates are really equally likely. All these considerations are swept under the rug by the logarithm of Eq. 11.1; essentially they contribute only additive constants to $S$, and do not affect the differences $\delta S$ that we care about. Curiously, quantum mechanics does provide some guidelines, for example a "unit cell" in phase space might be defined by the uncertainty principle $\Delta x\,\Delta p \geq \hbar/2$, and the precision in energy related to the observation time by $\Delta E\,\Delta t \geq \hbar/2$, but again these don't matter for most classical thermodynamics.

Return now to our two boxes of gas. More energy in $A_1$ increases $\Omega_1$ but at the expense of $\Omega_2$, and vice-versa, so there is an energy distribution that maximizes the product $\Omega_1\Omega_2$, or equivalently maximizes the total system entropy $S = S_1 + S_2$:

$$\frac{\partial(S_1+S_2)}{\partial E_1} = 0 = \frac{\partial S_1}{\partial E_1} - \frac{\partial S_2}{\partial E_2} = \frac{1}{T_1} - \frac{1}{T_2} \tag{11.3}$$

and we find the most likely state is the one where the temperatures are equal $T_1 = T_2$. In macroscopic systems, probability peaks are **very** sharp, so you can drop the phrase "most likely" and just say the subsystems reach thermal equilibrium.

### 11.1.1 Gibbs Paradox

Join a box of oxygen and a box of nitrogen, and remove the partition separating them. Entropy increases $\mathrm{d}S > 0$ as the gases mix irreversibly. Now do the same thing but with two boxes of oxygen – does entropy increase? Surely not, as the process is reversible (just slide the partition back into place). But if you labeled the molecules "L" and "R" with a Sharpie, doesn't the process look the same as the first case? This is the **Gibbs paradox**. It shows that the indistinguishable nature of molecules is critical to a correct counting of states.

## 11.2 Partition Function

Consider an "ensemble" of similarly-constructed systems, each of total energy $E$, and in each let $A_2$ be a heat bath at temperature $T$, meaning it can exhange energy with $A_1$ without (significantly) changing temperature, by virtue of being much larger. We want to know the likelihood, meaning the frequency in the ensemble, of finding $A_1$ in a particular state $r$ with energy $E_1 = E_r$. That likelihood is proportional to the number $\Omega_2(E - E_r)$ of possible microstates of the heat bath. Applying Eq. 11.2 we have $\ln(\Omega_2(E - E_r)) = \ln(\Omega_2(E)) - \beta E_r$, so the likelihood is proportional to $e^{-\beta E_r}$. We see that higher-energy states of $A_1$ are less likely because they leave less energy, and therefore less entropy, for the heat bath. Since probabilities must add to one,

$$P_r = e^{-\beta E_r} / \sum_s e^{-\beta E_s} \tag{11.4}$$

which is known as the **canonical distribution**, and $e^{-\beta E_r}$ is the **Boltzmann factor**. The (ensemble) mean energy in $A_1$ is $\bar{E}_1 = \sum_r P_r E_r$.

Even if there is no actual heat bath, a system $A_1$ at measured temperature $T$ is described by the canonical distribution. With the heat bath (subsystem $A_2$) gone, we can now drop the

subscript on $\bar{E}_1$, leaving this relation between temperature and mean energy:

$$\bar{E} = \sum_s E_s e^{-\beta E_s} / \sum_s e^{-\beta E_s} \tag{11.5}$$

For **classical** systems[1] with $N_Q$ quadratic degrees of freedom (DOF's), the **equipartition theorem** says that the mean energy is

$$\bar{E} = N_Q/(2\beta) = \tfrac{1}{2} N_Q k_B T \qquad \text{Equipartition Theorem} \tag{11.6}$$

The usual example is an ideal gas with kinetic energy from translation, rotation, and vibration. Translational motion of each molecule contributes 3 DOF's to $N_Q$, and for each (say $p_x$) you can confirm (see Appendix B.4.1) that Eq. 11.5 contributes

$$\left(\int_{-\infty}^{\infty} \frac{p_x^2}{2m} e^{-\beta p_x^2/(2m)} \mathrm{d}p_x\right) / \left(\int_{-\infty}^{\infty} e^{-\beta p_x^2/(2m)} \mathrm{d}p_x\right) = \frac{1}{2\beta} = \tfrac{1}{2} k_B T \tag{11.7}$$

This leads to a profound connection between the per-molecule specific heat (at constant volume) $c_V = \partial \bar{E}/\partial T$ and the molecular geometry: $c_V = \frac{1}{2} k_B (3 + n_R + 2n_V)$, where $n_R$ is the number of rotational modes and $n_V$ is the number of vibrational modes. A monatomic gas like helium has $n_R = n_V = 0$; a diatomic gas like oxygen ($O_2$) has $n_R = 2$ and $n_V = 1$, and so on. You add heat energy to the box (*e.g.* a 1-watt heater for 1 hour), measure how much the temperature rises, and now you can infer the shape of the molecules! Also note that, when you reduce the temperature below their quantum energy level spacing, vibrational modes "freeze out" causing $c_V$ to drop; and again when rotational modes freeze out. In practice at room temperature vibrational modes are usually frozen out but rotational are not, so $c_V = \frac{1}{2} k_B (3 + n_R)$.

Define the **partition function** $Z$ as[2]

$$Z \equiv \sum_s e^{-\beta E_s} \tag{11.8}$$

and then we have

$$\bar{E} = -\frac{\partial \ln(Z)}{\partial \beta} \tag{11.9}$$

For such systems, the entropy $S = k_B(\ln(Z) + \beta \bar{E})$ and the Helmholtz free energy $F = \bar{E} - TS = -k_B T \ln(Z)$.

## 11.3 Planck Distribution

Now let our "gas" be a box full of photons at temperature $T$. Each possible photon state has energy $\epsilon_i = \hbar\omega_i$ and occupation number $n_i$. The partition function is a sum over all possible sets of occupation numbers, unrestricted because photons are massless bosons that can be created or destroyed by the walls. We ignore the zero-point energy, which only matters if you change the size of the box – see the **Casimir effect**. We have

$$Z = \sum e^{-\beta \sum_i n_i \epsilon_i} = \prod_i \left(\sum_{n_i=0}^{\infty} e^{-\beta n_i \epsilon_i}\right) = \prod_i \left(\frac{1}{1 - e^{-\beta \epsilon_i}}\right) \tag{11.10}$$

[1] systems in which the quantum energy level spacing $\Delta E \ll k_B T$

[2] "Z" for *"Zustandsumme"*, German for "sum over states".

and thus

$$\ln(Z) = -\sum_i \ln(1 - e^{-\beta\epsilon_i}) \tag{11.11}$$

The mean occupation number of any given state

$$\bar{n}_i = -\frac{1}{\beta}\frac{\partial \ln(Z)}{\partial \epsilon_i} = \frac{1}{e^{\beta\epsilon_i} - 1} = \frac{1}{e^{\hbar\omega_i/k_BT} - 1} \tag{11.12}$$

which is the **Planck Distribution**.

## 11.4 Counting Waves in a Box

To enumerate states, impose periodic boundary conditions on the electromagnetic waves inside a cubic box of volume $L^3$. Then the wavenumber $\boldsymbol{k}$ is described by three integers $(n_x, n_y, n_z)$ where $k_x = 2\pi n_x/L$, etc. These lattice points have unit density in integer space, thus a density $(L/2\pi)^3$ in $\boldsymbol{k}$-space. We ask how many states lie between $\omega = c|\boldsymbol{k}|$ and $\omega + \mathrm{d}\omega$, which is a spherical shell in $\boldsymbol{k}$-space. Remembering that there are two polarizations, it is $8\pi\omega^2 \mathrm{d}\omega\, L^3/(2\pi c)^3$.

Multiply that number of states times the energy $\hbar\omega$ of each state, then by the mean occupation number from Eq. 11.12, and divide by volume $L^3$, to give **Planck's Law** for the spectral energy density $u(\omega, T)$:

$$u(\omega, T)\,\mathrm{d}\omega = \frac{\hbar}{\pi^2 c^3}\frac{\omega^3\,\mathrm{d}\omega}{e^{\hbar\omega/k_BT} - 1} \tag{11.13}$$

Defining $\eta \equiv \hbar\omega/k_BT$, we show the $\eta$-dependence in Fig. 11.1. The peak of the spectrum is approximately $\eta = 3$; that is, $\hbar\omega_{\max} \approx 3k_BT$. For example, the sun's surface at 5780 K peaks in the visible range around yellow and appears "white".

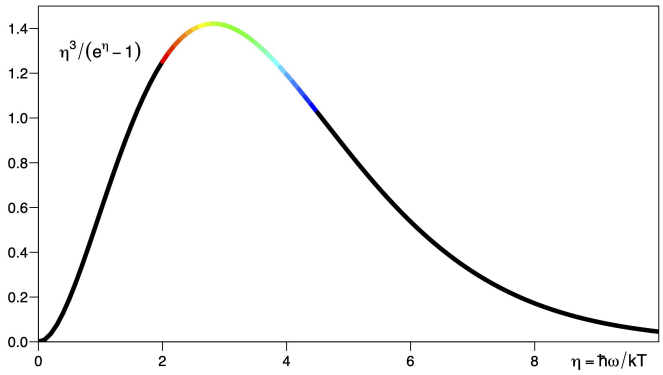



Figure 11.1: Blackbody Spectrum: Energy Density vs $\eta = \hbar\omega/k_BT$

We can integrate over $\omega$ for the total energy density:

$$\bar{u}(T) = \frac{\pi^2(k_BT)^4}{15(c\hbar)^3} \tag{11.14}$$

which is the **Stefan-Boltzmann law**. In a nutshell, as temperature rises, the typical photon energy rises like $\sim 3k_BT$, but the number of available modes rises like $T^3$ (because we live in 3 dimensions), so total energy density rises like $T^4$.

## 11.5 Black Hole Entropy

Learning thermodynamics by studying gases, we think of the maximum-entropy state as being spread out and uniform. But in gravitational systems, entropy increases as matter clumps up. According to **black hole thermodynamics**, a black hole of mass $M$ in fact has maximal entropy for its size: $S = A/4$, where $A = 4\pi R_S^2$ is the surface area of a sphere (event horizon) of **Schwarzschild radius** $R_S = 2M$. I've suppressed the physical constants in this **Bekenstein-Hawking formula** for $S$, but it really contains $k_B$ (Boltzmann's constant from thermodynamics), $G$ (Newton's gravitational constant), $\hbar$ (Planck's constant from quantum mechanics), and $c$ (the speed of light, or Einstein constant) – a remarkable mix of relativistic quantum gravitational thermodynamics!

From this point of view, the universe began in a low-entropy (uniform) state, and entropy increases as matter clumps to form black holes.

But systems with entropy have temperature. Eq. 11.2 tells us $1/T = \partial S/\partial E$; inserting $S = 4\pi M^2$ and $E = M$ gives the **Hawking temperature** $T_H = 1/(8\pi M)$. A black hole therefore glows with blackbody radiation, Eq. 11.13 with $T = T_H$, called **Hawking radiation**. This releases energy and eventually causes the black hole to evaporate.

Consider all the information that fell into the black hole during its lifetime – where did it all go? Evaporation of a black hole into pure thermal radiation seems to violate the quantum principle of unitarity, which roughly says information is never destroyed. This is the **black hole information paradox**. It's a topic of active debate, but consensus seems to be forming that the radiation differs subtly from pure blackbody radiation, and carries away the information.

Note that larger black holes have lower temperatures. Any black hole formed from stellar collapse has a temperature lower than a microKelvin, so it is cooler than the surrounding cosmic microwave background, making it a net absorber. Black hole evaporation is only relevant to either 1) the far distant (cold) future, or 2) tiny black holes that may have been formed in the early universe (**primordial black holes**).

## 11.6 Light Carries Negative Entropy: Non-Equilibrium Thermodynamics

People sometimes ask how life can evolve in the face of the Second Law of Thermodynamics, which says (roughly) that entropy (disorder) increases. The Second Law applies only to closed systems, and the Earth is not a closed system because light arrives from the sun and is emitted from the night side (mostly infrared). Energy flows in from the sun and flows out to space, balancing out. But the photons from the sun have relatively low entropy, while the ones flowing out to space have high entropy. We say the sun is not only a source of energy, but also of negative entropy (or "negentropy").

# Chapter 12

# Light is a Gauge Field

Physics is invariant under global rotation of quantum phase. We can promote that to a local ("gauge") symmetry by introducing a gauge field $A^\mu$. When that field gets a kinetic term in the Lagrangian, the result is electromagnetism.

## 12.1 Phase Invariance of the Lagrangian

In quantum mechanics, an electron is described by a wavefunction $\Psi = |\Psi|e^{i\phi}$ with an unobservable complex phase $\phi$. Differences in $\phi$ have observable effects, like interference patterns, but you can rotate the phase $\phi \to \phi + \theta$ everywhere with no effect; this is a **global symmetry**. We indicate the rotated quantities with a prime, so $\phi' = \phi + \theta$ and $\Psi' = \Psi e^{i\theta}$.

In quantum field theory, $\Psi$ is promoted to a field whose quantum fluctuations are electrons, but the same global symmetry applies. Dynamics are described by the Dirac **Lagrangian density** for non-interacting electrons:

$$\mathcal{L} = \bar{\Psi}(i\gamma^\mu \partial_\mu - m)\Psi \tag{12.1}$$

There's spin stuff going on here with the $\gamma^\mu$ which we can ignore for now; you only need to know that $\bar{\Psi}$ is still the complex conjugate of $\Psi$ (plus spin stuff), so $\bar{\Psi}' = \bar{\Psi}e^{-i\theta}$, and therefore the Lagrangian is invariant under global phase rotation.

Can we make phase rotation a local ("gauge") symmetry, where $\theta$ is a function of space and time? No, that screws up the derivative, because $\partial_\mu \Psi' = (\partial_\mu \Psi)e^{i\theta} + i(\partial_\mu \theta)\Psi e^{i\theta}$ by the chain rule. We can fix that by replacing the derivative $\partial_\mu$ with the **covariant derivative**

$$D_\mu \equiv \partial_\mu - iqA_\mu \tag{12.2}$$

where $A_\mu$ is an auxiliary field (a "connection") that we just made up with the transformation property $A'_\mu = A_\mu - \frac{1}{q}\partial_\mu \theta$. At this point $q$ is just a free parameter. Then you will find that $D_\mu \Psi' = (D_\mu \Psi)e^{i\theta}$, leaving this new Lagrangian gauge invariant:

$$\mathcal{L} = \bar{\Psi}(i\gamma^\mu D_\mu - m)\Psi \quad , \qquad D_\mu \equiv \partial_\mu - iqA_\mu \tag{12.3}$$

The difference between Eq. 12.1 and Eq. 12.3 is $J^\mu A_\mu$, where $J^\mu \equiv q\bar{\Psi}\gamma^\mu \Psi$.

You are perfectly free to make up auxiliary fields like this, no need to consult Nature, as long as they are not **dynamical**, *i.e.* as long as there is no **kinetic term** that associates energy with their derivatives in space or time. But now we are going to utter the magical incantation "Fiat Lux!" and add a kinetic term to our Lagrangian. The simplest expression that is both gauge invariant

and Lorentz invariant is $-F_{\mu\nu}F^{\mu\nu}$ where $F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$. How do we know how much to add? It doesn't matter, because you can always rescale $A_\mu$, and correspondingly $q$, so that the kinetic term gets a canonical normalization $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$. A field like $A_\mu$ is either dynamical or it isn't; we have transmuted the question "how dynamical?" into the question "how strong is the matter interaction $q$?". After rescaling, $A_\mu$ has become the electromagnetic potential and $q$ has become the particle charge. For the electron $q = -e$ and $e^2/(\hbar c) = 1/137$, so the interaction strength is respectable but still amenable to perturbation theory.

The resulting theory is called **quantum electrodynamics** (QED), with Lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\Psi}(i\gamma^\mu D_\mu - m)\Psi \quad , \qquad D_\mu \equiv \partial_\mu + ieA_\mu \tag{12.4}$$

The local phase transformation can affect other particles as well, such as the proton with $q_p = +e$. Each type of particle gets its own charge $q$. If you add interaction terms to the Lagrangian but want to preserve the gauge symmetry, the charges in each interaction term must sum to zero, *i.e.* the interactions must conserve charge. Interactions mediated through photons do so automatically. The question is, who assigns these $q$'s, and how are we so lucky that proton plus electron charge equals exactly zero? Possible answers must wait until our speculative discussion of Grand Unified Theories.

Emma Noether taught us that symmetries lead to conserved currents. As you might expect, electrical current is the conserved current $\partial_\mu J^\mu = 0$, where $J^\mu \equiv q\bar{\Psi}\gamma^\mu\Psi$ was defined above. The conserved Noether charge is electrical charge $Q$, which can be neither created nor destroyed. The interaction term $J^\mu A_\mu$ resulting from the covariant derivative leads to both the Lorentz Force Law and the inhomogeneous Maxwell's Equations.

To summarize, we took the existing global symmetry of electron phase rotation, made it local by adding an auxiliary field $A_\mu$, dynamicized the auxiliary field, and ended up with the full **gauge theory** of quantum electrodynamics. We obtained the conserved current $J^\mu$, the Lorentz Force Law, Maxwell's Equations, and the propagation of light. The gauge invariance of Eq. 7.13 is exactly the electron phase rotation symmetry, with $\Lambda = \theta/q$.

In later sections we will quantize the $A_\mu$ field, but you already know the quantum of light is a **photon**. Because $A_\mu$ has one Lorentz index, the photon has spin 1, and is called a **vector boson**. Because it arises from a gauge symmetry, it is a **gauge boson**.

| The photon is the massless gauge boson of quantum phase invariance. |
|---|

## 12.2 Lattice Gauge Theory

Suppose you discretize Wick-rotated spacetime as a 4D lattice on a computer. The links that connect each point to its 4 next-highest neighbors represent the $A_\mu$. For a **pure gauge** field $A_\mu = \partial_\mu\theta/q$, you assign a phase $\theta$ to each point, and then *e.g.* $qA_0 = \theta(t+1, x, y, z) - \theta(t, x, y, z)$. You'd find $F_{\mu\nu} = 0$ everywhere, and it wouldn't be very interesting.

Instead, you could directly assign $A_\mu$ to each link (4 DOF per point instead of 1), then you have the possibility of non-trivial fields and dynamics. The six independent closed loops you can form from each point, called **plaquettes**, represent $F_{\mu\nu}$. Convince yourself that $\partial_\mu A_\nu - \partial_\nu A_\mu$ is the line integral of $A$ around the $\mu\nu$ plaquette, Fig. 12.1.

Though usually applied to QCD (color force) rather than QED (electrodynamics), this is the basic setup for **lattice gauge theory**.

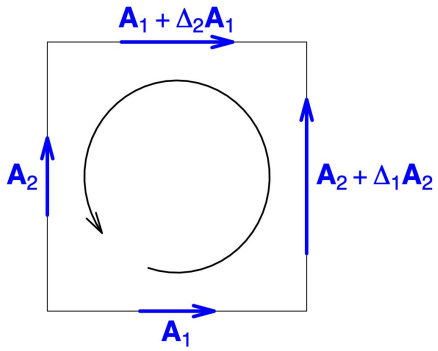



Figure 12.1: Plaquette loop $\oint \boldsymbol{A} \cdot \mathrm{d}\boldsymbol{s} = A_1 + (A_2 + \Delta_1 A_2) - (A_1 + \Delta_2 A_1) - A_2 = \Delta_1 A_2 - \Delta_2 A_1$

## 12.3 Aharonov-Bohm, Monopoles, Charge Quantization

You might try to build a theory of electromagnetism on the (more-or-less) observable $\boldsymbol{E}$ and $\boldsymbol{B}$ fields, contained in the tensor $F^{\mu\nu}$, rather than on $A^\mu$, which is not directly observable due to gauge invariance. If so, you'd be hard-pressed to explain the **Aharonov-Bohm effect**. Return again to the electron double-slit experiment, Fig. 9.1, and now put a solenoid between the slits and the screen, between the two electron paths. The $\boldsymbol{B}$ field is confined to the interior of the solenoid, which does not intersect the electron's possible paths to the screen, so you'd think it should not affect the electron. But the vector potential $\boldsymbol{A}$ circulates around the outside of the solenoid, affecting the phase of the electron wavefunction, and so the interference pattern shifts (experimentally verified!) when the solenoid is turned on. The phase shift is given by $\Delta\phi = e\Phi_B/\hbar$, where $\Phi_B = BA$ is the magnetic flux through the solenoid, the product of the field strength $B$ times the solenoid area $A$. Notably, no phase shift occurs if $\Delta\phi = 2\pi n$. Among other things, Aharonov-Bohm explains why magnetic flux is quantized in superconducting loops.

Suppose we want to add magnetic monopoles to our theory. The Maxwell Equation $\nabla \cdot \boldsymbol{B} = 0$ (from Eq. 6.3) seems to disallow them. But we can build a field configuration that looks like a monopole of magnetic charge $q_m$ carrying around a "tail" of flux $4\pi q_m$ that connects it to an opposite monopole somewhere else. For example, $A_\phi = q_m(\cos\theta - 1)$ and $A_t = A_r = A_\theta = 0$ has the right $\boldsymbol{B}$ field and an invisible flux tail along the $-\hat{x}$ axis. If the flux tail is quantized, it will not be detectable by Aharonov-Bohm phase shift. The quantization requirement suggests that the product of electron charge times monopole charge must equal $eq_m = n/2$. Quantized electric charge leads to quantized magnetic charge – or is it (as Dirac suggested) the other way around? The existence of even a single monopole in the universe would potentially explain the quantization of electric charge. Unfortunately, no monopole has ever been detected. Well actually Blas Cabrera may have detected one in 1982[15], but it was never repeated.

More interesting topological configurations[16] of $A^\mu$ fields occur in the non-Abelian gauge theories that we will study in the next chapter. The **sphaleron** is a static, localized configuration of the SU(2) 4-potential and the Higgs field. Taking $\theta_W = 0$, and using the gauge $A_0 = A_r = 0$, it looks like

$$\boldsymbol{A} = v\frac{f(\xi)}{\xi}\hat{r} \times \sigma, \qquad h = \frac{v}{\sqrt{2}}F(\xi)\hat{r} \cdot \sigma \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{12.5}$$

where $\xi \equiv rgv$, $g$ is the electroweak coupling, $v$ is the Higgs vev, $h$ is the Higgs field, $\sigma$ are the SU(2) generators, and $f(\xi)$ and $F(\xi)$ are functions that go from 0 to 1 as $\xi$ goes from 0 to $\infty$. It is believed that this sphaleron would have a mass of about 10 TeV, would be unstable, and its decay would violate baryon-number (but preserve $B - L$). Fortunately, sphalerons are very

difficult to make, and are probably only relevant to early-universe phase transitions. **Instantons** are topological $A^\mu$ configurations that are also localized in time.

# Chapter 13

# Light as a Model for Yang-Mills Theories

It turns out all fundamental forces are gauge theories of local symmetries.

## 13.1 U(1) Symmetry

The symmetry $\Psi' = \Psi e^{i\theta}$ is called a U(1) symmetry, because (in a degenerate sense) $e^{i\theta}$ is a $1 \times 1$ unitary matrix. A **unitary matrix** $U$ is a complex-valued matrix satisfying $U^\dagger U = UU^\dagger = 1$, where $U^\dagger \equiv (U^T)^*$. In words, the **Hermitian adjoint** $U^\dagger$ is found by taking both the transpose and the complex conjugate. The Hermitian adjoint of $e^{i\theta}$ is $e^{-i\theta}$. We saw above that $\bar{\Psi}\Psi$ is invariant; multiplication by a unitary matrix preserves the wavefunction's norm.

Following the development of **Yang-Mills theory** in 1954, maybe we can find some other symmetries, described by unitary matrices, that we can gauge?

The following statements are roughly equivalent, and specific to the U(1) symmetry:

1. U(1) is an **Abelian group**, also called commutative, meaning that if you apply two rotations to $\Psi$, it doesn't matter the order: $e^{i\theta}e^{i\phi}\Psi = e^{i\phi}e^{i\theta}\Psi$.

2. The photon carries no electric charge.

3. Photons do not interact with each other.

4. The classical theory can be described in terms of fields ($\boldsymbol{E}$, $\boldsymbol{B}$) that add linearly, *e.g.* the electric field of two charges is the sum of $\boldsymbol{E}$'s from each charge.

The other symmetry groups we will study (SU(n)) are non-Abelian, so do not expect them to have these nice features. But they do give rise to massless vector gauge bosons, like the photon. You might object that we don't know any other massless vector bosons, but, aha! we do, they're just hiding.

## 13.2 The Particle Zoo

Particles with integer spins are called **bosons**, and those with half-integer spins are **fermions**. The known fundamental bosons are the **graviton** (spin 2), the gauge vector bosons (spin 1), and the Higgs scalar boson (spin 0). All the known fundamental fermions are spin-1/2, and are thought of as the "matter" particles. To look for symmetries, first we must enumerate the known fundamental fermions.

They come in 3 generations, for reasons unknown (though the string theorists have some ideas), so let me describe the first generation for now: $[(u,d),(\nu_e,e)]$, with corresponding electric charges $[(+\frac{2}{3},-\frac{1}{3}),(0,-1)]$. There are two quarks (up and down), and two leptons (electron neutrino and electron), the difference being that only quarks feel the chromodynamic force (QCD). That force permanently binds quarks into either 3-quark combinations (hadrons) or quark-antiquark pairs (mesons). The lowest-energy hadrons are the nucleons that are found in all ordinary matter: the proton (uud) and the neutron (udd). (Fermi statistics prevent the existence of a ground-state uuu or ddd hadron.) Note the proton charge +1 and neutron charge 0 are consistent with these assignments. Electrons with charge -1, of course, orbit the nuclei to form atoms, while neutrinos with zero charge are nearly invisible to us.

The second generation looks like the first but heavier: $[(c,s),(\nu_\mu,\mu)]$, the charm and strange quarks, the muon neutrino and the muon. The third generation is even heavier: $[(t,b),(\nu_\tau,\tau)]$, the top and bottom quarks, the tau neutrino and the tau. The masses don't follow any clear pattern, but to pick the charged leptons for example: the electron mass is 0.511 MeV, the muon is 105.7 MeV, and the tau is 1,777 MeV. Particles from the heavier generations are short-lived, decaying quickly to first-generation versions, so we don't see them much outside of particle accelerators. When presented with evidence for multiple generations, specifically that the muon is a heavier electron, I.I. Rabi famously asked, "Who ordered that?" The generation structure is largely irrelevant to the symmetries we'll discuss, so we will stick to first generation as much as possible.

Neutrino masses, by the way, are unknown; they are certainly small and maybe zero. When the supernova **SN 1987A** was observed, **neutrino detector**s recorded a 13-second burst of neutrinos several hours before the light reached us. One presumes that they were emitted several hours earlier than the light was, and travelled for 168,000 years at so close to the speed of light that they stayed ahead of the photons. To be that **ultrarelativistic**, their mass has to be far smaller than their energy, something like 1 eV or less, which is to say a million times lighter than the electron. On the other hand, the phenomenon of **neutrino oscillation** requires that there be mass **differences** between generations of roughly $(0.01\,\text{eV})^2$, suggesting that at least two generations have positive neutrino masses around that size. We often assume, but don't know, that the neutrino masses increase in the same order of generations as the charged leptons.

These fermions all have antiparticles, for example the anti-electron is known as the **positron**, but the universe appears to contain far more matter than anti-matter. All the symmetries we will discuss apply to the antiparticles as well.

## 13.3 Quantum Chromodynamics

Quarks come in three "colors"[1], and there is a symmetry under a color rotation described by a $3\times3$ unitary matrix. There are 9 independent unitary $3\times3$ matrices, but one of them is just phase rotation of all three colors in unison (the matrix is $e^{i\phi}I$), and we've already done that, it's electromagnetism. The remaining 8 form a group $SU(3)_c$, where "S" for "Special" means they have unit determinant (excising the generator of electromagnetism), and the "c" subscript stands for "color". We gauge this symmetry and get **quantum chromodynamics** (QCD), the theory of the strong nuclear force. In place of the photon we get 8 gluons.

[1] We use "color" only as an analogy for something with 3-fold symmetry, nothing to do with real light and colors. We saw earlier that the perceived 3-ness of (real) color derives from three types of receptors in the human eye.

As SU(3) is non-Abelian, the gluons self-interact, in a way that makes the force effectively stronger at lower energies (larger distances). This results in **confinement** at low energies, where quarks must appear only in color-neutral combinations, and the flip side is **asymptotic freedom** at high energies, where the force is weak (perturbative) and more closely resembles QED. The crossover is roughly the **QCD scale** $\Lambda_{QCD}$, which is approximately the proton mass, which is approximately 1 GeV. In fact, $\Lambda_{QCD}$ sets the proton mass, as the proton is effectively a little bound ball of 3 "valence" quarks, plus virtual quarks and gluons, with 1 GeV of binding energy. Below $\Lambda_{QCD}$, quarks are very tightly bound in color-neutral triplets called hadrons, like protons and neutrons, or color-neutral quark-antiquark pairs called mesons, like the pion. You never see gluons, which also carry color, because they too are tightly bound to hadrons and mesons.[2]

### 13.3.1 Nuclear Physics

Let's appreciate how bold the idea of QCD was: it postulated a gauge theory of quarks and gluons even though quarks and gluons had never been seen, and could not be isolated! The important thing was that it explained patterns (the **eightfold way**) among the many hadrons and mesons that were produced in particle accelerators. Many physicists considered it more of a mathematical construction than a literal description of nature, until deep inelastic scattering experiments (1990 Physics Nobel for Kendall, Friedman, and Taylor) showed that quarks were real.

With QCD so strong in the systems of interest, the perturbative theory that worked well for QED was useless. **Lattice gauge theory**, described in the previous section, is an important tool for understanding things like the proton mass.

Why do nuclei bind together in the nucleus? Why do 6 protons (carbon) usually come with 6 neutrons ( $^{12}C$), but sometimes 7 or 8? Why is iron ( $^{56}Fe$) the most stable nucleus? These questions are the realm of **nuclear physics**. Now we consider nuclear forces to be residuals of the color force binding the nucleons, analogous to the chemical and intermolecular forces that are residual to the electromagnetic force binding atoms. But **nuclear physics** was well studied before QCD, and is a successful **effective field theory** unto itself. In the effective theory, hadrons interact by exchanging mesons (like pions), giving rise to a **Yukawa potential**. Indeed, pions are quite light because they are **pseudo-Goldstone bosons** of chiral symmetry, so they naturally play the role of force-carriers.

## 13.4 Weak Interactions

There is also a 2-fold symmetry ("weak isospin") among some pairs of particles, such as the neutrino and electron $(\nu, e)$, and the up and down quarks $(u, d)$. As with QCD we extract the U(1), so this is an SU(2) symmetry, with 3 gauge bosons called W for "weak".

Notably only left-handed chiralities of the particles form these isodoublets, which is a weird explicit parity violation. While the definition of **chirality** is not intuitive, in the limit of massless particles it is the same as **helicity**; a right-handed or positive-helicity fermion has its spin pointing in the same direction as its momentum. (Helicity applies to light as well; right-handed circularly polarized light consists of positive-helicity photons.) For massive fermions it's more complicated; helicity is not Lorentz-invariant, and mass eigenstates are not chirality eigenstates. Nonetheless, while discussing weak interactions we must distinguish $e_L$ from $e_R$, the former part of an isodoublet while the latter is a singlet. For this reason the symmetry is called $SU(2)_L$.

[2] There may exist gluon-only bound states, called **glueballs**.

The next weird thing about this symmetry is that it's broken by the Higgs scalar field. The Higgs is a spin-0 complex isodoublet, which means it has 4 real degrees of freedom. But its potential is such that it has lower energy by taking a **vacuum expectation value** of 246 GeV everywhere in space. A particular direction in $SU(2)_L$ space is singled out, breaking the symmetry. One Higgs degree of freedom becomes a massive scalar particle, the **Higgs boson**, finally detected in 2012 at LHC. The three remaining Higgs degrees of freedom are "eaten" by the 3 gauge bosons, which become massive particles $W^{\pm}$ and $Z^0$. You don't see weak vector bosons (except in particle accelerators) because they're heavy.

The astute reader will object that an isodoublet like $(\nu, e)_L$ cannot be viewed as a single particle by the $U(1)_{EM}$ of electromagnetism, because the components have different charges. We abandon electromagnetism for now, and reconstruct a different $U(1)_Y$ based on the "hypercharges" Y of Table 13.1. $U(1)_Y$ is actually the phase invariance symmetry we described before. But when

| $(u,d)_L$ | $(\nu_e,e)_L$ | $u_R$ | $d_R$ | $\nu_{eR}$ | $e_R$ |
|---|---|---|---|---|---|
| +1/6 | -1/2 | +2/3 | -1/3 | 0 | -1 |

Table 13.1: Hypercharges

$SU(2)_L$ is broken by the Higgs, part of it mixes with hypercharge symmetry to form electromagnetism. Strictly speaking, light is a mix of the hyperphoton with one of the $SU(2)_L$ gauge bosons, specifically it's the massless mixture. The other, massive, mixture is the $Z^0$, which as a result is somewhat heavier than the $W^{\pm}$.

What we call electrical charge $Q = Y + I_z$, where $I_z$ is $+\frac{1}{2}$ for the first component of a weak isodoublet, $-\frac{1}{2}$ for the second component, and 0 for an isosinglet. The electromagnetic coupling $e$ is a combination of the weak isospin and hypercharge couplings.

If QCD was bold, weak $SU(2)_L$ was positively audacious. We use the machinery of gauge symmetry to describe weak nuclear interactions where no symmetry is manifest, particles are changing in nature and crossing generations, and no gauge bosons were visible (until CERN discovered them in 1983). The very nature of electromagnetism was deconstructed and reconstructed. Symmetry breaking required a fundamental scalar particle, the Higgs, something never before seen. Many physicists thought the Higgs was a toy model standing in for another theory, like **Technicolor**, but nope, the Higgs boson was discovered at CERN in 2012 acting just like a scalar boson should. The electroweak theory completely replaced the 4-fermion effective theory that preceded it, and has proven wildly successful.

The $W^{\pm}$ and $Z^0$ mediate important interactions, such as muon decay and neutron decay. It's no coincidence that these are the ones that change the nature of the particles (muon to electron, or down quark to up quark), or that they are relatively slow or rare (because the bosons are so heavy). These interactions are known as the "weak (nuclear) force". The intimate connection established between electromagnetism and the weak force, resulting from mixing of $U(1)_Y$ and a broken $SU(2)_L$, was the **Third Unification**, and is called Electroweak Unification.

## 13.5 The Standard Model

Those are the gauge symmetries we know in nature. The overall symmetry (before being broken by the Higgs) is $SU(3)_c \otimes SU(2)_L \otimes U(1)_Y$, and the resulting Yang-Mills gauge theory is the **standard model** of Particle Physics.

It reached roughly its current form by about 1973, and arguably has yet to be improved upon. Several particles were predicted but only confirmed later: The $W^{\pm}$ and $Z^0$ in 1983, the top quark in 1995, the tau neutrino in 2000, and the Higgs boson in 2012. Strictly speaking it does not include $\nu_R$, but that's easy to add, as it is a singlet under all the symmetries, and we likely should in order to explain neutrino oscillations.

The Standard Model says nothing about gravity. It explains particle masses as interactions with the Higgs vev, but each of those interaction strengths is an unexplained parameter, and it does nothing to illuminate the 3-generation structure. In fact, adding mixing angles and gauge coupling strengths, the model requires 19 real parameters, plus any neutrino mass terms, plus the hypercharges of Table 13.1. It does not explain why the hypercharges come in integer multiples of 1/6, and therefore gives no insight on why hydrogen is exactly electrically neutral. Since 1973, the name of the game in particle physics has been to go "beyond the Standard Model" to address these deficiencies.

# Chapter 14

# Light Under Gravity: General Relativity

Gravity is the gauge theory of local Lorentz invariance.

## 14.1 Vierbein

At each point in spacetime, we set up **two** 4-dimensional coordinate systems[1]. The first uses generalized coordinates $x^\mu$ that are convenient in some global sense, such as the Schwarzschild coordinates $(t, r, \theta, \phi)$ around a spherical mass. The second is a local inertial ("free-falling") coordinate system with a **tetrad** of unit vectors $\hat{e}_m$. Note how generalized coordinates use Greek Lorentz indices $(\mu, \nu)$, while the tetrad uses Latin indices $(m, n)$. The tetrad metric $\eta_{mn} = \hat{e}_m \cdot \hat{e}_n = \text{diag}[+1, -1, -1, -1]$ is Minkowski (and therefore flat) by construction. The generalized coordinate metric $g_{\mu\nu} = \hat{x}_\mu \cdot \hat{x}_\nu$ is not flat; in our Schwarzschild example

$$g_{\mu\nu} = \text{diag}\left[\left(1 - \frac{2GM}{r}\right), -\left(1 - \frac{2GM}{r}\right)^{-1}, -r^2, -r^2 \sin^2\theta\right] \tag{14.1}$$

and its non-flatness represents gravity, in a way we will make more precise later. As in Eq. 7.4, the invariant interval $\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu$ between two points must be independent of our coordinate choice, so the metric is constrained to obey the Einstein equation which we will now derive.

The two sets of unit vectors are related by the **vierbein** $e^\mu_n$ as follows: $\hat{e}_n = e^\mu_n \hat{x}_\mu$. There is also the inverse vierbein $e^m_\nu$ defined by $e^\mu_m\, e^m_\nu = \delta^\mu_\nu$. Thus

$$g_{\mu\nu} = e^m_\mu\, e^n_\nu\, \eta_{mn} \tag{14.2}$$

so the vierbeins are kind of like square roots of the metric.

## 14.2 Covariant Derivative

Physics is invariant under **both** local Lorentz transformations (LLT) which are rotations of the tetrad $\hat{e}_n$ (see Eq. 7.1), **and** general coordinate transformations (GCT) which are redefinitions of $x^\mu$. It is the local Lorentz invariance that makes gravity a type of gauge theory.

Still following Collins, Martin, and Squires[17], we write a covariant derivative of the vierbein, analogous to Eq. 12.2, and require

$$D_\mu e^m_\nu = \partial_\mu e^m_\nu - \Gamma^\lambda_{\mu\nu}\, e^m_\lambda + \omega^m_{\mu\,n}\, e^n_\nu = 0 \tag{14.3}$$

[1] We follow the "Palatini formalism" described in Collins, Martin, and Squires[17]

There are **two** connections introduced here. $\Gamma^{\lambda}_{\mu\nu}$ is the **Christoffel symbol**, the connection associated with the GCT (notably named a "symbol" because it is not a proper tensor). $\omega_{\mu}^{m}{}_{n}$ is the **spin-connection** associated with the LLT (so named because it also shows up in the covariant derivative of a spinor field). Unlike $A_\mu$ from Eq. 12.2, however, neither becomes a dynamical field, but rather just end up as functions of $g_{\mu\nu}$.

If we impose symmetry on the lower indices of the Christoffel symbol ("zero-torsion"), it can be shown to be this function of the metric:

$$\Gamma^{\lambda}_{\mu\nu} = \tfrac{1}{2} g^{\lambda\rho} \left( \partial_\mu g_{\nu\rho} + \partial_\nu g_{\mu\rho} - \partial_\rho g_{\mu\nu} \right) \tag{14.4}$$

The spin-connection $\omega_{\mu}^{mn}$ has that single Lorentz index like $A_\mu$, so (see Eq. 7.8) we form an analog of the antisymmetric $F_{\mu\nu}$:

$$R^{mn}_{\mu\nu} = \partial_\mu \omega_\nu^{mn} - \partial_\nu \omega_\mu^{mn} + \omega_\mu^{m}{}_{k} \omega_\nu^{kn} - \omega_\nu^{m}{}_{k} \omega_\mu^{kn} \tag{14.5}$$

and by contracting with the vierbeins we form the **Ricci scalar**:

$$R \equiv R^{mn}_{\mu\nu} e^{\nu}_{m} e^{\mu}_{n} \tag{14.6}$$

## 14.3 Einstein Lagrangian and Stress-Energy Tensor

In electromagnetism we put $F^{\mu\nu}F_{\mu\nu}$ into the Lagrangian, making $A_\mu$ dynamical. Instead we form the **Einstein Lagrangian** from the Ricci scalar of Eq. 14.6. We introduce the vierbein determinant $e \equiv \mathrm{Det}(e^{m}_{k})$ to make the action $S = \int \mathcal{L} \mathrm{d}^4x$ invariant under GCT; for example if our units halved so that $x^{\mu\prime} = 2x^\mu$, you'd get a factor of 16 from $\mathrm{d}^4x$ and 1/16 from $e$ so the action remains unchanged. Finally, recall that just above Eq. 12.4 we eliminated the unknown constant in front of $F^{\mu\nu}F_{\mu\nu}$ by rescaling $A_\mu$, but here we do not rescale, so we introduce the physical constant $\kappa^2 = 8\pi G/c^4$ which will end up governing the strength of gravity's interaction with matter. The Einstein Lagrangian is:

$$\mathcal{L}^{(E)} = -eR/(2\kappa^2) \tag{14.7}$$

Since it is linear (not quadratic) in derivatives of $\omega_{\mu}^{mn}$, minimizing the action $S^{(E)}$ with respect to $\omega_{\mu}^{mn}$ just gives a formula for the spin-connection in terms of the vierbeins (a little too ugly to show here). Minimizing $S^{(E)}$ with respect to the vierbein then gives the Einstein equation (expressed below) in a vacuum.

We add the matter Lagrangian $\mathcal{L}^{(M)} \equiv eU$ where $e$ is the same vierbein determinant as in Eq. 14.7, and $U$ is a scalar (that depends on the type of matter). If we define the **stress-energy tensor** (aka **energy-momentum tensor**)

$$T_{\mu\nu} \equiv -e_{m\mu} \left[ \frac{\delta U}{\delta e^{\nu}_{m}} - e^{m}_{\nu} U \right] \tag{14.8}$$

then minimizing the total action $S^{(E)} + S^{(M)}$ with respect to the vierbein gives the full Einstein equation.

## 14.4 The Equations of General Relativity

LLT invariance allows us to remove all explicit references to the vierbeins, leaving only the metric $g_{\mu\nu}$ (including implict references via Eq. 14.4) as our primary, dynamical field. The **Reimann**

**curvature tensor** is

$$R_{\mu\nu\rho\sigma} = \tfrac{1}{2}\left[\partial_\nu\partial_\rho g_{\mu\sigma} - \partial_\mu\partial_\rho g_{\nu\sigma} - \partial_\nu\partial_\sigma g_{\mu\rho} + \partial_\mu\partial_\sigma g_{\nu\rho}\right] + \Gamma_{\lambda\mu\sigma}\Gamma^\lambda_{\nu\rho} - \Gamma_{\lambda\mu\rho}\Gamma^\lambda_{\nu\sigma} \tag{14.9}$$

and spacetime is "flat" only when $R_{\mu\nu\rho\sigma} = 0$. Notably the spacetime around a massive body, or in the presence of gravitational waves, is not flat. From this we obtain the **Ricci tensor** $R_{\mu\nu} \equiv R^\lambda_{\mu\nu\lambda}$ and the **Ricci scalar** $R \equiv g^{\mu\nu}R_{\mu\nu}$ (same as Eq. 14.6). Spacetime is "Ricci-flat" when $R_{\mu\nu} = 0$, as it must be anywhere away from matter. The Einstein equation is

$$R_{\mu\nu} - \tfrac{1}{2}\, g_{\mu\nu}\, R = -\kappa^2 T_{\mu\nu} \tag{14.10}$$

General Relativity is the gauge theory of local Lorentz invariance.

## 14.5 Particle Motion Under Gravity

As we did for electromagnetism, Eq. 12.2, we replace derivatives with covariant derivatives, and Newton's First Law $\ddot{x} = 0$ is replaced by

$$\frac{\mathrm{d}^2 x^\lambda}{\mathrm{d}\tau^2} + \Gamma^\lambda_{\mu\nu}\frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau} = 0 \tag{14.11}$$

where $x^\lambda(\tau)$ is a particle's path as a function of its proper time $\tau$. The "force" of gravity has been replaced by the Christoffel symbol $\Gamma^\lambda_{\mu\nu}$ defined in Eq. 14.4.

For a non-relativistic ($\tau \approx t$, $\gamma \approx 1$) apple falling towards our weak-static-field ($GM/r \ll 1$) Earth we use the Schwarzschild metric of Eq. 14.1, and the only relevant Christoffel symbol is

$$\Gamma^r_{00} \approx \tfrac{1}{2}\partial_r g_{00} \approx \frac{GM}{r^2} \tag{14.12}$$

giving us back Newtonian gravity, Eq. 2.2. Equivalently, in these approximations the gravitational potential $\phi = -GM/r$ derives from $g_{00}$ by $g_{00} = 1 + 2\phi$.

A more careful analysis[18] gives the gravitational potential as $\phi = \frac{1}{2}\ln(1-2GM/r)$, which you can Taylor expand to see the GR corrections. As discussed in Sec. 2.5, deviation from $1/r$ causes orbital precession, as verified for Mercury's orbit around the sun. For reference, at the Earth's surface, $GM/R \approx 7\times10^{-10}$ (and Earth's Schwarzschild radius $R_S = 2GM \approx 9\,\mathrm{mm}$). At the sun's surface, $GM/R \approx 2\times10^{-6}$ (and the sun's $R_S \approx 3\,\mathrm{km}$).

## 14.6 Light Under Gravity

For photon orbits it suffices to take the $m \to \infty$ limit of an ultrarelativistic particle. We must eliminate $\tau$ from our equations (*e.g.* calculate $r(\phi)$) since photons experience no proper time. For example, starlight grazing the sun's surface bends towards the sun by an angle $\Theta = 4GM/R$, as derived in MTW[19] Eq. (25.68), or Lightman[20] problem 15.6. For the sun that's $\Theta = 8\times10^{-6} = 1.7\,\mathrm{arcsec}$. Arthur **Eddington** observed this effect in 1919 during a solar eclipse.

As a historical aside, in 1911 Einstein incorrectly predicted the deflection to be half the correct value. Bad weather during eclipses, faulty equipment, and World War I prevented an accurate measurement long enough for Einstein to correct himself in 1915.

The bending of starlight makes astronomical bodies, particularly galaxies, act like the converging lens of Fig. 5.9. One way we track dark matter is by its gravitational lensing.

Photons can follow stable circular orbits around a (non-spinning, uncharged) black hole only at $r = 3GM$.

Gravitational time dilation: A clock ticks 1-second intervals in its own frame; observed from far away the tick time (in seconds) is $\mathrm{d}t = 1/\sqrt{g_{00}}$. For Schwarzschild, that's $\mathrm{d}t = (1 - 2GM/r)^{-1/2}$: the clock in a gravitational well runs slow. Light of a given frequency is like a clock, so as it rises from the well it must decrease in frequency by the same **gravitational redshift** factor. In the weak-field limit with $g_{00} = 1 + 2\phi$, we have $\Delta\omega/\omega = -\Delta\phi$. Since photon energy $E = \hbar\omega$, the photon loses energy as it rises. A rising massive particle loses kinetic energy (slows down), but a rising photon loses energy by redshifting. For example, light from an atomic transition on the surface of the sun is redshifted a factor of $2 \times 10^{-6}$ by the time it reaches us.

An electron sitting on the surface of the Earth surely doesn't radiate (where would the energy come from?). But to a free-falling observer, it is an accelerating charge, so it should, leading to a potential **paradox**. Conversely, a falling charge must radiate due to its acceleration, but a comoving observer sees only a Coulomb field (the radiation field is beyond his Rindler horizon[21]). Under General Relativity, we must recognize that even the very existence of radiation is observer-dependent.

# Chapter 15

# Light and the Cosmic Microwave Background

> Our whole universe was in a hot, dense state
> Then nearly fourteen billion years ago expansion started, wait!
>
> "The Big Bang Theory" theme song

Light is key to our understanding of the large-scale structure of spacetime.

## 15.1 The Big Bang Theory

The Big Bang theory (not the TV show) states that the universe began small, dense, and hot; then expanded and cooled. Why do we believe this? I'll give four arguments of increasing sophistication: Olbers's paradox, the Hubble expansion, the cosmic microwave background (CMB), and primordial nucleosynthesis.

### 15.1.1 Olbers's Paradox

Before Hubble's observations, the default assumption was that the universe is infinite in space and time, homogeneous and static. In such a universe, in any direction that you pointed a telescope, there would be a bright star somewhere along the way. "Wait," you argue, "maybe most stars would be obscured by dust?" But after enough time, the dust would thermally equilibrate with the stars, so the dust would glow hot too. **Olbers's paradox** says that in such a universe, the night sky should be blazingly bright and hot. Thus the very darkness of space suggests that the universe is finite in either extent or duration, maybe both.

### 15.1.2 Hubble Expansion

Edwin Hubble observed that the further objects are from Earth, the faster they are receding from us. This is only true at very large scales, *i.e.* galactic superclusters and larger, not for bound systems like our solar system or our galaxy. A distant galaxy's recession speed is inferred from from its Doppler redshift; really what he saw was that more distant galaxies are redder. The relation is linear; if $a$ is the distance to a galaxy, then $\dot{a} = Ha$ where $H$ is the **Hubble constant**. Note it is "constant" throughout space but not time, so "Hubble constant" isn't really

such a great name, sorry. The units of $H$ are inverse time, so its current observed value can be written $H = 1/(14.4\,\text{Gy})$, where Gy is a billion years. Even without any further calculation, this measurement immediately suggests the universe is roughly 14 billion years old.

It's not an indication that Earth is particularly unpopular and everyone is avoiding us! Rather, everything in the universe (at large scales) is moving away from everything else; that is, the whole universe is expanding. When you're part of an explosion, no matter where you are, everything else is moving away. A better analogy is an inflating balloon with dots marked on its surface - it's better because no dot can claim to be at the "center" of the expansion; there is no center. If everything is expanding outwards, then in past times everything was closer together.[1] Gravity has slowed the expansion, so $H$ used to be a bit larger, and when you account for that the age of the universe comes out to be about 13.8 billion years.

### 15.1.3 Cosmic Microwave Background

Expanding fluids cool down. It's the principle behind refrigerators[2], cold weather on mountain tops, and the adiabatic cooling of the early universe.

And so with the early universe, there is a thermodynamic relation between the time, density, and temperature. It depends on the contents, both matter and light. Early on, the universe was so hot that everything was in a chaotic plasma, light constantly interacting with quarks and electrons. But then the quarks settled into protons and neutrons, and then nuclei, and when the universe cooled to around $3,000\,\text{K}$ the electrons hooked up with the nuclei to form neutral atoms. Suddenly the universe became (relatively) transparent, and the $3,000\,\text{K}$ black-body radiation (mostly) decoupled from the matter. The black-body radiation continued to expand and cool, so that it now appears as a (nearly) uniform $2.73\,\text{K}$, called the **Cosmic Microwave Background** (CMB), because black-body radiation at that temperature peaks in the microwave part of the electromagnetic spectrum.

The story of the CMB discovery is worth looking up. In short, Wilson and Penzias of Bell Labs were investigating an annoying noise in their microwave antenna, which was intended for communication with satellites. They tried everything to eliminate the noise, including scraping off all the bird poop, until finally recognizing that they were observing a universal phenomenon that had been predicted in 1948 by Alpher and Herman using the Big Bang theory.

### 15.1.4 Primordial Nucleosynthesis

The universe at time $t$ is characterized by a length scale $a \sim t^{1/2}$ and a temperature $T \sim t^{-1/2}$ during the early, radiation-dominated era. The growth of $a$ is the expansion ($a$ might today represent the distance between galaxies), while the reduction in $T$ is the cooling. Their time-dependence derives from general relativity and the thermodynamics of a photon gas. Various types of matter (quarks, later nucleons, electrons) start out in thermal equilibrium with the photon bath, then "freeze out" as the processes that create and destroy particles stop keeping up. A careful modeling of the various freeze-outs gives a remarkably accurate prediction for the present-day abundances of several light isotopes.

[1] Not necessarily. The Steady-State theory holds that new matter is being spontaneously created all the time, countering the dilution of the observed expansion. But then you lose all the successful predictions from the hot, dense early state.

[2] Strictly speaking, common refrigeration uses the **Joule-Thomson effect**, in which an expanding fluid cools as it breaks its own intermolecular bonds, rather than adiabatic cooling.

In the first microseconds after the Big Bang, quarks run free in a hot "quark-gluon plasma". At about $t = 10^{-5}$ s they permanently settle into protons and neutrons, collectively called nucleons. At the beginning of the nucleosynthesis era, about $t = 10$ s, the ratio of protons to neutrons is about 6:1, as both are in thermal equilibrium but neutrons are slightly heavier, and thermodynamics tells us that $n_p/n_n = e^{\Delta m/kT}$, with $\Delta m = 1.3$ MeV and $kT = 0.7$ MeV at the time that processes like $n + \nu_e \leftrightarrow p + e^-$ freeze out.

Between $t = 10$ s and $t = 20$ min (that is 20 minutes), while the temperature $T > 10^9$ K, nuclear fusion processes occur easily and reversibly, so that various light isotopes are in thermal equilibrium. After $t = 20$ min, these light nuclei freeze out. This is the period of **primordial nucleosynthesis**, also known as **big bang nucleosynthesis**.

Free neutrons decay with a lifetime of about 15 minutes (while neutrons bound in nuclei last forever), so during these 20 minutes a few neutrons decay to protons. As a result, by the end the ratio of protons to neutrons is more like 7:1. For every 16 nucleons, there are 2 neutrons, which by the end of nucleosynthesis are most likely to have ended up (along with 2 protons) as a stable helium nucleus $^4$He. Other stuff happens, but most of the matter that remains in the universe is this one helium and 12 protons (hydrogen). Here then is our first prediction: about 25% of the mass of the universe is helium, and the rest is mostly hydrogen. And that's about right! Though helium also gets produced in stars, that turns out to be a small correction. Helium plays little role on Earth, where our gravity can't even hold onto it for long, but it's where most of the primordial neutrons end up.

You know how hard it is to get four people to do something simultaneously? So it goes with 2 protons and 2 neutrons forming helium. Mostly the reaction proceeds through intermediate states of deuterium $^2$H, tritium $^3$H, and helium-3 $^3$He. We focus now on deuterium, which has one proton and one neutron (so chemically it is a heavy isotope of hydrogen), and is stable, though only barely. Anywhere that's hot enough for deuterium to form, the deuterium then quickly forms helium (deuterium's binding energy is only 2.2 MeV, while helium's is 28.3). So deuterium is constantly created and destroyed inside stars, but very little ever gets out. Similarly during primordial nucleosynthesis, except that at the end of the 20 minutes the universe cools rapidly enough to freeze out some deuterium, and that primordial abundance remains stable even to the present! Numerical simulations put deuterium at about .007% of the mass of the universe, another successful prediction. In other words, there's about 1 deuterium for every 38,000 hydrogen. On Earth it's about 5x that, 1 deuterium for every 6,420 hydrogen, most of it in the form of "heavy water" in the sea. If we ever get controlled nuclear fusion to work, this primordial deuterium floating around in the ocean is a likely source of fuel; we will finally complete the reactions that were interrupted by cooling just minutes after the Big Bang.

A similar calculation applies to helium-3 $^3$He, and there's a trickier one for lithium $^7$Li. Along with deuterium, these depend somewhat sensitively on a parameter $\eta = n_B/n_\gamma$, the baryon-to-photon ratio, which is hard to derive from first principles, as it depends on the origin of matter/antimatter asymmetry. CMB measurements suggest $\eta = 6 \times 10^{-10}$. The Schramm plot, Fig. 15.1, shows the predicted abundances as a function of $\eta$. The lithium prediction seems off - we don't see as much lithium in the universe as we would expect - but the others are excellent.

Heavier elements like carbon and iron do not form during primordial nucleosynthesis; rather they are born much later in the hearts of dying stars. Also remember that we have described the formation of nuclei, but it is still too hot for the electrons to bind to them until much later.

As a purely historical aside, much of this work was laid out in the famous **Alpher-Bethe-Gamow paper** published on April Fools' Day of 1948. Ralph Alpher was a Ph.D. student at George Washington University, working under George Gamow on "The Origin of Chemical Ele-

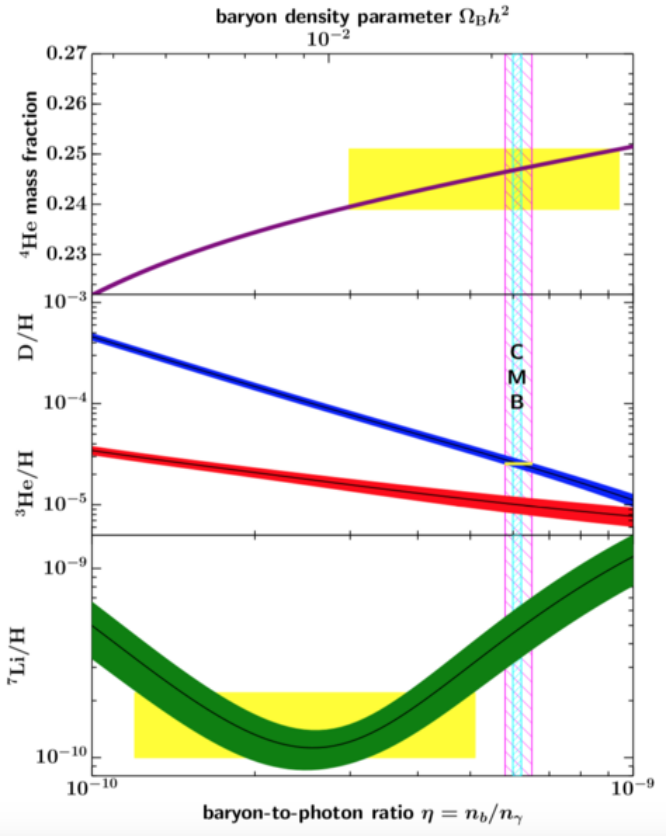



Figure 15.1: Schramm Plot

ments". Gamow apparently added the name of Hans Bethe, his friend at Cornell, to the paper primarily to make it sound like $\alpha\beta\gamma$ (the first three Greek letters: alpha, beta, gamma). Later, Hans Bethe made major contributions to the field, but his inclusion was initially a wordplay. Alpher, it should be noted, was not amused.

## 15.2 History of the Universe

Before $t = 10^{-6}$ s, that is when temperatures were hotter than anything we can explore on Earth, the history is highly speculative (as I indicate by the font color). Note the intimate connection here between high-energy physics and early-universe cosmology. Here are some ideas of what may have occurred:

- $t < 10^{-43}$ s (the Planck time), any description would require a more solid theory of quantum gravity than we have. Does the coordinate $t$ simply begin at zero, the way that there is nothing "north" of Earth's north pole? Or was there something before the Big Bang? Hartle and Hawking suggested that the universe is a quantum fluctuation of the spacetime metric $g^{\mu\nu}$, and the boundary conditions constrain it to minimum entropy in its initial state. It's all angels on a pin for now.

- $t = 10^{-36}$ s: If there is a GUT (Grand Unified Theory) such as SU(5), then that symmetry breaks around this time. GUT symmetry breaking might explain baryogenesis, the creation of an excess of baryons (matter) over anti-baryons (antimatter) - or it might not. GUT breaking also should produce magnetic monopoles, which is unfortunate because we don't observe any. Much as we love the idea, there is little evidence for any of this.

- $t = 10^{-32}$ s: In 1979 Alan Guth suggested the theory of **inflation**, which said that the universe expanded exponentially during the first $10^{-32}$ s. This would solve the **monopole problem**, by diluting away any GUT monopoles that might have been produced. It also solves the **horizon problem**, which is that distant parts of the universe that were never causally connected are nevertheless at the same temperature. And it solves the **flatness** problem, which is that the curvature of space seems to be zero or very close to it. The mechanism for inflation is that some new field (the "inflaton") gets stuck in a metastable state (a "false vacuum"); general relativity says that the pervasive positive energy density in the universe then gives the desired exponential expansion. But nobody has ever detected this inflaton field or explained why it gets stuck.

- $t = 10^{-14}$ s?: If there is supersymmetry, then at some time it must have been broken. As supersymmetry has not yet been seen, we don't know at what energy scale that happens, if it happens at all.

- $t = 10^{-12}$ s: Electroweak symmetry breaking occurs, separating the weak nuclear force from electromagnetism. Possibly baryogenesis occurs during this phase transition, but we're not sure. At this point, though, energies of $\sim 100$ GeV (or equivalently temperatures of $\sim 10^{15}$ K) bring the physics into the realm of what can be studied in high-energy particle accelerators.

From this point forward, our understanding gets pretty solid. After electroweak symmetry breaking (mediated by the Higgs field, whose scalar particle was discovered in 2012), we know the full set of light particles (quarks, leptons, gauge bosons) that inhabit the thermal bath and remain

relativistic (say $m < 1\,\text{TeV}$). Since we do not really understand baryogenesis (the emergence of matter over antimatter), one unknown parameter is the baryon-to-photon ratio, which CMB measurements put at $\eta = 6 \times 10^{-10}$. Regardless of what came before, with this we can model events going forward with confidence:

- $t = 10^{-6}\,\text{s}$: The quark-gluon plasma condenses into protons and neutrons. The hot plasma that pervades the universe then contains these nucleons, plus electrons, muons, neutrinos, and photons. Most of the thermal energy is in photons and other relativistic particles, making this the "radiation-dominated era".

- $t = 20\,\text{min}$: Primordial nucleosynthesis, described above. By its conclusion, the universe is about 75% hydrogen, 25% helium, 0.1% deuterium, and $10^{-9}$ lithium.

- $t = 50,000\,\text{y}$: The matter-dominated era begins. Energy density of radiation (photons) goes like $a^{-4}$ while energy density of matter goes like $a^{-3}$, so by about 50,000 years after the Big Bang the majority of the universe's energy is now in matter. In this context, "matter" means both ordinary (baryonic) matter and dark matter. The temperature is about $T = 16,000\,\text{K}$ at this point. Once matter dominates, $a \sim t^{2/3}$ and $T \sim t^{-2/3}$, which differ from the radiation-dominated behavior.

- $t = 379,000\,\text{y}$: Electrons bind to nuclei to form neutral atoms, mostly hydrogen and helium. There is a transition from a hot charged plasma which interacts strongly with light, to a bath of neutral atoms that are largely transparent, so light and matter decouple. One might say the light separated from the darkness. The thermal bath of light is at about $T = 3,000\,\text{K}$, a scale set by the **Rydberg energy** $Ry = \alpha^2 m_e c^2/2 = 13.6\,\text{eV}$, and as that light cools to the present day it becomes the cosmic microwave background (CMB) at $T = 2.73\,\text{K}$.

- $t = 2 \times 10^8\,\text{y}$: The first-generation ("Population III") stars form, and when they die they form all the heavier elements that are necessary for our rocky planet and carbon-based life.

- $t = 3 \times 10^9\,\text{y}$: Dark energy, whose energy density does not depend on $a$ at all, begins to dominate the expansion equation. We don't know anything about dark energy (it may just be a term in Einstein's equation), so I call this speculative. However, cosmologists require it to explain their observations that expansion is accelerating again.

- $t = 13.8 \times 10^9\,\text{y}$: People evolve on planet Earth.

Fig. 15.2, copied from Collins, Martin, & Squires[17], shows the universe's history on log scales of time, size, energy density, temperature, and energy.

## 15.3 General Relativity and Thermodynamics

It isn't really all that mysterious: stuff flies apart, expanding stuff cools down, and gravitational attraction slows the expansion. But to derive the expansion rate $a(t)$ and temperature $T(t)$ we need relativity and thermodynamics.

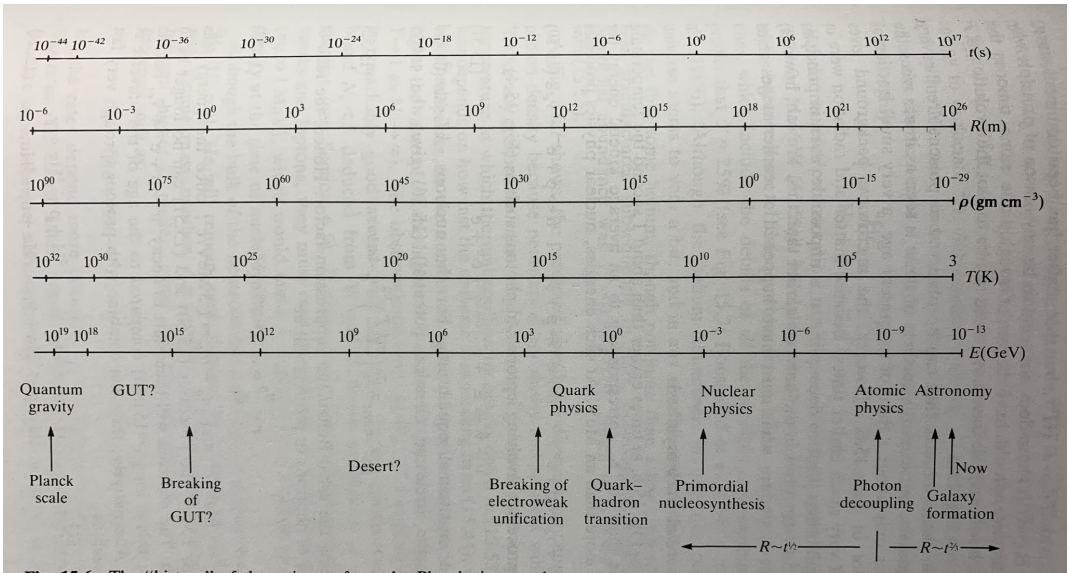



**Fig. 15.6** The "history" of the universe from the Planck time to the present, showing how the size of the presently observable universe $R$, the average density $\rho$, the temperature $T$, and the energy per particle $kT$, have varied with time $t$ according to the hot big bang model. Some of the major "events" and the dominant type of physics in each epoch are indicated.

Figure 15.2: History of the Universe

### 15.3.1 Expanding Space and the Robertson-Walker Metric

A central tenet of general relativity is that you can define any coordinate system, even a non-inertial one, and describe your system's dynamics in those coordinates with the same fundamental laws. Non-inertial frames give rise to gravity-like forces. A familiar example is a rotating frame, in which centrifugal and Coriolis forces appear - Coriolis forces on the rotating Earth, for example, produce hurricanes.

To describe our expanding universe, we use expanding (aka "comoving") spatial coordinates. In these coordinates, stuff is and remains "at rest". In the present day, I mean the distant galaxies are (roughly) at fixed $(X, Y, Z)$ values. For thermodynamics, I mean the thermal average motion is zero (though individual particles still exhibit Brownian motion, and light always moves at $c$). It's a convenient choice, because the stress-energy tensor is diagonal, *i.e.* no momentum. People say "space itself is expanding", and that sounds cool, but it really describes our chosen coordinate frame rather than any new rubbery property of space.

The invariant interval $\mathrm{d}s^2$ between events must be the same regardless of coordinate choice. In ordinary Cartesian coordinates (with $c = 1$), absent gravity, $\mathrm{d}s^2 = \mathrm{d}t^2 - (\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2)$. For our new expanding coordinates, $\mathrm{d}s^2 = \mathrm{d}t^2 - a^2(t)(\mathrm{d}X^2 + \mathrm{d}Y^2 + \mathrm{d}Z^2)$. The value of the expansion parameter $a(t)$ is not important (just a change of length units!) but its derivatives are; for now think of it as the (expanding) distance to your favorite Hubble-flow galaxy. In tensor notation we write $\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}X^\mu \mathrm{d}X^\nu$, with:

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -a^2(t) & 0 & 0 \\ 0 & 0 & -a^2(t) & 0 \\ 0 & 0 & 0 & -a^2(t) \end{pmatrix} \tag{15.1}$$

which is called the **Robertson-Walker metric**[3]. For shorthand I'll write $g_{\mu\nu} = \mathrm{diag}_4[1, -a^2]$,

[3] Also known as "FLRW" for Friedmann, Lemaître, Robertson, and Walker

indicating a $4 \times 4$ diagonal matrix with the spatial entry repeated 3 times. (I've also assumed space is flat, based on empirical evidence.)

The stress-energy tensor $T^\mu_\nu$ is, in general, a spacetime-dependent 4-tensor containing energy density, momentum density, shear stress, and pressure. The beauty of the Robertson-Walker (RW) coordinates is that $T^\mu_\nu$ is diagonal (no momentum or shear) and uniform everywhere. In my notation $T^\mu_\nu = \mathrm{diag}_4[\rho, -p]$, where $\rho$ is energy density, and $p$ is pressure. The **equation of state** for ("cold") non-relativistic matter is $p = 0$, while for relativistic matter and light it is $p = \rho/3$.

John Wheeler said, "Matter tells spacetime how to curve, and spacetime tells matter how to move." The first of these is the Einstein Field Equation:

$$R^\mu_\nu - \tfrac{1}{2}\, g^\mu_\nu\, R = -\kappa^2 T^\mu_\nu \tag{15.2}$$

where the left side is a complicated function of the metric $g_{\mu\nu}$ and its derivatives; $\kappa^2 = 8\pi G/c^4$ is a constant related to Newton's constant in Eq. 2.2, and $T^\mu_\nu$ is the stress-energy tensor. Thus Eq. 15.2 is a differential equation for $g_{\mu\nu}$ that depends on the matter and radiation present. It is the analog of Eq. 7.12 ($\nabla_\mu F^{\mu\nu} = J^\nu$) in electromagnetism. In our Robertson-Walker metric, it's just an equation for $a(t)$ that depends on $\rho$ and $p$. I've omitted the cosmological constant $\Lambda$, as appropriate for the early universe, but I will discuss it in SubSec. 15.3.5, and again I will assume space is flat, see SubSec. 15.4.2. We get the flat-space **Friedmann equations**:

$$H \equiv \frac{\dot a}{a} = \kappa \sqrt{\frac{\rho}{3}} \quad , \qquad \frac{\ddot a}{a} = -\frac{\kappa^2}{6}(\rho + 3p) \tag{15.3}$$

We emphasize again that you could derive these equations from ordinary Newtonian mechanics, no "expanding space" needed, though with less rigor.

### 15.3.2 Adiabatic Cooling

Let's take a moment to understand why a gas cools when it expands. Suppose you have a plastic bag at sea level, containing some air but not fully inflated. The bag delimits a fixed set of $N$ air molecules, and transmits pressure between the outside and the inside, but does not itself impose a force on the contents (in contrast to, say, a balloon). I'm thinking of a bag of snacks on an airplane, for example, ignoring the actual snacks. Then you rise in altitude, and see the bag has expanded. The pressure of the air inside the bag always matches the air outside (as long as the bag hasn't reached its maximum capacity). But to expand the bag, against the air pressure all around it, took work. By energy conservation, after the air performed that work it must be cooler. Now take away the bag (and the snacks), as it was only a conceptual crutch. **All** the air at altitude cooled down as it expanded into its surroundings, which is why mountain tops are cold.

We write $\mathrm{d}E = -p\,\mathrm{d}V$: the energy loss (cooling) equals pressure times change in volume. With equation of state $p = w\rho$, this becomes $\mathrm{d}(\rho a^3) = -w\rho\,\mathrm{d}(a^3)$, whose solution is:

$$\rho \sim a^{-3(1+w)} \qquad \text{when} \quad p = w\rho \tag{15.4}$$

For cold matter $p = 0$, so $\rho \sim a^{-3}$ as you'd expect: energy density is the same as number density which varies inversely as the volume. For radiation $p = \rho/3$, so $\rho \sim a^{-4}$. If you like, you can say that the number density of photons varies inversely as the volume, but also their wavelength stretches with space, reducing each photon's energy. Or you can just call it adiabatic cooling of the photon gas.

### 15.3.3 The Radiation-Dominated Era

By "radiation" we mean light and other relativistic particles, also called "hot matter". These dominate the energy density and dynamics of the early universe.

The equation of state for radiation is $p = \rho/3$. That's because, in the rest frame of the thermal bath, $T_{xx} = \rho\langle v_x^2\rangle$, but for relativistic particles $v_x^2 + v_y^2 + v_z^2 = 1$ and so by isotropy $\langle v_x^2\rangle = 1/3$.

From the Friedmann equations $\dot{a}/a \sim \sqrt{\rho}$, and we have $\rho \sim a^{-4}$ from adiabatic cooling, thus $a \sim t^{1/2}$. The universe expands as the square root of time.

The energy density of a relativistic gas at temperature $T$ is:

$$\rho = \frac{g}{2\pi^2}\int_0^\infty \frac{Ep^2\mathrm{d}p}{e^{E/kT} \mp 1} \tag{15.5}$$

where the top sign is for bosons, the bottom for fermions, $g$ is the number of spin states, and $p = E$. A units analysis should immediately convince you that the result must go like $(kT)^4$. The integral for fermions turns out to be 7/8 as big as the one for bosons. When radiation dominates, then, the total energy density is

$$\rho = \tfrac{1}{2}\, g_*\, a_{SB}\, T^4 \tag{15.6}$$

where $a_{SB}$ is the Stefan-Boltzmann constant, $g_* = g_b + \frac{7}{8}g_f$, and $g_b$, $g_f$ are the **total** number of spin degrees of freedom (DOFs) of relativistic bosons and fermions respectively.

Under the Standard Model, for temperatures $kT > 1\,\mathrm{TeV}$, we have all the gauge bosons (photons, gluons, $W^\pm$ and $Z^0$), the Higgs, and three families of quarks and leptons, giving $g_* = 106.75$. As the temperature drops below each particle's mass, those particles drops out of the mix, reducing $g_*$. The binding of quarks into nucleons also reduces $g_*$.

Knowing that $\rho \sim T^4$ from the Boltzmann distribution, and $\rho \sim a^{-4}$ from adiabatic cooling, and $a \sim t^{1/2}$ from Friedmann, we find $T \sim t^{-1/2}$. Temperature drops as the square root of time. Filling in all the constants:

$$\rho = \frac{3}{32\pi G}\, t^{-2} \quad , \qquad T = \frac{45c^5\hbar^3}{16\pi^3 G g_* k^4}\, t^{-1/2} \tag{15.7}$$

### 15.3.4 The Matter-Dominated Era

Since $\rho \sim a^{-4}$ for radiation but $\rho \sim a^{-3}$ for cold (non-relativistic) matter, cold matter began to dominate the expansion of the universe after around 50,000 years. Coincidentally, this is close to (but slightly earlier than) the time when photons decouple. Of course all gases have some pressure, but it is negligible compared to $\rho$ in the Einstein field equations, so we set $p = 0$.

Combining $\rho \sim a^{-3}$ with the Friedmann equation $\dot{a}/a \sim \sqrt{\rho}$ gives $a \sim t^{2/3}$ in the matter-dominated era. Again I have assumed flat space, based on observation; otherwise the equations could give different behavior such as reaching a maximum $a$ and then shrinking into a "Big Crunch". Observations suggest there will be no Crunch.

Meanwhile, after the photon bath decouples from everything else, it cools at $T \sim t^{-2/3}$ because matter controls the expansion. The $3,000\,\mathrm{K}$ bath at decoupling ($t_d = 3.79 \times 10^5\,\mathrm{y}$) becomes a $2.73\,\mathrm{K}$ bath today ($t_0 = 13.8 \times 10^9\,\mathrm{y}$), thus the CMB. To be clear, the decoupling time is derived this way: $(2.73/3,000) = (t_0/t_d)^{-2/3}$. Neutrinos also decouple and form their own bath, at a slightly different $T$ due to the 7/8 effect, but that's probably impossible to observe.

### 15.3.5 The Dark-Energy Era

Based on observed redshifts of very distant galaxies, we may need to modify Eq. 15.2. You can add a cosmological constant term $+\Lambda g^\mu_\nu$ on the left-hand side, or equivalently postulate a "dark energy" contribution to $T^\mu_\nu$ on the right-hand side, defined by $\rho_\Lambda = \Lambda/\kappa^2$ and equation of state $p_\Lambda = -\rho_\Lambda$. The second view, which is in favor these days, treats dark energy like a substance with negative pressure, causing the universe to accelerate outward, a sort of universal antigravity. Notice that negative adiabatic cooling (Eq. 15.4 with $w = -1$) keeps the energy density $\rho_\Lambda$ constant even as space expands, consistent with the first view that $\Lambda$ is a just a physical constant. Because it does not dissipate, if there really is a $\rho_\Lambda > 0$ then it inevitably dominates over both radiation ($\rho_\gamma \sim a^{-4}$) and matter ($\rho_M \sim a^{-3}$). We may already be in that era now.

## 15.4 The Large-Scale Structure of Spacetime

Galaxies and galactic clusters are non-homogeneous, bound systems, for which RW coordinates are not useful. Starting at scales around $10^8$ ly, Hubble flow dominates the motion of matter. To first-order, we can view it classically: galaxies move apart due to initial conditions, they slow down due to mutual attraction, and their light is Doppler redshifted. Alternatively, we can say they are at rest in "expanding space", and their light is redshifted as wavelengths expand.

As distances (or times) approach $R_c \approx 10^{10}$ ly, however, the large-scale structure renders our classical intuition useless. By way of analogy, a Mercator projection of the Earth shows a reasonably accurate picture of the US, though we note how flight paths appear curved; but Greenland is oddly large, and Antarctica is unrecognizable. Similarly, at the largest scales (or earliest times) our universe appears to violate the simplest rules of special relativity: very distant galaxies recede at speeds greater than $c$, while light from a galaxy that once was $10^9$ ly away only now reaches us $10^{10}$ y later. General Relativity still holds, of course, but our intuition from overlaying flat Minkowski coordinates fails badly. RW coordinates are needed to understand the large-scale structure.

On a related note, a core principle of special relativity is that all inertial frames are equally valid, but now we say that (at any point in spacetime) the CMB singles out one particular frame to be "at rest": only in this frame does the universe look isotropic. Notably, Earth is **not** at rest, but moves about 370 km/s with respect to the CMB, and it varies as Earth orbits the sun, the sun orbits the galaxy, etc. Our motion makes the CMB appear to have a large dipole moment (blueshifted in front of us, redshifted behind), but you will usually see that subtracted out. For most cosmology discussions, when we say "Earth" we really mean a hypothetical observer here-ish but at rest with respect to the CMB.

### 15.4.1 Spacetime is Curved

4D spacetime curvature is described by the **Ricci tensor** $R^\mu_\nu$ which appears in Eq. 15.2. For the flat-space RW metric of Eq. 15.1, and applying the Friedmann equations Eq. 15.3, the Ricci tensor is:

$$R^\mu_\nu = \mathrm{diag}_4[3\ddot{a}/a,\ (\ddot{a}a + 2\dot{a}^2)/a^2] = \tfrac{1}{2}\kappa^2\,\mathrm{diag}_4[-\rho - 3p,\ \rho - p] \tag{15.8}$$

We note as promised that the curvature is large in the early universe, when $\rho$ is large, and approaches infinity at the singularity $t = 0$.

We can further calculate the **Ricci scalar** $R = g^\nu_\mu R^\mu_\nu$ (that is, the trace of the matrix $R^\mu_\nu$):

$$R = 6(\ddot{a}a + \dot{a}^2)/a^2 = \kappa^2(\rho - 3p) \tag{15.9}$$

We note that $R$ vanishes in the radiation-dominated era, but is positive for matter-dominated and dark-energy-dominated eras. $R$ vanishes for radiation because $\mathrm{Tr}\{T^\mu_\nu\} = 0$.

Assuming matter-domination ($p = 0$), and $\rho = \rho_c = 3H^2/\kappa^2$ the critical density for flat space, the Ricci scalar is simply $R = 3H^2$. If you can thereby loosely define a "radius of curvature" for the 4D spacetime, it is of order the age of the universe.

Finally, we confirm that Eq. 15.2 is satisfied:

$$\tfrac{1}{2}\kappa^2\,\mathrm{diag}_4[-\rho-3p,\,\rho-p] - \tfrac{1}{2}\kappa^2(\rho-3p)\,\mathrm{diag}_4[1,1] = -\kappa^2\,\mathrm{diag}_4[\rho,-p] \tag{15.10}$$

### 15.4.2 Space is Flat

Separately you can ask about the curvature of 3D space at a given time $t$, and in particular the present time $t_0$. Empirically space appears to be flat and we don't really know why: maybe inflation, maybe it's an initial condition. I have assumed flatness to simplify equations starting with Eq. 15.1. Let's discuss what makes us believe space is flat, and the implications.

The **CDM Model** was the prevalent theory when I was in school, and it has an elegant simplicity, so let's start there. Consider a matter-dominated universe of spatial curvature radius $R_S$, then $-1/R_S^2 = (2\ddot{a}a + \dot{a}^2)/a^2$. Actual measurements of $\dot{a}/a$ (Hubble constant) and $\ddot{a}/a$ (deceleration) show that $-1/R_S^2$ is consistent with zero, *i.e.* space is flat. You can see that the Friedmann equations as written in Eq. 15.3 (with $p = 0$) have that assumption baked right in; otherwise additional terms would have appeared. The fluctuation spectrum of the CMB also indicates flatness.

By convention we define the **deceleration parameter** $q = -\ddot{a}a/\dot{a}^2 = \rho\kappa^2/(6H^2)$. Space is flat if $q = 1/2$, and that happens if the energy density $\rho$ takes a particular critical value $\rho_c = 3H^2/\kappa^2 = 9.47\times10^{-30}\,\mathrm{g/cm^3}$. We define $\Omega = \rho/\rho_c$ as the ratio of actual energy density to the critical value.

In this matter-dominated model, if $\Omega = 1$, there's just enough matter around that its mutual attraction slows expansion down to asymptotically zero, not quite enough to make the universe start contracting. It is analogous to a rocket whose engines cut exactly at escape velocity – it keeps rising, but ever more slowly – and we recall from freshman physics that the rocket's positive kinetic energy and negative gravitational binding energy cancel, so $E = 0$. So in this model, there is a beautiful connection between the curvature of space and the fate of the universe: positively curved space ($\Omega > 1$) reaches a maximum size and then contracts; while flat ($\Omega = 1$) and negatively curved space ($\Omega < 1$) expand forever. To repeat: flat space $\leftrightarrow$ asymptotic expansion $\leftrightarrow \Omega = 1$.

There's a problem: the energy density of baryonic matter is known, not just from counting stars but from primordial nucleosynthesis, and its contribution is only $\Omega_b = .04$. So we postulate the existence of **cold dark matter** (CDM), some unknown stuff that acts like baryonic matter as far as gravity is concerned, but doesn't participate in nucleosynthesis. And dark matter definitely exists! We see its gravitational effects in the rotation curves of galaxies, and gravitational lensing of light; it seems to live in large spherical blobs around most galaxies. We don't know how much CDM there is, but the flat-space CDM model would suggest that $\Omega_{DM} = .96$, so that the total matter density $\Omega_M = \Omega_b + \Omega_{DM} = 1$.

The flat ($\Omega_M = 1$) matter-dominated FLRW universe with $\Lambda = 0$, expanding asymptotically, is called the Einstein-de Sitter model.

Recent studies of very distant supernovae do not support the Einstein-de Sitter model, and instead suggest a **$\Lambda$-CDM** model that includes both dark matter (CDM) and dark energy ($\Lambda$). We still believe space is flat ($\Omega = 1$), but now the contributions are: $\Omega_b = .04$, $\Omega_{DM} = .29$, and

$\Omega_\Lambda = .67$.[4] We lose that beautiful connection between flat space and asymptotic expansion; rather the dark energy will always eventually cause runaway exponential expansion.

### 15.4.3 The Observable Universe

A flat universe could be infinite in spatial extent, but only a finite portion has emitted light that could have reached us by now. Returning to the Einstein-de Sitter model (flat, $\Omega_\Lambda = 0$, matter-dominated), we set $a(t) = (t/t_0)^{2/3}$ so that $a(t_0) = 1$ today, and therefore a light-year locally means the same thing in comoving coordinates $X^\mu$ as Minkowski coordinates $x^\mu$. Since $\mathrm{d}t = a(t)\mathrm{d}X$ for light, the total comoving distance light has travelled is

$$\Delta X = \int_0^{t_0} \frac{\mathrm{d}t}{a(t)} = t_0^{2/3} \int_0^{t_0} t^{-2/3}\mathrm{d}t = 3t_0 \tag{15.11}$$

We say the observable universe has a radius of 41.4 billion light years (comoving distance).

## 15.5 Astronomical Observations

In recent years, neutrino and gravitational-wave astronomy have started to take off, but still most of our knowledge of the cosmos comes from light passing through telescopes and spectrometers. Space-based telescopes have greatly expanded our ability to see across the full observable universe, in x-ray, visible, infrared, and radio frequencies.

### 15.5.1 Dark Matter

If we see a galaxy edge-on, stars on (say) the left side will be blue-shifted as they are heading towards us, and stars on the right will be red-shifted, with velocity depending on a star's distance from the galactic center. In freshman physics we learned that if a galaxy's mass is concentrated at the center, then $|v| \sim 1/\sqrt{r}$. But Vera Rubin and others found that **galactic rotation curves** are actually nearly flat, and velocities are higher than expected, suggesting a large amount of unseen mass distributed throughout the galaxy. That mass is far more than the visible stars, and more than the total baryonic matter expected in the universe from $\eta = n_B/n_\gamma = 6 \times 10^{-10}$, so we call it **dark matter**.

Gravity bends light, and therefore a galaxy can act as a **gravitational lens**, magnifying and distorting the image of stars behind it. It has become routine to map the mass in a galaxy based on the lensing effect. Again we see huge amounts of non-baryonic mass, distributed in a giant sphere centered on the galactic center. The sphere is sometimes called a "halo", but it is emphatically **not** flattened to a disk the way most of the ordinary matter is. That suggests that the dark matter does not interact much, because if it did the stuff at the poles would eventually settle down to the orbital plane.

It is thought that fluctuations in dark matter density may have happened early, with the denser regions attracting baryonic matter to create galaxies. The self-interacting baryonic matter could lose energy (but not angular momentum) to form a galactic disk, and further clump into stars and planets, while the dark matter halo remained unchanged.

The **Bullet Cluster** is the exception that proves the rule. As one galactic cluster tore through another, the baryonic matter interacted, slowing down and heating up to produce X-rays, while

[4] Note these are rough values and still changing.

the dark matter halos passed right through each other. In Fig. 15.3, the pink color shows X-ray emissions, while blue represents the mass implied by gravitational lensing.

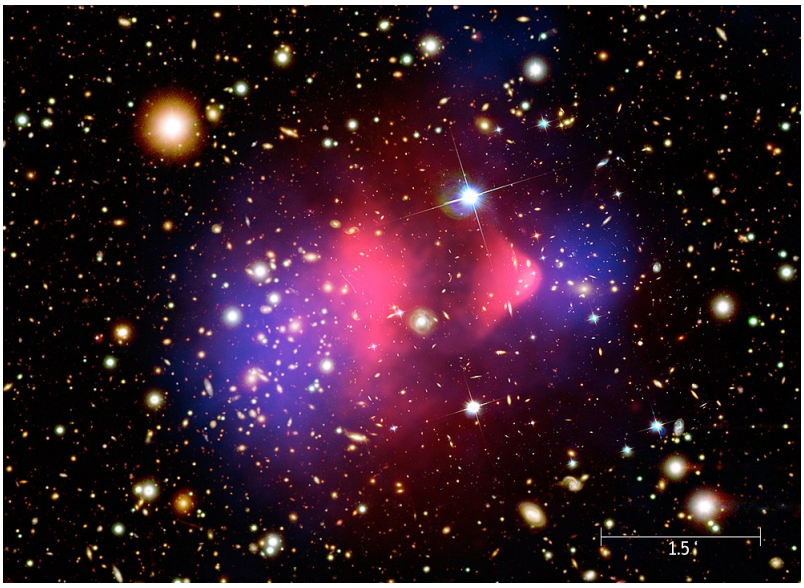



Figure 15.3: Bullet Cluster

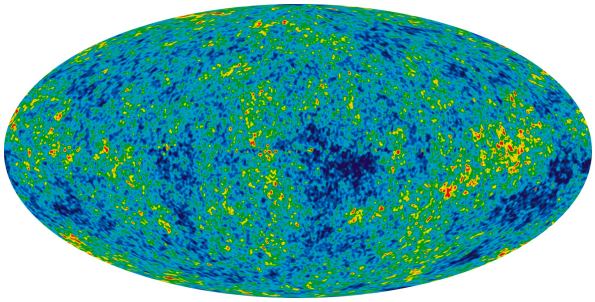
Nothing in the Standard Model could be the dark matter. One way we know this is that all ordinary matter (even neutrinos!) interacts enough that halos would form a disk, not a sphere. Another is that Primordial Nucleosynthesis predicts the abundances of all known particles to be in agreement with observation, and far too little to be dark matter.

So far efforts to detect dark matter have failed, but they all rely on the hope that maybe it is not *completely* dark (non-interacting). Maybe it's truly impossible to see except by its gravitational effects. Based on galactic and galactic cluster halos, there is at least 5x as much dark matter as baryonic matter (and could be more), so $\Omega_{DM} > .20$.

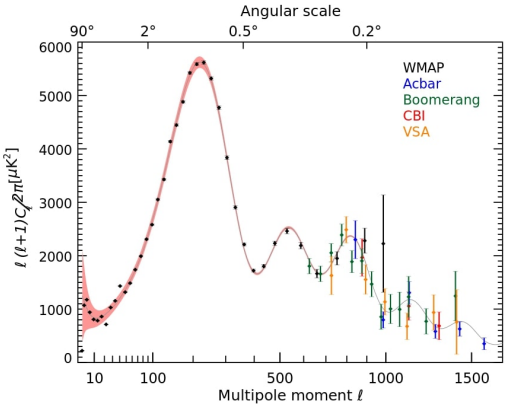
### 15.5.2 CMB Anisotropy

The $3,000\,\mathrm{K}$ thermal photon bath that existed at decoupling has cooled (by adiabatic cooling or expanding space, as you prefer) to $2.73\,\mathrm{K}$ today, and no matter what direction you point your radio telescope you will see a Planck spectrum at that temperature. Except that it's a little hotter (blueshifted) in the direction of Earth's motion, a little cooler (redshifted) in the opposite direction. Subtract that out, and there are still tiny fluctuations, less than 1 part in $10^5$, which look like Fig. 15.4a. Decompose them into spherical harmonics (as we saw in 9.5) and plot the power against $l$ (starting at $l = 2$, since $l = 0$ is the $2.73\,\mathrm{K}$ average and $l = 1$ is the dipole due to Earth's motion), see Fig. 15.4b. These very detailed data have been collected mainly by space-based radio telescopes, notably COBE, WMAP, and Planck.

What an amazingly rich structure in the power spectrum Fig. 15.4b! Perhaps even more surprising, the solid line based on a theoretical model is a perfect fit up to at least $l = 1000$. The model is based on acoustic oscillations in the photon-baryon plasma prior to decoupling. The location of the first peak (around $l = 200$ or $1°$) confirms for us that space is flat ($\Omega = 1$). The second gives the baryon-to-photon ratio $\eta = n_B/n_\gamma = 6 \times 10^{-10}$ as discussed in SubSec. 15.1.4. The third informs $\Omega_{DM}$. The ratio of peak $l$-values (roughly 1:2:3) suggests adiabatic density perturbations, as predicted by inflation.



(a) WMAP full-sky map of CMB anisotropy




(b) Power spectrum

Figure 15.4: CMB Anisotropy

There is even more information to be seen in CMB polarization, both E-modes and B-modes. In 2014 there was a brief excitement that BICEP2 had detected B-modes from primordial gravitational waves predicted by certain models of inflation, but alas, it was only dust.

### 15.5.3 Redshift Surveys

Our knowledge of the expansion $a(t)$ after photon decoupling comes primarily from redshifts $z(x)$ of distant galaxies, where $x$ is the distance to a galaxy and

$$z = \frac{\lambda_0 - \lambda_1}{\lambda_1} = \frac{a(t_0)}{a(t_1)} - 1 \tag{15.12}$$

The second relation assumes the emitter galaxy is in the Hubble flow. The measured redshift is unambiguous, *e.g.* if sodium D lines emitted at $\lambda_1 = 589\,\text{nm}$ are detected at at $\lambda_0 = 700\,\text{nm}$ then we say the emitter galaxy is at redshift $z = 0.19$, and the expansion parameter at the time of emission was $a(t_1) = a(t_0)/1.19$.

What is $x$, and how does it relate to $t_1$? First there is the general question of how astronomers calculate an object's distance, using a **cosmic distance ladder**, which we will not describe in detail. Second, in curved spacetime there are competing definitions of "distance", including luminosity distance, angular diameter distance, comoving distance, and light travel time distance.

Saul Perlmutter of the **Supernova Cosmology Project**, and Adam Riess and Brian Schmidt of the **High-Z Supernova Search Team**, won the 2011 Nobel Prize in Physics for their use of distant type Ia supernovae to measure $\Omega_M$ and $\Omega_\Lambda$. These supernovae are "standard candles", meaning their intrinsic luminosity $L$ is known (we think!), so their apparent brightness $F$ gives their luminosity distance $x_L = \sqrt{L/(4\pi F)}$. For a flat FLRW universe the relation between $z$ and $x_L$ should be:

$$x_L = \frac{c}{H_0}\,(1+z)\int_0^z \frac{\mathrm{d}z'}{\sqrt{\Omega_\Lambda + \Omega_M(1+z')^3}} \tag{15.13}$$

So the best fit for $z(x_L)$ reveals parameters $\Omega_\Lambda$ and $\Omega_M = \Omega_b + \Omega_{DM}$. Their results showed (surprisingly!) that dark energy already dominates, with $\Omega_\Lambda = .67$. Fig. 15.5 shows recent data for $\Omega_\Lambda$ vs $\Omega_M$.

These survey results boil down to high-redshift supernovae being dimmer than expected, so one worries that we are being fooled by intervening dust ("extinction"), or maybe supernovae were just dimmer in the early universe ("evolution"), *e.g.* due to lower metallicity. There also remains a small but nagging discrepancy between $H_0$ from supernovae and $H_0$ from CMB, so the final word is probably not yet written.

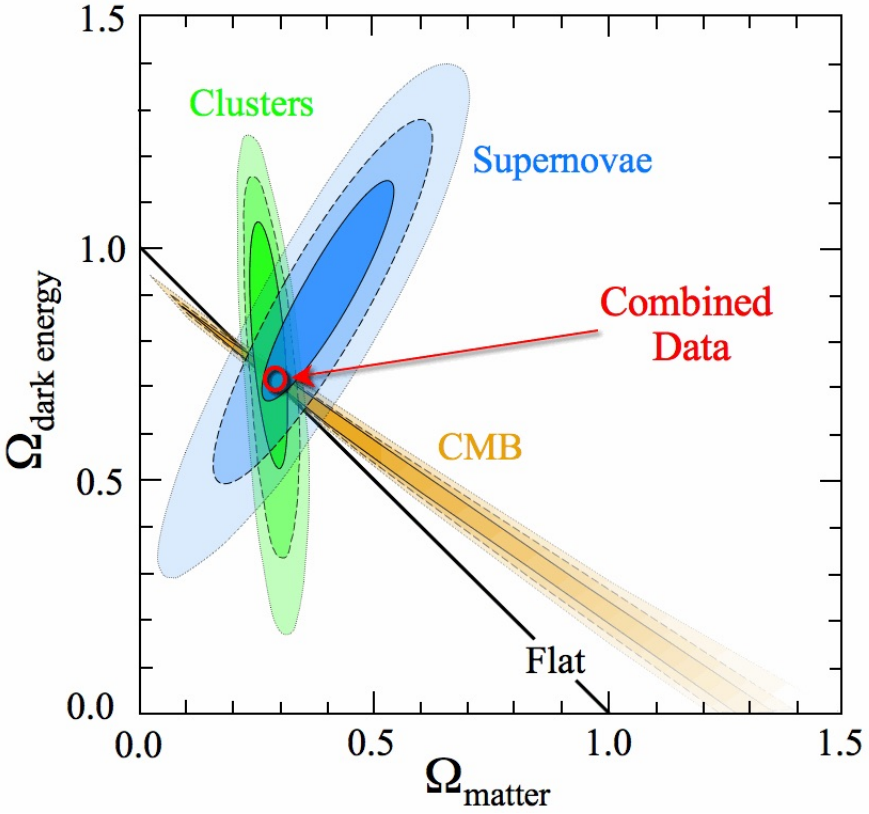



Figure 15.5: Best Fit for $\Omega_\Lambda$ and $\Omega_M$

# Chapter 16

# Light is a Quantum Field

A **field** $\Phi(\boldsymbol{x}, t)$ is an infinite set of dynamical variables, one at each point $\boldsymbol{x}$ in $D$-dimensional space. The variables may be scalars, spinors, vectors, or other sets of numbers. For example, the temperature $T$ at every point in a room is a real scalar field in $D$=3. We have already seen the electromagnetic potential $A^\mu$ is a 4-vector field in $D$=3. Now we will claim there is also an "electron field" everywhere, and each electron (and positron) in the universe is an excitation of it. A triumph of field theory is the ability to describe both photons and electrons with the same formalism.

In classical and quantum mechanics, a particle's position $\boldsymbol{x}$ is the dynamical variable, also called the generalized coordinate $q(t)$. Now in **field theory** $\Phi$ plays the role of the generalized coordinates, and $\boldsymbol{x}$ is just an index for them (along with any spinor or vector indices). In the language of Eq. 10.7, the Lagrangian density is a function of $\Phi$ and its derivatives.

## 16.1 Scalar Field

We begin with a scalar field $\Phi(\boldsymbol{x}, t)$, which (classically) is just a number at each point in spacetime. As described so far, time and space play very different roles: the field is indexed by spatial coordinates and evolves in time. But we learned in Sec. 9.8 that relativity is necessary if we are to understand particle creation and annihilation. So we immediately impose Lorentz invariance on the Lagrangian density $\mathcal{L}$, thus requiring the $\boldsymbol{x}$-derivative terms to look like the $t$-derivative terms. This is so highly constraining that the only allowed derivative term is $\frac{1}{2}\partial_\mu\Phi\,\partial^\mu\Phi$, which is henceforth called the **kinetic term**. By analogy with the electromagnetic source term $J^\mu A_\mu$ derived from Eq. 12.4, we also allow for a spacetime-dependent source $J(\boldsymbol{x}, t)$. We have

$$\mathcal{L} = \frac{1}{2}\partial_\mu\Phi\,\partial^\mu\Phi - \frac{1}{2}m^2\Phi^2 - \tilde{V}(\Phi) + J\Phi \tag{16.1}$$

where $\tilde{V}(\Phi)$ (the "interaction term") contains cubic and higher powers of $\Phi$.

The **classical field theory** is solved using the relativistic Euler-Lagrange Equation, Eq. 10.7. For $J = 0$ and $\tilde{V} = 0$ ("free field"), Euler-Lagrange gives the **Klein-Gordon equation** $(\Box + m^2)\Phi = 0$, whose solutions are waves $\Phi = e^{\pm ik\cdot x}$ where $k^\mu = (\omega, \boldsymbol{k})$ and $\omega^2 \equiv k^2 + m^2$. Localized sources $J$ can initiate (or terminate) these waves. Nonzero $\tilde{V}$ allows these waves to interact. That's all we'll say about the classical theory; we move on now to quantization.

### 16.1.1 D=0 and Canonical Quantization

Consider the free field in $D$=0 (no spatial dimensions): $\mathcal{L} = \frac{1}{2}\dot{\Phi}^2 - \frac{1}{2}m^2\Phi^2$. You have already solved this problem! It's mathematically identical to the unit-mass harmonic oscillator Lagrangian $L = \frac{1}{2}\dot{x}^2 - \frac{1}{2}\omega^2 x^2$, but with some (potentially confusing) name changes. We revisit SubSec. 9.4.2, setting the parameter $m \to 1$, renaming $\omega \to m$, and renaming the dynamical variable $x \to \Phi$. The quantized energy levels $(n + \frac{1}{2})\hbar\omega$ become $n$ "particles" of energy $mc^2$ (on top of the vacuum energy $\frac{1}{2}mc^2$). The raising operator $a^\dagger$ is renamed a "particle creation operator", and the lowering operator $a$ now annihilates particles. Now you know why the world is full of particles: they are the field-theory analog of the equally-spaced energy levels in a harmonic oscillator. The Heisenberg operator Eq. 9.8 becomes

$$\Phi = \frac{1}{\sqrt{2m}}(a^\dagger e^{imt} + a e^{-imt}) \tag{16.2}$$

| Harmonic oscillator | $D$=0 free scalar field | Description |
|---|---|---|
| $x$ | $\Phi$ | Dynamical variable |
| $p$ | $\dot{\Phi}$ | Canonical momentum |
| $m$ | 1 | Kinetic term coefficient |
| $\omega$ | $m$ | Quadratic coefficient in $V$; energy level spacing |

Returning to $D$=3, **canonical quantization** applied to relativistic fields is called **second quantization**. That is, first quantization quantizes the phase space $(x, p)$ of a single, immutable particle; while second quantization describes how particles arise from fields. We won't go through it, but you can now almost guess this result for the field in Fourier space:

$$\Phi = \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3}\frac{1}{\sqrt{2\omega}}(a^\dagger_{\boldsymbol{k}} e^{ikx} + a_{\boldsymbol{k}} e^{-ikx}) \tag{16.3}$$

where $\omega \equiv \sqrt{\boldsymbol{k}^2 + m^2}$. There are creation and annihilation operators for each 3-momentunm $\boldsymbol{k}$. Note the exponents are dot-products of 4-vectors: $ikx \equiv i\omega t - i\boldsymbol{k}\cdot\boldsymbol{x}$.

In the harmonic oscillator problem, position $x$ is implicitly real, so the solution in Eq. 16.3 is Hermitian: $\Phi^\dagger = \Phi$. We call it a "real scalar field", and the particle is its own antiparticle, which implies it cannot have an electric charge or other non-zero quantum number. An example is the neutral pion $\pi^0$. If we drop the hermiticity constraint, we have a "complex scalar field" whose antiparticles are created and annihilated by different operators $b^\dagger$ and $b$, in which case

$$\Phi = \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3}\frac{1}{\sqrt{2\omega}}(a^\dagger_{\boldsymbol{k}} e^{ikx} + b_{\boldsymbol{k}} e^{-ikx}) \tag{16.4}$$

An example is the charged pion $\pi^\pm$. The Lagrangian is then written

$$\mathcal{L} = \frac{1}{2}\partial_\mu\Phi^\dagger\,\partial^\mu\Phi - \frac{1}{2}m^2|\Phi|^2 - \tilde{V}(|\Phi|) + (J\Phi + J^\dagger\Phi^\dagger) \tag{16.5}$$

and the interaction potential $\tilde{V}$ contains only even powers of $|\Phi|$.

We have motivated **Relativistic Quantum Field Theory** (QFT) from canonical quantization, but we have not talked about how to calculate interactions. For that, we next re-derive QFT using Feynman's path-integral approach, Eq. 10.2, and the source-expansion approach ("sorcery") of Julian Schwinger. See the first chapter of Zee[13] for a highly readable and much more thorough description.

## 16.2 Vacuum-to-Vacuum Transition Amplitude

We quantize our field theory by appling Eq. 10.2. If you are studying, say, a collision of two particles to produce four, you might be inclined to set $|q_i\rangle$ to the two-particle initial state, and $\langle q_f|$ to the four-particle final state. Instead, we are going to create and annihilate states with the source $J$. Now we can start with a vacuum $|0\rangle$, create two particles with $J^2$, have them interact, annihilate four particles with $J^4$, and end up again at the vacuum. For the price of introducing the new parameter $J(\boldsymbol{x},t)$, we can now describe all of physics with a single **vacuum-to-vacuum transition amplitude**:

$$Z(J) \equiv \langle 0|e^{-i\hat{H}T}|0\rangle = \oint D[\Phi]\, e^{iS} \tag{16.6}$$

Here the closed path integral $\oint D[\Phi]$ must begin and end on the ground state (vacuum). Recall that $S = \int \mathcal{L} \mathrm{d}^4x$, where $\mathcal{L}$ is Eq. 16.1.

You could (in principle) expand $Z(J) = \sum_n \frac{1}{n!} J(x_1)\cdots J(x_n)\, G^{(n)}(x_1,\cdots,x_n)$ in powers of $J$ and the $n$-point Green's functions $G^{(n)}$, so $Z(J)$ is also called the **generating functional** for these Green's functions. In our $2 \to 4$ example we'd want to calculate $G^{(6)}$.

There is deep relation between this $Z$ and the partition function $Z$ of thermodynamics, see Eq. 11.8. The relation is even clearer when we **Wick rotate** to imaginary time by setting $t = i\tau$ and integrating over $\tau$ instead of $t$, which turns $e^{iS}$ into $e^{-S_E}$. You can study finite-temperature quantum field theory in thermal equilibrium at temperature $k_B T \equiv 1/\beta$ by imposing periodic boundary conditions on imaginary time $\Psi(\tau+\beta) = \Psi(\tau)$.

### 16.2.1 The Free Particle

For $\tilde{V} = 0$ (free particle), the relativistic Euler-Lagrange Eq. 10.7 gives the Klein-Gordon equation $(\Box + m^2)\Phi = 0$. The classical solutions are $\Phi = e^{ik\cdot x}$ where $k^\mu = (\omega, \boldsymbol{k})$ and $\omega^2 \equiv k^2 + m^2$.

When we quantize the free particle using Eq. 16.6, $Z(J)$ is a solvable Gaussian integral[13]. After an integration by parts that turns $(\partial\Phi)^2 \to -\Phi\partial^2\Phi$, we have

$$Z(J) = \oint D[\Phi]\, e^{i\int \mathrm{d}^4x(-\frac{1}{2}\Phi\partial^2\Phi - \frac{1}{2}m^2\Phi^2 + J\Phi)} \tag{16.7}$$

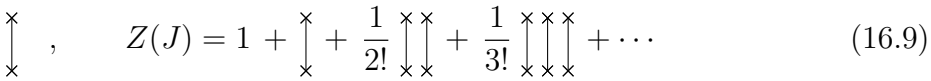
We now translate from the language of spacetime functions and integrals to the language of linear algebra: functions $\Phi$ and $J$ are (infinite-dimensional) vectors, the differential operator $A = (-\partial^2 - m^2)$ is a matrix, and integrals $\int \mathrm{d}^4x$ form dot products. Then we can use Eq. B.12 from Appendix B.4.1, and we don't even need the square root prefactor, just call it $Z(0)$:

$$Z(J) = \oint D[\Phi]\, \exp\left(\frac{i}{2}\Phi A\Phi + iJ\Phi\right) = Z(0)e^{iW(J)} \quad \text{where} \quad W(J) = -\frac{1}{2}JA^{-1}J \tag{16.8}$$

The **propagator** $A^{-1}$ is the **Green's function** of $A = (-\partial^2 - m^2)$, see Appendix B.3.2. That motivates us to "rotate" to a basis where $A$ is diagonal, which is actually Fourier space. Henceforth $J(\boldsymbol{x},t)$ is replaced by its Fourier transform $J(k)$, the propagator $A^{-1} = 1/(k^2 - m^2)$, and those implicit spacetime integrals $\int \mathrm{d}^4x$ are replaced by 4-momentum integrals $\int \mathrm{d}^4k/(2\pi)^4$.

Let's represent this result with little diagrams[22, 23]. The source $J$ is an "X", and the propagator $A^{-1}$ is a line:

$$iW(J) = \quad , \qquad Z(J) = 1 + \quad + \frac{1}{2!} \quad + \frac{1}{3!} \quad + \cdots \tag{16.9}$$



Think of the diagram for $W$ as a particle popping into existence at $x$, moving from $x$ to $y$, then disappearing at $y$. Or in Fourier space, it pops into existence with 4-momentum $k$ and exits with the same $k$. In the free theory, that's all a particle can do. The full amplitude $Z$ has any number of these pop-up particles, disconnected from one another, briefly existing between the initial vacuum and the final vacuum. We say $Z = e^{iW}$ contains "disconnected diagrams", while $W$ contains the sole "connected" diagram, so $W(J)$ is called the "connected generating functional". $W$ is analogous to the Helmholtz free energy $F$ in thermodynamics, where $e^{-\beta F} = Z = \Sigma_i e^{-\beta E_i}$.

There's a technicality worth mentioning. Implicitly we said a particle could pop into existence at any $x$ and exit at any $y$, but causality really requires $y^0 \geq x^0$. If you learn field theory using "canonical quantization", this is explicit in the time-ordering of operators. We realize this constraint by modifying the propagator to $1/(k^2 - m^2 + i\epsilon)$, forcing us to do the $k^0$ contour integration (Appendix B.5) in a particular way, and thus defining which pole is enclosed. Rather than having a particle propagate backwards in time, we will see an antiparticle propagating forwards in time. The $i\epsilon$ only matters in some loop diagrams.

In case you are disturbed by my mathematical sloppiness, here's the proper expression for $W$:

$$W(J) = -\frac{1}{2}\int \frac{\mathrm{d}^4k}{(2\pi)^4} J^*(k) \frac{1}{k^2 - m^2 + i\epsilon} J(k) \tag{16.10}$$

It bears repeating that the diagram in Eq. 16.9 is simply a shorthand for this expression, but at the same time it invokes a mental picture of a particle being created, propagating, and being destroyed. All that is left is to add interactions!

### 16.2.2 Quartic Interaction

Take the example of a quartic interaction potential $\tilde{V} = \frac{\lambda}{4!}\Phi^4$, so our Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\partial_\mu\Phi\,\partial^\mu\Phi - \frac{1}{2}m^2\Phi^2 - \frac{\lambda}{4!}\Phi^4 + J\Phi \tag{16.11}$$

The basic trick of **perturbative field theory** is that we expand $Z(J,\lambda)$ in both $J$ and $\lambda$. Mostly particles just propagate along like the free particle, but "occasionally" they interact.

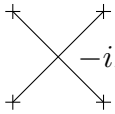
We do the same integration by parts as for Eq. 16.7, so we have the same quadratic operator $A = (k^2 - m^2)$ in Fourier space. The new trick is to replace $\Phi \to (\partial/\partial(iJ))$ in the interaction term $\tilde{V}$, by placing it left of $\frac{i}{2}\Phi A\Phi + iJ\Phi$, since $(\partial/\partial(iJ))(\frac{i}{2}\Phi A\Phi + iJ\Phi) = \Phi$. We end up with an exponential of $J$-derivatives acting on the solution $Z(J,0)$ that we already found:

$$Z(J,\lambda) = \exp\left[\frac{-i\lambda}{4!}\left(\frac{\partial}{\partial(iJ)}\right)^4\right] Z(0,0)\exp\left[iW(J,0)\right] \tag{16.12}$$

where $W(J,0)$ was given in Eq. 16.8, $W = -\frac{1}{2}JA^{-1}J$. Now Taylor expand the exponentials as usual, and choose the number of powers of $J$ and $\lambda$ you want.

Suppose we want the $J^4\lambda^1$ term, for example to calculate $2 \to 2$ scattering. We take the $\lambda^1$ term of the first exponential and the $W^4$ term of the second, to get

$$\frac{-i\lambda}{4!}\left(\frac{\partial}{\partial(iJ)}\right)^4 \frac{1}{4!}\left(\frac{-i}{2}JA^{-1}J\right)^4 \sim \frac{-i\lambda}{4!}(JA^{-1})^4 \quad = \quad -i\lambda \; + \quad \text{disconnected} \tag{16.13}$$



There are four $J$-derivatives acting on eight $J$'s in the first expression; think about the many different ways they can match up. First, there are 4! ways to match each derivative with a different $W$, neatly canceling (by design) the 4! in the denominator of $\tilde{V}$. Diagrammatically that's like chopping an "X" off each of four propagators and joining them together in a $\lambda$ vertex, as shown. Also the two $J$'s in each $W$ cancels the $\frac{1}{2}$ in the definition of $W$, in the same sense that $(\partial/\partial J)(\frac{1}{2}J^2) = J$. Second, two derivatives might act on the same $W$, which chops off both X's and turns that propagator into a loop. Third, that could happen twice, leaving a figure-eight floating around with no attachment to sources. But the second and third options lead to disconnected diagrams, which we will omit for the same reason as before: they can be generated from the connected generating functional $W$ by exponentiation $Z = e^{iW}$. Thus Eq. 16.13 shows the sole connected diagram at $\mathcal{O}(\lambda)$ for $2 \to 2$ scattering.

If you recall the definition of the $n$-point Green's functions, we have $G^{(4)} = (-i\lambda)(A^{-1})^4$ to first order in $\lambda$.

Any connected $n$-point diagram will have $n$ external propagators ("legs") corresponding to the $(JA^{-1})^n$ terms, and a delta function on the sum of the external momenta corresponding to 4-momentum conservation. We **truncate** the legs (removing the X's in the diagram), omit the delta function (by imposing 4-momentum conservation implicitly), and multiply by $i$ (as in $iW$), to produce the **matrix element** $\mathcal{M}$. To first order in $\lambda$, the 4-point matrix element for our quartic model $\mathcal{M} = \lambda$.

The rules to calculate $\mathcal{M}$, then, are:

1. Draw every connected diagram with the right number of external legs, to the desired power of $\lambda$. Assign a 4-momentum to every line, enforcing 4-momentum conservation.
2. Write $-i\lambda$ for every vertex.
3. Write $i$ times the propagator: $iA^{-1} = i/(k^2 - m^2 + i\epsilon)$ for every internal line.
4. If there are identical particles, include a symmetry factor (not discussed here).
5. If there are loops, integrate over the unconstrained 4-momenta $\int \mathrm{d}^4k/(2\pi)^4$.
6. Multiply each diagram by $i$, then add all diagrams to get $\mathcal{M}$.

Armed with these rules, and some table of integrals, anyone can calculate matrix elements from the TinkerToys of Feynman diagrams. Julian Schwinger famously said that "Feynman brought quantum field theory to the masses."

### 16.2.3 Decay Rates and Cross Sections

The most common processes for which we need matrix elements are decays $1 \to N$, and scattering cross-sections $2 \to N$. We present without proof the formulas you need, in terms of incoming and outgoing 3-momenta $\boldsymbol{p}_k$ and energies $E_k$.

If a particle of mass $M$ decays with a total decay rate $\Gamma$, then the number of particles at rest in a box will go like $e^{-\Gamma t}$ over time, so $\Gamma$ has units of inverse time. The particle lifetime is $\tau = 1/\Gamma$ (and its **half-life** is $\tau \ln(2)$). $\Gamma$ is the sum of all the partial decay rates $\Gamma_j$ from various decay modes, and each $\Gamma_j$ depends on $|\mathcal{M}_j|^2$ and kinematic factors. The differential decay rate is

$$\mathrm{d}\Gamma_j = |\mathcal{M}_j|^2 \frac{1}{2M} \prod_{k=1}^{N} \frac{\mathrm{d}^3\boldsymbol{p}_k}{(2\pi)^3 2E_k} (2\pi)^4 \delta^4(P - \Sigma_k p_k) \tag{16.14}$$

where the delta function ensures that 4-momentum is conserved: $\sum_k (E_k, \boldsymbol{p}_k) = (M, 0)$. There's an additional symmetry factor needed if you have identical final-state particles.

For a decay with $N = 2$, let $|\boldsymbol{p}|$ the magnitude of the 3-momentum of either outgoing particle, then

$$\Gamma_j = |\mathcal{M}_j|^2 \frac{|\boldsymbol{p}|}{8\pi M^2} \tag{16.15}$$

If two particles collide, the overall chance that they interact is measured by their **cross-section** $\sigma$ with units of area. We are particularly interested in "inelastic" collisions, meaning the particles change nature (rather than just momenta). Think classically of a cloud of incoming particles with **flux** $\Phi = \rho v$ (where $\rho$ is spatial density and $v$ is velocity) impinging on a target of cross-section $\sigma$, then the number of interactions per unit time is $\Phi\sigma$. The differential cross-section for incoming particles $i$ and $j$ is

$$\mathrm{d}\sigma = |\mathcal{M}|^2 \frac{1}{4\sqrt{(\boldsymbol{p}_i \cdot \boldsymbol{p}_j)^2 - m_i^2 m_j^2}} \prod_{k=1}^{N} \frac{\mathrm{d}^3\boldsymbol{p}_k}{(2\pi)^3 2E_k} (2\pi)^4 \delta^4(p_i + p_j - \Sigma_k p_k) \tag{16.16}$$

Again, you also need a symmetry factor if you have identical final-state particles.

When $N = 2$, in the center-of-mass frame, let $|\boldsymbol{p}_i|$ be the momentum of either incoming particle, and $|\boldsymbol{p}_f|$ the momentum of either outgoing particle, then

$$\mathrm{d}\sigma = |\mathcal{M}|^2 \frac{1}{64\pi^2 E_{CM}^2} \frac{|\boldsymbol{p}_f|}{|\boldsymbol{p}_i|} \mathrm{d}\Omega \tag{16.17}$$

### 16.2.4 Spin

We have spent a lot of time on scalar (spin-0) fields, but the electron is spin-1/2 and the photon is spin-1, so we will discuss these next. Table 16.1 summarizes the Lagrangians and propagators for spin up to one.

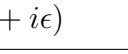
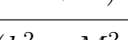
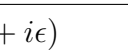
| Spin | Free Lagrangian | $i \times$ Propagator | Diagram |
|---|---|---|---|
| 0 | $\frac{1}{2}\partial_\mu\Phi\,\partial^\mu\Phi - \frac{1}{2}m^2\Phi^2$ | $i/(k^2 - m^2 + i\epsilon)$ |  |
| 1/2 | $\bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi$ | $i(\not{k} + m)/(k^2 - m^2 + i\epsilon)$ |  |
| 1 | $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2 A_\mu A^\mu$ | $i(-g^{\mu\nu} + k^\mu k^\nu/M^2)/(k^2 - M^2 + i\epsilon)$ |  |
| 1∗ | $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ | $-ig^{\mu\nu}/(k^2 + i\epsilon)$ |  |

Table 16.1: Propagators for Spins up to 1. "1*" is the massless photon in Feynman gauge.

We mention in passing that spin-3/2 particles, like the $\Omega^-$ baryon or the hypothetical gravitino, were described by Rarita and Schwinger using a combined 4-vector and Dirac spinor. A massless spin-2 particle like the graviton is represented by a tensor $g^{\mu\nu}$. In all cases, constraints (like gauge fixing) are imposed to keep the number of degrees of freedom correct; so for example the graviton has only 2 physical polarizations (not 10).

See Zee[13] chapter I.5 for a fascinating discussion of how the propagator numerators for scalar, vector, and tensor bosons (which can be derived from a sum over polarization vectors) lead to forces that are attractive, repulsive, and attractive, respectively. The attractive force mediated by

(scalar) pion exchange holds nuclei together. The repulsive force between like charges mediated by (vector) photons also binds opposite charges into atoms. And the attractive force of gravity, mediated by (tensor) gravitons, holds stars and planets together.

## 16.3 Spin 1/2

To represent a massive spin-1/2 particle like the electron, the field must be a (4-component) Dirac spinor as introduced in Sec. 9.8. With each external electron in a Feynman diagram, we associate a 4-spinor $u$, which can take two possible spin values $u_+$ and $u_-$. External positrons (anti-electrons) can be $v_+$ or $v_-$. By convention,[1] in the particle rest frame, these are

$$u_+ = \sqrt{2m}\begin{pmatrix}1\\0\\0\\0\end{pmatrix} \qquad u_- = \sqrt{2m}\begin{pmatrix}0\\1\\0\\0\end{pmatrix} \qquad v_+ = \sqrt{2m}\begin{pmatrix}0\\0\\1\\0\end{pmatrix} \qquad v_- = \sqrt{2m}\begin{pmatrix}0\\0\\0\\1\end{pmatrix} \tag{16.18}$$

The fermion version of Eq. 16.4 (canonically quantized field) is

$$\Psi = \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3}\frac{1}{\sqrt{2\omega}}\sum_s \left[d^\dagger(\boldsymbol{k},s)v(\boldsymbol{k},s)e^{ikx} + b(\boldsymbol{k},s)u(\boldsymbol{k},s)e^{-ikx}\right] \tag{16.19}$$

where $b$ destroys electrons and $d^\dagger$ creates positrons.

To make Lorentz-invariant terms, we use the gamma matrices of Eq. 9.21. There we explicitly showed (in the Dirac basis) the four matrices $\gamma^\mu = (\gamma^0, \boldsymbol{\gamma})$ and introduced the Feynman slash notation $\not{p} \equiv p_\mu\gamma^\mu$. You can read more about gamma matrices in Appendix B.7. In the electron rest frame, the requirement that the bottom two components of $u$ vanish can be written $(\gamma^0 - 1)u = 0$, as you can see from Eq. 9.21, and so the Lorentz-invariant statement is $(\not{p} - m)u = 0$, which is the famous **Dirac equation**. Similarly for the positrons $(\not{p} + m)v = 0$.

If spinors $u$ and $v$ are "column vectors" in spinor space, we define cospinors $\bar{u} \equiv u^\dagger\gamma^0$ and $\bar{v} \equiv v^\dagger\gamma^0$ which are "row vectors". That extra $\gamma^0$ in the Dirac adjoint is necessary for hermiticity. A cospinor times a spinor (row vector dotted with column vector), *e.g.* $\bar{u}u$, is a "spin-scalar" *i.e.* has no spin indices. Then $\bar{u}u$ and $\bar{v}v$ are Hermitian spin-scalar Lorentz scalars, while $\bar{u}\gamma^\mu u$ and $\bar{v}\gamma^\mu v$ are Hermitian spin-scalar Lorentz vectors.

By contrast, a spinor times a cospinor (outer product of column vector and row vector), *e.g.* $u\bar{u}$, is a 4x4 spin matrix, in fact must be some combination of the gamma matrices. Of particular interest is the sum $\Sigma_s u\bar{u}$ over the two possible spin states:

$$\sum_s u\bar{u} = \not{p} + m \quad , \quad \sum_s v\bar{v} = \not{p} - m \tag{16.20}$$

In most experiments, the spins of the incoming and outgoing fermions are not known, so to calculate $|\mathcal{M}|^2$ you average over initial spins and sum over final spins. "Average" is just half the sum, so equivalently you sum over all external fermion spins, with a factor of $\frac{1}{2}$ for each incoming one.

We saw above that the **Dirac equation** $(\not{p} - m)\Psi = 0$ constrains the 4-dimensional spinor $u$ to just two physical spin states, where $\Psi$ is the field of Eq. 16.19. We also saw in Sec. 9.8 that

[1] Some authors[13, 24] set $\bar{u}u = 1$, but then they need different Lorentz-invariant phase space elements for bosons and fermions. We choose the other convention[25] with $\bar{u}u = 2m$, allowing us to use Eqs. 16.14 and 16.16 unaltered.

Dirac wrote $(\not{p}-m)\Psi=0$ to make a linear operator equation that satisfies the relativistic relation $E^2=p^2+m^2$. Both are true! The Dirac equation derives from the **Dirac Lagrangian** which we saw in Eq. 12.1:

$$\mathcal{L}=\bar{\Psi}(i\not{\partial}-m)\Psi \tag{16.21}$$

The propagator is the inverse (Green's function) of the quadratic term in the Lagrangian, so $A^{-1}=1/(\not{p}-m)$. Nobody wants gamma matrices in the denominator, so multiply top and bottom by $(\not{p}+m)$ to get $A^{-1}=(\not{p}+m)/(p^2-m^2+i\epsilon)$, where I've added the $i\epsilon$ for causality just as in the scalar case.

There's another approach to deriving the fermion propagator. Start with the scalar propagator, and then multiply by the sum over spins of the bilinears: $\Sigma_s u\bar{u}=(\not{p}+m)$. We will find this works for the photon as well.

I've omitted some interesting details about fermion fields, such as their representation as anticommuting **Grassmann numbers**. One result of theoretical interest is that, while bosonic fields have positive infinite zero-point energy (like the $+\frac{1}{2}\hbar\omega$ of harmonic oscillators), fermionic fields have negative infinite zero-point energy. Of more practical interest, closed fermion loops in Feynman diagrams must be assigned an extra $-1$.

## 16.4 Spin 1

Massive spin-1 particles like the $Z^0$ require a Lorentz 4-vector field $A^\mu$, but (much like the spinors) that's too many degrees of freedom, as there are only 3 spin directions (or polarizations). In the rest frame, the 3 polarizations are along $(\hat{\boldsymbol{x}},\hat{\boldsymbol{y}},\hat{\boldsymbol{z}})$, so we set $A^0=0$, and the Lorentz-invariant version is $k_\mu A^\mu=0$, or equivalently $\partial_\mu A^\mu=0$.

The vector boson Lagrangian is

$$\mathcal{L}=-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}+\tfrac{1}{2}M^2A_\mu A^\mu \tag{16.22}$$

where $F_{\mu\nu}\equiv\partial_\mu A_\nu-\partial_\nu A_\mu$ and $M$ is the boson mass. The action $S=\int\mathcal{L}\mathrm{d}^4x$ can be put into quadratic form using an integration by parts as we did for Eq. 16.7, giving

$$S=\int\mathrm{d}^4x\tfrac{1}{2}A_\mu\left[(\partial^2+M^2)g^{\mu\nu}-\partial^\mu\partial^\nu\right]A_\nu \tag{16.23}$$

The propagator must be the inverse (Greens' function) of $[-(k^2-M^2)g^{\mu\nu}+k^\mu k^\nu]$, which is

$$A^{-1}=\frac{-g_{\mu\nu}+k_\mu k_\nu/M^2}{k^2-M^2+i\epsilon} \tag{16.24}$$

and I've put in $i\epsilon$ for causality.

In the rest frame, the propagator numerator is the diagonal matrix $\mathrm{diag}[0,1,1,1]$, and we note that is equal to the sum of the outer products of the three polarization vectors $\Sigma_i\epsilon^{(i)}\otimes\epsilon^{(i)}$. In fact that's a Lorentz-invariant statement, reminiscent of the fermion propagator numerator $\Sigma_s u\bar{u}$. On a related note, $k^\mu$ dotted with the propagator vanishes. Recall the propagator represents a free particle created at a source and destroyed at a sink; if you associate polarizations with the source and sink, then the amplitude is just the scalar propagator times $\Sigma_i\epsilon_\mu^{(i)}\epsilon_\nu^{(i)}$.

To describe the massless photon we just set $M=0$. Now you may worry about the $k_\mu k_\nu/M^2$ term, but in fact you can drop it. A simple way to see that is to recall current conservation $\partial_\mu J^\mu=0$, which implies the $k_\mu J^\mu$ terms vanish in $W=\frac{1}{2}JA^{-1}J$. The more formal result is called

the **Ward-Takahashi identity**, and is related to gauge invariance. For our purpose, we "choose the Feynman gauge" and write

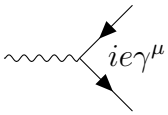
$$iA^{-1} = \frac{-ig_{\mu\nu}}{k^2 + i\epsilon} \tag{16.25}$$

## 16.5 QED

The QED Lagrangian, repeating Eq. 12.4, is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\Psi}(i\partial\!\!\!/ - eA\!\!\!/ - m)\Psi \tag{16.26}$$

The only vertex of QED is therefore



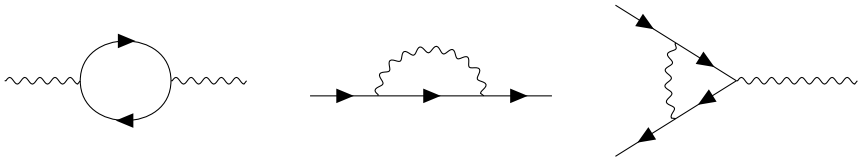

(16.27)

The wavy line is a photon $\gamma$, while the solid lines represent an electron/positron. A different charged fermion can be represented if you replace $(-e)$ with $(Qe)$ in the vertex, for example $Q = +2/3$ for quarks (u,c,t), or $Q = -1/3$ for quarks (d,s,b).

As shown, reading time as passing left-to-right, this is a photon turning into an electron-positron pair. A real photon can't do that, it would violate momentum conservation, so this must be a virtual photon, *i.e.* a propagator in a larger diagram, with $k^2 \neq 0$. If you flip the diagram horizontally, it's an electron-positron collision into a virtual photon (for the same reason), something that happens a lot in $e^+e^-$ colliders. Turned in other orientations, it could be an electron or a positron emitting or absorbing a photon. The matrix element contribution $ie\gamma^\mu$ is the same in every case.

### 16.5.1 Renormalization

There are loop diagrams that renormalize the photon mass, electron mass, and interaction strength:



(16.28)

Fortunately the first diagram vanishes: we would not want the photon to acquire mass from QED! But the other two diagrams tell us that measured values of $m_e$ and $e$ may not be the "raw" values in the Lagrangian. There is a whole art of **renormalization** regarding these corrections which we won't get into. We must conclude, however, that $\alpha = e^2/(\hbar c)$ depends on the energy scale of interactions. $\alpha = 1/137$ is true at low (electron-mass scale) energies, but it gets larger (stronger) at higher energies.

The Standard Model, Sec. 13.5, has three symmetry groups $SU(3)_c \otimes SU(2)_L \otimes U(1)_Y$, and therefore gauge three forces (strong, weak, and hypercharge). Remember electromagnetism is a mix of weak and hypercharge. There are therefore three (squared) couplings $\alpha_c$, $\alpha_L$, $\alpha_Y$, each of which "runs" in a way we can calculate if we know what particles can exist inside the loops. Weak and hypercharge (and therefore electromagnetic) couplings get stronger with energy. The strong coupling $\alpha_c$, however, gets weaker with increasing energy, a phenomenon known as **asymptotic**

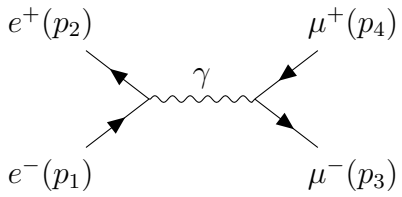
**freedom**, which allows us to analyze the theory perturbatively at energies above the **QCD scale** $\Lambda_{QCD} \approx 150\,\text{MeV}$.

It's speculated that the couplings run towards a common value at very high energies, which is an argument for Grand Unified Theories. For example, $SU(5)$ would have a single coupling at high energies. Calculations require knowledge of what heavy particles may exist beyond the known ones, and convergence improves when supersymmetric partners are included in the loops, which is an argument supporting Supersymmetry.

Having acknowledged the "running" of $m_e$ and $e$, we can henceforth omit diagrams like Eq. 16.28 from our calculations.

### 16.5.2 Electron-Positron Annihilation

We now calculate[25] the cross-section for $e^+e^- \to \mu^+\mu^-$ to lowest order in QED. This represents a real experiment, for example at the linear collider SLAC at Stanford, or the circular LEP collider at CERN, where electrons and positrons were slammed together and all sorts of stuff could come out, including a muon-antimuon pair. The diagram (with time flowing left-to-right) is:




(16.29)

$p_1$ and $p_2$ are incoming 4-momenta (of the electron and positron, respectively), while $p_3$ and $p_4$ are outgoing 4-momenta (of the muon and anti-muon, respectively) Do not be confused by the arrows, which track lepton number not momentum, so for the antiparticles momentum is opposite to the corresponding arrow. The **Mandelstam variables** are defined as: $s \equiv (p_1 + p_2)^2$, $t \equiv (p_1 - p_3)^2$, and $u \equiv (p_1 - p_4)^2$. Note that $\sqrt{s}$ is the center-of-mass energy achieved by the collider (*e.g.* SLAC's $\sqrt{s} = 91\,\text{GeV}$). The virtual photon's 4-momentum is $k = p_1 + p_2$, so the photon propagator is $-ig^{\mu\nu}/s$.

The matrix element from the Feynman rules is

$$\begin{aligned} \mathcal{M} &= i\,\bar{v}_2(ie\gamma_\mu)u_1\,\frac{-ig^{\mu\nu}}{s}\,\bar{u}_3(ie\gamma_\nu)v_4 \\ &= -\frac{e^2}{s}(\bar{v}_2\gamma_\mu u_1)(\bar{u}_3\gamma^\mu v_4) \end{aligned} \tag{16.30}$$

and its complex conjugate (using a different dummy Lorentz index) is

$$\mathcal{M}^* = -\frac{e^2}{s}(\bar{u}_1\gamma_\nu v_2)(\bar{v}_4\gamma^\nu u_3) \tag{16.31}$$

There are no other diagrams to add, so we can proceed to square the matrix element. We assume that the spins of the incoming particles ($e^+$, $e^-$) are unknown, so for each one we average over two possible spins. The spins of the outgoing particles ($\mu^+$, $\mu^-$) are not measured, so for each we sum over their spins. Thus we want $\frac{1}{4}\Sigma_s|\mathcal{M}|^2$, where the sum is over the $2^4$ possible spin states

of the four fermions.

$$\begin{aligned}\frac{1}{4}\Sigma_s|\mathcal{M}|^2 &= \frac{e^4}{4s^2}\Sigma_s(\bar v_2\gamma_\mu u_1\bar u_1\gamma_\nu v_2)(\bar u_3\gamma^\mu v_4\bar v_4\gamma^\nu u_3)\\ &= \frac{e^4}{4s^2}\Sigma_s\,\mathrm{Tr}\{\gamma_\mu u_1\bar u_1\gamma_\nu v_2\bar v_2\}\,\mathrm{Tr}\{\gamma^\mu v_4\bar v_4\gamma^\nu u_3\bar u_3\}\\ &= \frac{e^4}{4s^2}\,\mathrm{Tr}\{\gamma_\mu(\not p_1+m)\gamma_\nu(\not p_2-m)\}\,\mathrm{Tr}\{\gamma^\mu(\not p_4-M)\gamma^\nu(\not p_3+M)\}\end{aligned} \tag{16.32}$$

Each parenthetical term in the first line is a spin-scalar, so we are free to put a spin-Trace around it, after which we can permute its components: $\mathrm{Tr}\{ABCD\}=\mathrm{Tr}\{BCDA\}$, to get the second line. That procedure ("Casimir's Trick") may seem abstruse, but it lets us use Eq. 16.20 to eliminate the $u$'s and $v$'s in line three in favor of gamma-matrix traces, which you can look up in Appendix B.7. Here $m$ is the electron mass, and $M$ is the muon mass, but now for simplicity let's ignore them, assuming $(m,M)\ll\sqrt{s}$. Then the first Trace is (approximately)

$$p_1^\rho p_2^\sigma\,\mathrm{Tr}\{\gamma_\mu\gamma_\rho\gamma_\nu\gamma_\sigma\}=4\left[p_1^\mu p_2^\nu+p_1^\nu p_2^\mu-(p_1\cdot p_2)g^{\mu\nu}\right] \tag{16.33}$$

and similarly for the second. In the center-of-mass frame, with $2E=\sqrt{s}$, the only relevant variable is the angle $\theta$ between the incoming electron and the outgoing muon, *e.g.* we write $p_1=E(1,0,0,1)$ and $p_3=E(1,\sin\theta,0,\cos\theta)$, and with some algebra we get

$$\frac{1}{4}\Sigma_s|\mathcal{M}|^2=e^4(1+\cos^2\theta) \tag{16.34}$$

From Eq. 16.17 and using $\alpha=e^2/(4\pi)$, we calculate the differential and total cross-section:

$$\mathrm{d}\sigma=\frac{\alpha^2}{4s}(1+\cos^2\theta)\mathrm{d}\Omega \tag{16.35}$$

$$\sigma=\frac{4\pi\alpha^2}{3s} \tag{16.36}$$

### 16.5.3 Counting Quark Colors

The same calculation applies if we replace the muon with a quark (with mass $M\ll\sqrt{s}$) of charge $Qe$, except the cross section gets multipled by $N_cQ^2$ where $N_c$ is the number of quark colors. The subsequent hadronization of the quark/antiquark is messy but unitary, so the quark-level calculation applies. Consider the ratio

$$R\equiv\frac{\sigma_{e^+e^-\to\mathrm{hadrons}}}{\sigma_{e^+e^-\to\mu^+\mu^-}}=N_c\left[(\tfrac{2}{3})^2+(\tfrac{-1}{3})^2+(\tfrac{-1}{3})^2+(\tfrac{2}{3})^2+(\tfrac{-1}{3})^2+\cdots\right] \tag{16.37}$$

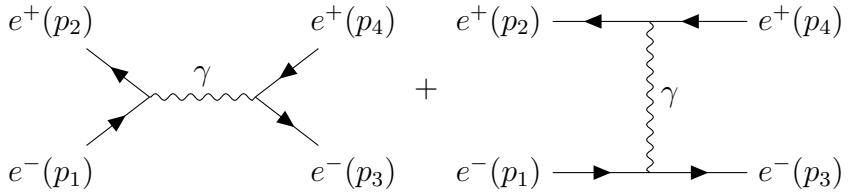
as a function of energy $\sqrt{s}$. The terms in square brackets represent contributions from (u,d,s,c,b) quarks (in order of increasing mass), and you should only include the quarks kinematically allowed for given $\sqrt{s}$. Thus $R$ is piecewise flat, with jumps up as $\sqrt{s}$ passes twice the quark mass for each flavor. A numerical comparison with experiment confirms that $N_c=3$.

### 16.5.4 Bhabha Scattering

**Bhabha scattering** is the process $e^+e^-\to e^+e^-$. The annihilation diagram Eq. 16.29 with an "s-channel" photon contributes, but so does a scattering diagram with a "t-channel" photon.

Analogous to the constructive/destructive interference of the double-slit experiment, you add the amplitudes before squaring, so you need to be particularly careful with signs and $i$'s! We won't do the calculation, but here are the diagrams:

$$\mathcal{M}_{\text{Bhabha}} = \quad + \tag{16.38}$$

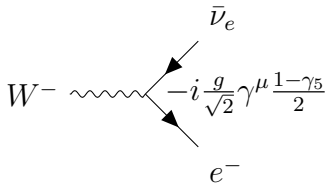



## 16.6 Electroweak Interactions

Recall that the $SU(2)_L$ gauge theory described in Sec. 13.4 introduces 3 new vector bosons. After symmetry breaking and mixing with $U(1)_Y$, these vector bosons $W^\pm$ and $Z^0$ are massive, with $M_W = 80.38\,\text{GeV}$ and $M_Z = 91.19\,\text{GeV}$. The $W^\pm$ bosons change one member of a doublet into the other, *e.g.* an electron into a neutrino. Only left-handed fermions form doublets, so we need a $\frac{1-\gamma_5}{2}$ in the interaction. The relevant portion of the Lagrangian for $W$ interactions is:

$$\mathcal{L} = -\frac{g}{\sqrt{2}}\bar{\nu}_e W_\mu \gamma^\mu \frac{1-\gamma_5}{2} e \tag{16.39}$$

where I have now renamed $u$ and $\bar{v}$ as $e$ and $\bar{\nu}_e$ to show which particles those spinors represent. Here $g = e/\sin\theta_W$ is the weak coupling constant. This gives us a new vertex

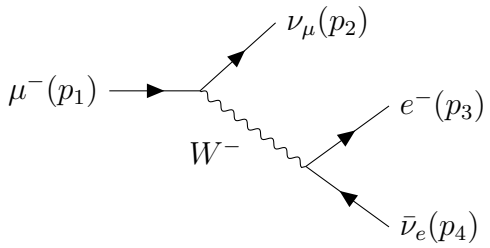



(16.40)

There is also a new vertex involving the $Z^0$ that looks like Eq. 16.27 except for more complicated gamma-matrix structure, but I won't need that here.

### 16.6.1 Muon Decay

The muon decays by the weak process $\mu^- \to \nu_\mu e^- \bar{\nu}_e$, whose diagram looks similar to Eq. 16.29:




(16.41)

Of course since this is a decay, we calculate $\Gamma$ rather than $\sigma$. Incoming 4-momentum $p_1 = (m_\mu, \mathbf{0})$ in the muon rest frame, and this must equal the sum of the outgoing 4-momenta. The photon has been replaced by a $W^-$, so the propagator denominator is dominated by $M_W^2$ rather than $s$

(which is exactly why it's "weak"). We define the **Fermi constant** $G_F \equiv \sqrt{2}g^2/(8M_W^2)$. The gamma-matrix traces work out to:

$$\frac{1}{2}\Sigma_s|\mathcal{M}|^2 = 64G_F^2(p_1\cdot p_4)(p_2\cdot p_3) \tag{16.42}$$

and integrating over final momenta,

$$\Gamma = G_F^2 m_\mu^5/(192\pi^3) = 1/(2.2\,\mu\text{s}) \tag{16.43}$$

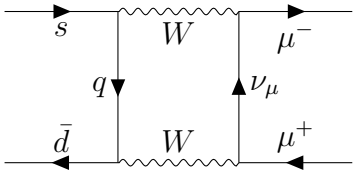
Before we ever knew about $W$'s, Fermi postulated the 4-fermion interaction represented by Eq. 16.41, with coupling strength $G_F/\sqrt{2}$. As long as you work well below the $W$ mass scale, that's a perfectly reasonable **effective theory**, even though it is non-renormalizable. There's a whole science to building up effective theory Lagrangians.

### 16.6.2 Prediction of the Charm Mass

The neutral kaon $K_L$ is, for our purpose, a bound state of a strange quark and a down antiquark (more accurately, $(s\bar{d} + d\bar{s})/\sqrt{2}$). The rare kaon decay mode $K_L \to \mu^+\mu^-$ is a **flavor-changing neutral current** (FCNC) process with quarks running around an internal loop. For this story, we treat up, down, and strange quark masses as small, the charm quark mass $m_c$ is our interest, and top and bottom quarks are too heavy to contribute.

In the limit $m_c \to 0$, diagrams with loops of u,d,c,s cancel out, due to the **GIM mechanism** (named for Glashow, Iliopoulos, and Maiani[26]: when quarks are massless you can rotate your set of quarks from the mass eigenstate basis to the weak eigenstate basis where the cancelation is manifest). So it must be that the $K_L \to \mu^+\mu^-$ decay rate $\Gamma$ is proportional to some power of $m_c$. In 1974, Gaillard and Lee[27] did that calculation:

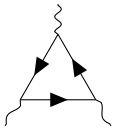
$$\sum_{q\in\{u,d,c,s\}} \quad \sim m_c^n \tag{16.44}$$




to predict a charm mass $m_c \lesssim 2\,\text{GeV}$, soon thereafter confirmed by experiment. They successfully predicted the mass of a particle that had never been seen, based on its suppression of a decay that was rarely seen. Remarkable!

### 16.6.3 The Gauge Anomaly

There is a quantum inconsistency in electroweak theory known as the **gauge anomaly**, aka **triangle anomaly**, aka **local chiral anomaly**, due to this loop:



(16.45)

The calculation is beyond our scope, but this loop breaks gauge symmetry unless $\sum_i Q_i = 0$, where we sum the electric charges of all fermions in the theory. In the real world, this sum does vanish (within each family) when you include the quarks (with multiplicity 3). In a sense, cancellation

of the gauge anomaly requires hydrogen to be electrically neutral. It is a philosophical question whether that *explains* hydrogen neutrality or just reinforces the mystery. When the $\tau$ lepton was discovered (1975), gauge anomaly cancellation strongly suggested a third family of quarks must also exist, later confirmed as the bottom (1977) and top (1995).

Feynman diagrams can only describe perturbations around a vacuum. Non-vanishing **anomalies** that violate a symmetry of the tree-level theory suggest that multiple vacua exist.[28] They cannot be probed with perturbative diagrams, but may be connected by non-perturbative configurations like the sphaleron of Eq. 12.5. These are very relevant to finite-temperature physics, *e.g.* early-universe baryogenesis when the temperature was comparable to the sphaleron mass.

## 16.7 The Higgs Sector

The Higgs field is a complex SU(2)-doublet scalar field with four real degrees of freedom; you could write $\binom{a+bi}{c+di}$, or just $\Phi = (a, b, c, d)$ for our purpose. The Lagrangian is like Eq. 16.11 but with a sign flip on the mass term $m^2 \to -\mu^2$. The classical ground state will have no dependence on space or time, so we just study the potential

$$V = -\frac{1}{2}\mu^2|\Phi|^2 + \frac{\lambda}{4!}|\Phi|^4 \tag{16.46}$$

The "Mexican Hat" potential shown in Fig. 16.1 gives the idea, though it shows only two dimensions instead of four. The potential takes a minimum at $|\Phi| = v = \sqrt{6\mu^2/\lambda}$, but the direction in $\Phi$-space is undetermined. The Higgs field breaks the SU(2) symmetry by choosing a direction at random, and to minimize energy it must be the same choice throughout the universe. Whatever direction that is, we henceforth call it $\Phi = \binom{v+0i}{0+0i}$. The Higgs **vacuum expectation value** (vev) is $v = 246\,\text{GeV}$.

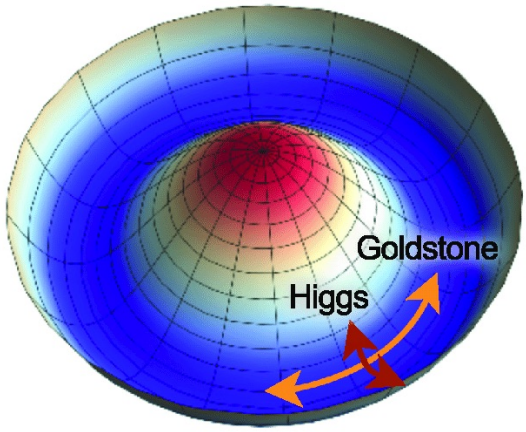



Figure 16.1: Higgs Mexican Hat Potential

The original Lagrangian has an interaction term $\mathcal{L}_i = \lambda_e \Phi \bar{L}_L e_R$, where $L_L = \binom{\nu_e}{e}$ is the left-handed SU(2) lepton doublet containing both neutrino and electron, while $e_R$ is the right-handed electron singlet. When the Higgs takes a vev, this becomes $\mathcal{L}_i = \lambda_e(v + h)\bar{e}_L e_R$, which is both a Dirac mass $m_e = \lambda_e v$ for the electron, and a Higgs-electron interaction vertex $\lambda_e$. Since we know both $v$ and $m_e$, we automatically know the interaction strength $\lambda_e \approx 2 \times 10^{-6}$.

Remember that scalar particles are field fluctuations around a minimum of $V$, indexed with a momentum $\boldsymbol{k}$, and their squared mass is $V''(\Phi)$ at the minimum. There is a lovely theorem

by Goldstone that says when a four-dimensional $\Phi$ takes a non-zero value and breaks a **non-gauge** symmetry, fluctuations about the minimum in the radial direction produce a massive boson, while fluctuations in the axial directions (where the potential is flat) produce three three massless **Goldstone bosons**.[2] There's an even lovelier theorem that says when a **gauge** symmetry is broken, the (massless) gauge bosons "eat" the (massless) Goldstone bosons to become massive. So what we see in real life is a massive Higgs scalar and the massive vector bosons $W^{\pm}$ and $Z^0$.

Back in the 90's, I would have had to put the Higgs field in the "Speculative" chapter. It was clear that symmetry breaking led to massive weak vector bosons $W^{\pm}$ and $Z^0$ that were seen in 1983 at CERN. But some people thought the Higgs mechanism was only a placeholder for a more complicated theory like **Technicolor**. The $W^{\pm}$ and $Z^0$ masses told us that $v = 246\,\text{GeV}$, but did not tell us $\mu$ or $\lambda$ individually, so the Higgs particle mass $m_h^2 = 2\mu^2$ was unknown, and the search for the Higgs (or evidence of Technicolor) lasted for decades. Finally in 2012 CERN found the Higgs scalar particle with mass $m_h = 125\,\text{GeV}$ and all the right properties. It is still possible that the 125 GeV Higgs is part of a larger scalar sector, as would be predicted for example by Supersymmetry.

[2]The theorem is realized, though imperfectly, in chiral symmetry breaking, where the pseudo-Goldstone bosons are the pions $\pi^{\pm}$ and $\pi^0$.

# Chapter 17

# Light May Be... Some Speculative Theories

I hear babies cry / I watch them grow
They'll learn much more / Than I'll ever know
And I think to myself / What a wonderful world

---

*What a Wonderful World*, Louis Armstrong
lyrics by George David Weiss and Robert Thiele

Until now this book has covered established physics, but now we shift gears and look at some speculative theories: Grand Unified Theories (GUT's), Supersymmetry, and String Theory. As a postdoc I worked on the first two in some depth, so this section has a kind of "90's vibe" – when motivation for these ideas was so strong that we were convinced it was just a matter of time (and accelerator funding) before they would be experimentally validated. Unfortunately, that has not come to pass as of this writing (2024). Nevertheless, the motivations remain, and light plays an interesting role in each.

## 17.1 Grand Unified Theories

At the end of Sec. 13.5 we were left wondering how to explain the simple charge ratios of various elementary particles, and specifically why the hydrogen atom (3 quarks and an electron) is exactly neutral (to at least 1 part in $10^{25}$!). What if the three independent symmetries of $SU(3)_c \otimes SU(2)_L \otimes U(1)_Y$ were part of a single larger symmetry, broken by something like a Higgs mechanism? Such a **Grand Unified Theory** (GUT) would require only a single coupling constant instead of three. By unifying quarks and leptons, it could explain why their hypercharges are related, but also would allow one to turn into the other.

Group theory tells us that there is a good candidate in SU(5), and we will briefly describe the SU(5) **Georgi-Glashow model** which was proposed in 1974. It is the original and simplest GUT, combining quantum chromodynamics with electroweak theory, and that would make it the **Fourth Unification**.

As you'd expect, 5 chiral fermions fit into a fundamental representation $\Psi$ that transforms as $\Psi' = U\Psi$, and we call this the $\bar{5}_F$. But there is also another representation, called a $10_F$, that's an antisymmetric tensor that transforms like ??? The known fermions of a single generation fit into

the $\bar{5}_F$ and the $10_F$:

$$\bar{5}_F = \begin{pmatrix} d_R^1 \\ d_R^2 \\ d_R^3 \\ e \\ -\nu_L \end{pmatrix} \quad , \qquad 10_F = \begin{pmatrix} 0 & u_R^3 & -u_R^2 & u_L^1 & d_L^1 \\ & 0 & u_R^1 & u_L^2 & d_L^2 \\ & & 0 & u_L^3 & d_L^3 \\ & & & 0 & e_R \\ & & & & 0 \end{pmatrix} \tag{17.1}$$

where the quark superscripts 1-3 are color indices. If there is a $\nu_R$, it lives separately in an SU(5) singlet. The Standard Model Higgs appears in a $5_H$ representation: $5_H = (T_1, T_2, T_3, H^+, H^0)^T$, where the $T_i$ are predicted new scalars. The gauge bosons, as always, live in an adjoint $24_G$ representation; after breaking they become the gluons, weak bosons, hyperphoton, plus some new $X$ and $Y$ bosons that mediate proton decay. Finally, another set of scalar fields in an adjoint representation $24_H$ break the $SU(5)$ symmetry when they take the vacuum expectation value $v\,\mathrm{diag}[-1/3, -1/3, -1/3, 1/2, 1/2]$.

Presumably the SU(5) symmetry breaking occurs at a much higher energy than the electroweak scale, so there are not a lot of testable predictions for current accelerators. Unfortunately, there are two predictions: proton decay and magnetic monopoles, and no evidence has been seen for either. A lot of effort has been put into figuring out how to evade these apparent failures.

Here's the beautiful thing: that breaking mechanism automatically gives the Standard Model particles their correct hypercharges. There are no arbitrary $q$'s to be assigned here; the $U(1)_Y$ of hypercharge, and ultimately the $U(1)_{EM}$ of electromagnetism, fall out of broken SU(5) in a completely deterministic way. The SU(5) model solves the mystery of why hydrogen is electrically neutral.

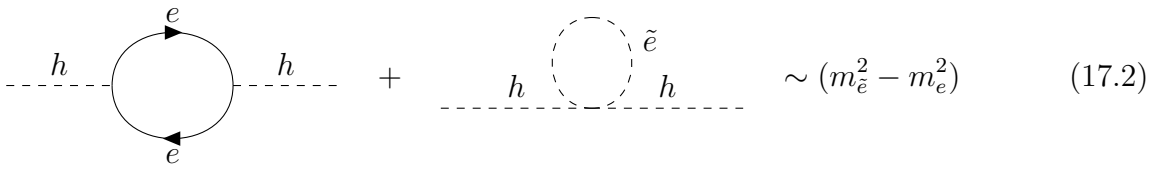
## 17.2 Light May Be the Superpartner of Dark Matter

The **Coleman-Mandula theorem** of 1967 "proved" that spacetime and internal symmetries cannot mix. For example, the generators of $SU(3)_c \otimes SU(2)_L \otimes U(1)_Y$ symmetries are Lorentz scalars. But then the **Wess-Zumino model** of 1974, containing a scalar and a fermion of equal mass, exhibited the first example of **supersymmetry**, a symmetry whose generators transform as spinors, in apparent contradiction to Coleman-Mandula. It was eventually understood that anticommuting generators ("supercharges") provide a loophole to Coleman and Mandula's no-go theorem, which made assumptions about commutation relations. I mention this story because it shows how a well-crafted no-go theorem can contain, in its axioms, the seeds of its own demise. In an alternate history, someone *could have* noticed this loophole and invented supersymmetry in 1967.

Supersymmetry is a symmetry between bosons and fermions, so along with photons (spin 1) it has photinos (spin $\frac{1}{2}$); along with electrons (spin $\frac{1}{2}$) it has selectrons (spin 0), and so on. No such pairs are observed in nature, nor so far (as of 2024) in colliders, so supersymmetry must be broken at some energy scale above the reach of current experiments. Why propose such a theory? It would solve several outstanding theoretical problems:

1. The Higgs boson mass$^2$ gets quadratic corrections from bosons (positive) and fermions (negative). Without a symmetry, it would be wildly "unnatural" that the sum of the bare mass$^2$ and the quadratic corrections would be anywhere as small as $(125\,\mathrm{GeV})^2$. This is the **naturalness problem**. With supersymmetry, there is a cancellation (like Eq. 16.44) between

the electron and selectron contributions to $m_h^2$:




(17.2)

and this is strong reason to expect the supersymmetry scale is not much higher than the weak scale (Higgs mass).

2. The $SU(5)$ Grand Unified Theory requires the 3 low-energy couplings $g_1$, $g_2$, and $g_s$ to run together towards a single value at the GUT scale. Numerically this works better with the additional particles in the spectrum. Also, the higher resulting GUT scale predicts less proton decay, which is good because we haven't seen any proton decay.

3. The lightest unseen superpartner may be the dark matter. In particular, the photino is an excellent candidate. Actually the superpartner neutral fermions (photino, zino, and sneutrino) mix, and the result is **neutralinos**. But the lightest neutralino in many models is mostly photino, so what I said is mostly right.

## 17.3 Kaluza-Klein Theory

In Chapter 14 we showed that electromagnetism and general relativity are both theories of connections. In fact, Kaluza showed that you can obtain both theories at once if you imagine that spacetime is 5-dimensional.

Write down the equation for the metric tensor in 5-dimensional General Relativity. It can be separated into 4D General Relativity, plus a 4D vector which is essentially $A^\mu$, plus a scalar called the **dilaton**. If we assume that the extra spatial dimension is "curled up" into a very small loop, the $A^\mu$ part reproduces Maxwell's Equations. Essentially, "position" in this small and periodic spatial dimension plays the role of "phase" in quantum mechanics.

The dilaton is a problem, so we don't believe Kaluza-Klein theory describes our real world, but the concept of tiny extra spatial dimensions which manifest as forces lives on in string theory. String theory is only mathematically consistent in 10 dimensions, so people have spent a lot of time thinking about how 6 dimensions can curl up (into what are called **Calabi-Yau manifolds**). It would have been great it people had found that there is one "best' Calabi-Yau manifold and that it corresponds to (say) SU(5) symmetry. However, there are actually a LOT of ways 6 dimensions can curl up, and nobody has found a way to derive the Standard Model from them.

## 17.4 The Holographic Principle

In Eq. 11.1 we saw that entropy in traditional thermodynamics is extensive (scales with volume), but in Sec. 11.5 we learned that for black holes it scales with area. It's as though the information content of the black hole resides on its Schwarzschild surface rather than in its bulk. This turns out to be a feature of gravitational systems, known (by analogy to SubSec. 5.4.1) as the **holographic principle**.

In 1997, Juan Maldacena showed that a 5D theory of gravity could be encoded in (was equivalent to) a 4D conformal field theory (CFT) on the surface of the 5D space. Since the space had to be

Anti-de Sitter (AdS, meaning closed and finite), this became known as AdS/CFT correspondence. Type IIB string theory on $\mathrm{AdS}_5 \otimes S_5$ is equivalent to $N$=4 supersymmetric Yang–Mills theory on its boundary. Maldacena's article became the most-cited paper in high-energy physics.

Picture this toy model: a finite spherical space ("bulk") containing a black hole at $r = 0$ and a single particle at $(r, \theta, \phi)$. Now imagine the spherical-shell surface of this universe containing a bound pair of particles, with center of mass at $(\theta, \phi)$, and distance $r$ separating them. Imagine you can make the dynamics of the bound pair on the surface reproduce the behavior of the single particle in the bulk. A being in this world would not know if he really lived in the bulk or on the surface. That's how I picture AdS/CFT.

How can information from 5D spacetime "fit" into a mere 4D surface? As with actual holography, there must be some limit to the "resolution" of information in 5D, imposed by gravity itself. AdS/CFT provided a detailed realization of the more general holographic principle proposed by 't Hooft and Susskind. Applied back to the real world, there is a maximum amount of information that can be stored in a 3D region, and (counterintuitively!) it's proportional to the surface area of that region. A black hole ($S = A/4$) saturates this limit.

## 17.5 Light May Be An Open String

### 17.5.1 Spacetime Coordinates as Fields

Let's re-write basic Newtonian mechanics once again. This time we'll think of a particle's spacetime coordinates $X^\mu$ as fields that live on the particle worldline. Note this is different than before – usually we talk about matter fields that live in spacetime, but now it's the spacetime coordinates that are the fields! The worldline of a particle can be parametrized by one variable $\tau$; one particularly intuitive choice is to take $\tau$ as the proper time (the "internal clock"), but it doesn't have to be. Then the particle's motion is described by $X^\mu(\tau)$. For example, if a particle just sits at the origin forever and we choose $\tau$ to be proper time (same as observer's time), then $X^\mu(\tau) = (\tau, 0, 0, 0)$. For a free particle, these fields obey $\partial_\tau^2 X^\mu(\tau) = 0$, which is Newton's First Law. The fact that they hold for *any* parametrization $\tau$ gives us Lorentz invariance.

A **string** existing through time has a 2-dimensional worldsheet, which can be parametrized by two variables $\tau$ and $\sigma$. Think of them as proper time and position along the string, respectively, though again they can be more general, as long as one is time-y and one is space-y. By analogy, the spacetime coordinate fields for a free string should obey $(\partial_\tau^2 - \partial_\sigma^2)X^\mu(\tau, \sigma) = 0$, which is a wave equation. Thus a free string, in addition to moving through space at fixed velocity (like a particle), can also vibrate. The wave equation allows for both left-moving and right-moving waves.[29]

### 17.5.2 String Theory

A string can be closed or open. In each case, boundary conditions constrain the possible $X^\mu$ waves. We quantize the wave amplitudes, just as we did for the harmonic oscillator, to get creation operators that represent quanta of energy (mass).

For the closed string, periodic boundary conditions require the same number of left- ($\alpha$) and right-moving ($\tilde{\alpha}$) quanta, so the first excited state is $\zeta_{\mu\nu}\alpha^\mu_{-1}\tilde{\alpha}^\nu_{-1}|0; p\rangle$. As it turns out, the ground state $|0; p\rangle$ is tachyonic ($M^2 < 0$) and must be removed, but the first excited state is massless. The two Lorentz indices show it has spin 2. So it's a massless spin-2 particle, and Weinberg showed using the Coleman-Mandula theorem that any such particle has to be the graviton $h^{\mu\nu}$. Closed

strings vibrating in their lightest physical mode are gravitons.

For the open string, boundary conditions constrain left- and right-movers so the first excited state is $\zeta_\mu \alpha^\mu_{-1}|0;p\rangle$. Again this is a massless state (the tachyonic ground state was removed), but of spin 1. Open strings in their lightest physical mode are photons.

The ends of an open string are constrained to some hyperdimensional surface, a "d-brane" (short for "membrane in d dimensions"). If there's just one d-brane, the vector boson (photon) is associated with a $U(1)$ symmetry. If there are $N$ parallel d-branes, you can attach the string ends to different d-branes, and end up with the $N^2$ vector bosons of a $U(N)$ symmetry. With 3 parallel d-branes intersecting 2 other parallel d-branes, you get a $U(3) \otimes U(2)$ theory which is essentially $SU(3) \otimes SU(2) \otimes U(1)$ plus a global $U(1)$ *e.g.* $(B - L)$, baryon number minus lepton number. Fermions come from strings that connect one set of d-branes to another set. So, roughly speaking, the gauge group structure and matter content of the Standard Model can be built up by suitable choices of d-branes that form the open-string boundary conditions.

One, perhaps embarrassing, feature of string theory is that it does not work in $D$=4 dimensions (*i.e.* $\mu = 0..3$). A cancellation is required that only happens for certain values of $D$; in superstring theory we need $D$=10. We already saw in Kaluza-Klein theory that an extra space dimension can be "curled up" and thus hidden away, so here we need to "curl up" 6 dimensions. Mathematicians had already studied the properties of certain curled-up 6-dimensional spaces called **Calabi-Yau manifolds**, and in a sense the possible 4D vacua of string theory are related to the properties of Calabi-Yau manifolds. The unfulfilled dream of string theory is that a better understanding of these curled-up dimensions will lead to the d-brane structure required to produce the gauge structure and particle content of the Standard Model.

There are higher, massive excitations of both the closed and open strings, but the associated masses are huge (Planck scale), so these are not particles seen in our low-energy world. However, the existence of this ladder of excitations is critical to the cancellation of divergences that makes this quantum theory of gravity finite, and that is perhaps string theory's greatest success.

Unification of electroweak and strong forces with gravity is the **Fifth Unification**.

## 17.6 It From Bit

Special Relativity tells us that information cannot travel faster than $c$. Quantum entanglement may have caused you doubt, but as we discussed in Sec. 8.7, you still can't make a superluminal communicator. Gauge theories arise as we promote global symmetries to local, on the theory that information about a transformation must obey causality. Entropy is a measure of information content, and indeed in addition to thermodynamic entropy and black hole entropy, there is **computational entropy**. The black hole information paradox asks where does the information go when matter falls into a singularity that later evaporates. The holographic principle suggests that information lives on the Schwarzschild surface, encoded somehow as a manifold of Planck-sized 2D bits. The **ER = EPR** conjecture proposes that entangled black holes (an Einstein–Podolsky–Rosen pair) are connected by a wormhole (aka Einstein–Rosen bridge), unifying fundamental concepts of gravity and quantum.

All this suggests that **information theory**, causality, and locality may play deeper roles in physics than we realize. John Wheeler called the idea "It from Bit".

# Chapter 18

# Light is Inevitable: Conclusion

> [T]he end of all our exploring will be to arrive where we started and know the place for the first time.
>
> T.S. Eliot

Aristotle notwithstanding, modern physics really began with Newton's insight that the same force of gravity explains both falling apples and planetary orbits. Another great unification came two centuries later, when Maxwell's Equations Eq. 6.3 tied together the forces of electricity and magnetism. Gravity and electromagnetism explain most everyday phenomena, as atoms consist of electrons electromagnetically bound to nuclei, while chemistry derives from residual forces of these bindings. A "strong nuclear" force was needed to explain how protons and neutrons stick together in nuclei (overcoming their Coulomb repulsion), and a "weak nuclear" force was needed to explain radioactivity, for a total of four fundamental forces.

The early 20th century saw the development of quantum mechanics, as waves (like light) displayed particle-like properties, and particles (like electrons) displayed wave-like properties. Quantum mechanics postulates a complex-valued wavefunction, whose squared magnitude gives the probability for finding a particle in a given position or state, while the phase is not directly observable. *Differences* in phases lead to important phenomena like interference, but physics is invariant under an overall phase shift. This global phase invariance can be elevated to a local symmetry by introducing a compensating auxiliary field $A^\mu$. The addition of a kinetic term in the Lagrangian turns this field into the electromagnetic 4-potential, with real observable consequences like Coulomb forces and light. Gauge invariance of $A^\mu$ follows directly from the electron phase rotation symmetry. Electromagnetism is fully explained as a $U(1)$ gauge theory, and the photon is the gauge boson of quantum phase invariance. Arguably, the very nature of quantum mechanics makes the existence of light inevitable.

The early 20th century also brought a radical change in our view of space and time with special relativity, and then (incorporating gravity) general relativity. The Schrödinger view of quantum was incompatible with special relativity, as unitary evolution ensures that particle number never changes, while relativity tells us that matter can convert to energy. Again, the classical field theory of electromagnetism pointed the way towards relativistic quantum field theory, which naturally reconciles quantum theory and special relativity. QFT provides a unifying framework for describing both bosons (forces) and fermions (matter), and explains why they have both wave and particle properties.

As we grew to understand how a gauge theory derives from a symmetry, we learned that all four fundamental forces could be explained this way. Gravity is the gauge theory of local Lorentz

invariance. The strong nuclear force derives from quantum chromodynamics, which is a gauge theory of $SU(3)_c$, the symmetry of the three quark colors, mediated by gluons. The weak nuclear force is an $SU(2)_L$ gauge theory, but with symmetry breaking through the Higgs mechanism. The mixing of the $SU(2)_L \otimes U(1)_Y$ gauge theories led to the theory of electroweak interactions, another successful unification.

Redshifted light from distant galaxies showed that the universe is expanding, implying that it began in a hot, dense state. Thermodynamics, general relativity, and nuclear physics allow us to trace its history back to the Big Bang, and to understand the present matter and light (CMB) densities as remnants of earlier thermodynamic equilibria. Astronomical observations thus constrain our models of particle physics, though the nature of dark matter remains a mystery.

The $SU(3)_c \otimes SU(2)_L \otimes U(1)_Y$ Standard Model, Sec. 13.5, thoroughly explains essentially all non-gravitational phenomena. It has some conceptual weaknesses, though, such as the need to quantize charges by hand, a plethora of input parameters including coupling constants and particle masses, and the uncomfortable quadratic divergences of the Higgs mass. We hope to find a greater symmetry that underlies the Standard Model, such as an $SU(5)$ Grand Unified Theory (GUT) with symmetry breaking, that would remove some of these issues, and maybe provide dark matter as a bonus. Unifying these forces with gravity requires an even greater conceptual leap, such as string theory. The search for physics "beyond the Standard Model" is a search for ever more unifying principles.

# Appendix A

# Physical Constants and Units

Here are a few physical constants in SI (Système International) and CGS units:

| Symbol | Name | SI | CGS |
|---|---|---|---|
| $c$ | Speed of light | $3 \times 10^8\,\mathrm{m/s}$ | $3 \times 10^{10}\,\mathrm{cm/s}$ |
| $k_B$ | Boltzmann's constant | $1.381 \times 10^{-23}\,\mathrm{J/K}$ | $1.381 \times 10^{-16}\,\mathrm{erg/K}$ |
| $\hbar$ | Reduced Planck constant | $1.055 \times 10^{-34}\,\mathrm{kgm^2/s}$ | $1.055 \times 10^{-27}\,\mathrm{gcm^2/s}$ |
| $G$ | Newton's constant | $6.674 \times 10^{-11}\,\mathrm{m^3/(kg\,s^2)}$ | $6.674 \times 10^{-8}\,\mathrm{cm^3/(g\,s^2)}$ |

When it comes to electromagnetism, there are several subvariants of the metric system; see Jackson's Appendix A. Engineers who actually measure current in "amps" and magnetic fields in "tesla" will prefer SI, where the constant $k_c$ that appears in Eq. 6.1 is $k_c = 1/(4\pi\epsilon_0)$, and $\epsilon_0$ is the "permittivity of free space". SI also has a magnetic constant $\mu_0$, the "permeability of free space". If you learn Maxwell's Equations in SI units, you will have the thrill exactly once (as Maxwell did) of discovering that $1/\sqrt{\epsilon_0\mu_0}$ is numerically equal to $c$, but then you have to carry around these two extra constants for the rest of your life.

Theoretical physicists prefer the more elegant CGS Gaussian units. We define a new unit of electric charge, the "esu", to make $k_c = 1$, eliminating the need for $\epsilon_0$. We redefine magnetic field strength to eliminate $\mu_0$, allowing $B$ and $E$ to live together in the 4-tensor $F^{\mu\nu}$. $c$ shows up directly in Maxwell's Equations, ruining the surprise perhaps but emphasizing the inherently relativistic nature of electromagnetism.

If you encounter SI relations, you can convert pretty easily to CGS Gaussian by taking $c = 1$ (in both systems), $\epsilon_0 = 1/(4\pi)$, and $\mu_0 = 4\pi$.

A recurring theme in metrology is that dimensionful physical constants may indicate profound relations between seemingly disparate quantities. Finding the speed of light $c$ embedded in Maxwell's Equations, independent of observer, showed (Chapter 7) that time and space can rotate into one another, so it makes sense to give them the same units. Setting $c = 1$ means that we measure distance in light-seconds rather than meters or centimeters. Since $E = mc^2$, mass and energy then share the same units, such as the electron-volt eV (the energy it takes to move an electron across one volt of potential). For example, the mass of the electron is $511\,\mathrm{keV}$.

In thermodynamics, the mean energy per degree of freedom is $\frac{1}{2}k_BT$. Setting Boltzmann's constant $k_B = 1$ thus gives $T$ units of energy, eliminating the "kelvin". We also learn that entropy is a pure number: the log of the number of possible microstates.

The fundamental constant of quantum mechanics is $\hbar$, with units of angular momentum (which are also the units of action $S$ and of phase space volume $\Delta x \Delta p$). You may find it convenient to memorize $\hbar c = 1973\,\mathrm{eV \cdot Å}$, where the Angstrom $\mathrm{Å} = 10^{-10}\,\mathrm{m}$. When we set $\hbar = 1$, time and

distance now take on units of $\mathrm{eV}^{-1}$, that is, the inverse of energy. Angular momentum becomes a dimensionless number, and the "quantum" aspect is that it is often an integer (or half-integer). Also electric charge becomes dimensionless; if $e$ is the absolute charge on a proton or electron, we find $\alpha \equiv e^2/(\hbar c) = 1/137$ (the "fine structure constant"). The phrase "natural units" means $c = \hbar = 1$.

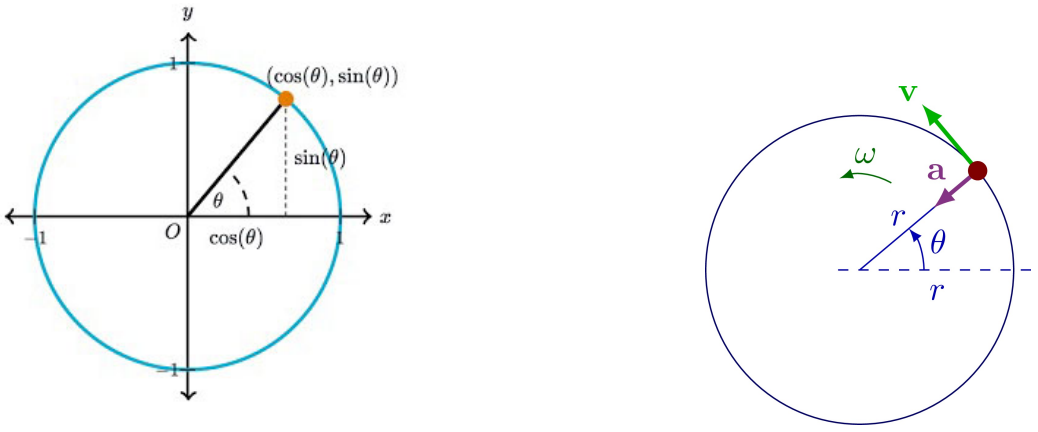
If we keep going and set Newton's constant $G = 1$, all human-defined units disappear. Everything is measured in units of Planck time, Planck distance, Planck energy. These are not very convenient unless you're doing quantum gravity, so maybe we won't go that far. Arguably the most bizarre thing about the universe is that it is so much larger (in time and space) than the Planck time (or length), by about $10^{61}$.

# Appendix B

# Math

## B.1 Trigonometry and Circular Motion

Choose a point on the unit circle (Fig. B.1a) an angle $\theta$ from the x-axis. Its coordinates are $(x = \cos\theta, y = \sin\theta)$. We generally measure $\theta$ in radians, where a full circle is $2\pi$ radians (same as 360°). Pythagoras tells us $\cos^2\theta + \sin^2\theta = 1$. We define $\tan\theta \equiv \sin\theta/\cos\theta$.




(a) The Unit Circle

(b) Circular Motion

Figure B.1: The Unit Circle, and Circular Motion

**Uniform circular motion** occurs if $\theta = \omega t$, Fig. B.1b (no longer requiring $r = 1$ here). $\omega = v/r$ is the angular velocity. For *any* vector $\boldsymbol{z}$ that rotates at $\omega$ without changing magnitude $z$, its derivative has magnitude $\omega z$ and points 90° counterclockwise of $\boldsymbol{z}$. When $\boldsymbol{z} = \boldsymbol{r}$ (position), the velocity $v = \omega r$ as we already knew. Applying that rule again to $\boldsymbol{z} = \boldsymbol{v}$, the acceleration $a = \omega v = \omega^2 r$ and points inward. That's why a central attractive force can produce a circular orbit.

## B.2 Complex Numbers

A complex number $z = x + iy = re^{i\theta}$, where $i^2 = -1$ by definition. We say $x = \Re z$ (the "real part") and $y = \Im z$ (the "imaginary part"); and also $r = |z|$ (the "magnitude" or "norm"), and $\theta$ is the "phase". $z$ can be naturally plotted on the plane, with Cartesian coordinates $(x, y)$ and

radial coordinates $(r, \theta)$. Now $(-i)^2 = -1$ as well, so the whole structure of complex algebra could have been built instead with $i \to -i$, an operation called **complex conjugation** and denoted by a star: $z^* = x - iy = re^{-i\theta}$. Note that $x = (z + z^*)/2$, $y = (z - z^*)/(2i)$, and $r = \sqrt{zz^*}$.

As a curiosity we note the existence of **quarternions** which have 3 new elements satisfying $i^2 = j^2 = k^2 = -1$. The relations between them are $ij = k$ but $ji = -k$, and cyclic permutations. They are tempting tools for quantum mechanics because of their anticommutivity, and for relativity because of their similarity to space and time dimensions.

Octonions have 7 new numbers, and are of interest in group theory. Compared to the reals, complex numbers lack ordering (there's no meaning to $z_1 < z_2$), quarternions also lack commutivity, and octonions also lack associativity.

## B.3 Differential Calculus

Given a function $f(x)$, its derivative $\frac{\mathrm{d}f}{\mathrm{d}x}$ represents the slope of $f(x)$ at any given $x$, *i.e.* how much it changes $\mathrm{d}f$ over a small interval, divided by the size of the interval $\mathrm{d}x$. Since $\frac{\mathrm{d}f}{\mathrm{d}x}$ is itself a function, we can take its derivative as well, and we call that the second derivative (or curvature) $\frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f}{\mathrm{d}x}\right)$. By convention we often write $f' \equiv \frac{\mathrm{d}f}{\mathrm{d}x}$ and $f'' \equiv \frac{\mathrm{d}^2 f}{\mathrm{d}x^2}$. If the independent variable is $t$ (time), we sometimes write $\dot{f} \equiv \frac{\mathrm{d}f}{\mathrm{d}t}$ and $\ddot{f} \equiv \frac{\mathrm{d}^2 f}{\mathrm{d}t^2}$ instead. If $f$ is a function of several variables, use the partial derivative $\frac{\partial f}{\partial x}$.

The **chain rule** says that $\frac{\mathrm{d}}{\mathrm{d}x}(fg) = f'g + fg'$. There's also a usefule version for ratios: $\frac{\mathrm{d}}{\mathrm{d}x}(f/g) = (f'g - fg')/g^2$.

Here's a second-order differential equation and its (real) solution:

$$f'' = -k^2 f \quad : \quad f(x) = A\sin(kx) + B\cos(kx) \tag{B.1}$$

The solution could be re-written as $f(x) = A\sin(kx + \phi)$, but either way there are two constants to be determined from boundary conditions. Here's another:

$$f'' = \kappa^2 f \quad ; \quad f(x) = Ce^{\kappa x} + De^{-\kappa x} \tag{B.2}$$

Of course, those two examples are the same if $\kappa = ik$, and you can derive one solution from the other using the Euler identity $e^{ikx} = \cos(kx) + i\sin(kx)$.

### B.3.1 Grad, Div, Curl

The nabla $\nabla$ is a vector differential operator $(\partial_x, \partial_y, \partial_z)$ in 3 dimensions. where $\partial_x \equiv \mathrm{d}/\mathrm{d}x$, etc.

The gradient $\nabla\phi$ of a scalar function $\phi$ is a vector of its derivatives, having the components $(\partial_x\phi, \partial_y\phi, \partial_z\phi)$. At any point in space, $\nabla\phi$ represents the direction in which $\phi$ is increasing most rapidly. Mathematicians call such a construct (a vector defined at each point in space) a **tangent vector** on $\mathbb{R}^3$.

The divergence of a vector field $\nabla \cdot \boldsymbol{E}$ is a scalar function of $\boldsymbol{x}$, with value $\nabla \cdot \boldsymbol{E} = \partial_x E_x + \partial_y E_y + \partial_z E_z$. It's the extent to which vectors "diverge" (or radiate from) from a point. Electric fields, for instance, diverge from charges, and nowhere else.

The curl of a vector field $\nabla \times \boldsymbol{B}$ is a vector function of $\boldsymbol{x}$, with components

$$\nabla \times \boldsymbol{B} = (\partial_y B_z - \partial_z B_y,\ \partial_z B_x - \partial_x B_z,\ \partial_x B_y - \partial_y B_x) \tag{B.3}$$

It's the extent to which vectors circulate around a point. Magnetic fields, for instance, curl around currents.

It's useful to know that the curl of a gradient vanishes: $\nabla \times (\nabla\phi) = 0$. You can prove it by just writing out the definitions, but here's some intuition. The gradient $\nabla\phi$ points in the direction that $\phi$ increases most rapidly, and its value is that maximal rate of increase. The curl measures how much a vector circulates around a point. A positive curl of the gradient would require that $\phi$ be increasing all the way around a closed loop, but of course it cannot because $\phi$ must return to its starting value.

The divergence of the curl also vanishes: $\nabla \cdot (\nabla \times \boldsymbol{A}) = 0$.

The Laplacian $\nabla^2 = \nabla \cdot \nabla = \partial_x^2 + \partial_y^2 + \partial_z^2$ is a scalar operator that can act on scalars or vectors. For example, $\nabla^2\phi = \partial_x^2\phi + \partial_y^2\phi + \partial_z^2\phi$.

Choose any closed surface $S$ that encloses a volume $V$. The **divergence theorem** states that $\int_V (\nabla \cdot \boldsymbol{E})\mathrm{d}V = \int_S (\boldsymbol{E} \cdot \hat{n})\mathrm{d}S$. The left side is a volume integral of the divergence, and the right side is a surface integral of the **flux** (the component of $\boldsymbol{E}$ normal to the surface). This theorem relates the differential and integral forms of Gauss's Law.

In 4 spacetime dimensions, similar operators appear, but the Einstein summation convention makes the nabla notation unnecessary. 4-gradient is $\partial_\mu\phi$. 4-divergence is $\partial_\mu A^\mu$. There is no curl in 4D. The Laplacian is replaced by the **d'Alembertian** operator $\Box\phi \equiv \partial^\mu\partial_\mu\phi$.

### B.3.2 Green's Function

The tools of linear algebra, Appendix B.6, apply to linear differential operators like the Laplacian $\nabla^2$ (an operator $L$ is *linear* if $L(af + bg) = aL(f) + bL(g)$, where $f$ and $g$ are functions). That's explicit when you discretize a finite region of 1-dimensional space, so that $f(x)$ is really just a finite vector, but it's also true for the infinite-dimensional **Hilbert space** of continuous functions on 3D space. The inverse $L^{-1}$ of a linear differential operator is called its **Green's function** $G$, defined by $LG(\boldsymbol{x}) = \delta^3(\boldsymbol{x})$, analogous to the matrix equation $AA^{-1} = I$.

We illustrate with the example $L = \nabla^2$, for which the Green's function $G(\boldsymbol{x}) = \frac{-1}{4\pi r}$, with the usual shorthand $r = |\boldsymbol{x}|$. That is, $\nabla^2\left(\frac{-1}{4\pi r}\right) = \delta^3(\boldsymbol{x})$. Recall Poisson's equation from electrostatics $\nabla^2\Phi = -4\pi\rho$, see Eq. 6.5. If someone hands you $\Phi$ then it's easy to calculate $\rho$, but what if they hand you $\rho$ and ask for $\Phi$? The answer is the **convolution**:

$$\Phi(\boldsymbol{x}) = G * (-4\pi\rho) = \int \mathrm{d}^3\boldsymbol{y}\frac{\rho(\boldsymbol{y})}{|\boldsymbol{x} - \boldsymbol{y}|} \tag{B.4}$$

which is no surprise, it's just the sum of Coulomb potentials $Q/r$ from every charge. The general rule is that if $L\phi = f$ then $\phi = G * f$.

## B.4 Integral Calculus

Integration is the inverse of differentiation. If $g(x) = \frac{\mathrm{d}}{\mathrm{d}x}f(x)$, then $f(x) = \int g(x)\mathrm{d}x$. Sometimes we write that as $f(x) = \int \mathrm{d}x\, g(x)$, whichever way seems clearer. Since you can add any constant $C$ to $f(x)$ without changing its derivative, sometimes we say the **indefinite integral** is $f(x) + C$.

If you put limits on the integral sign, you get the **definite integral** $\int_a^b g(x)\mathrm{d}x = f(b) - f(a)$. This is also the area under the curve $g(x)$ from $x = a$ to $x = b$.

Honestly a lot of integrals are done just by recognizing $g(x)$ as the derivative of some $f(x)$, maybe with a clever change of variables. For example, $\int x^n\mathrm{d}x = \frac{1}{n+1}x^{n+1}$ which you can see just by differentiating the right-hand side.

The technique of **integration by parts** uses the chain rule. If you find yourself with a definite integral of the form $\int_a^b u\,\mathrm{d}v$ for some functions $u(x)$ and $v(x)$, and you know instead how to do $\int_a^b v\,\mathrm{d}u$, then use

$$\int_a^b u\,\mathrm{d}v = (uv)|_a^b - \int_a^b v\,\mathrm{d}u \tag{B.5}$$

where $(uv)|_a^b \equiv u(b)v(b) - u(a)v(a)$.

### B.4.1 The Gaussian Integral

We will show that the **Gaussian integral**

$$I = \int_{-\infty}^{\infty} e^{-ax^2}\mathrm{d}x = \sqrt{\frac{\pi}{a}} \tag{B.6}$$

The strategy is to write $I^2$, using a second dummy variable $y$, then convert the resulting 2D Cartesian integral to polar coordinates:

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{d}x\,\mathrm{d}y\; e^{-a(x^2+y^2)} = 2\pi\int_0^{\infty} re^{-ar^2}dr = \frac{\pi}{a} \tag{B.7}$$

A normalized Gaussian probability distribution with mean $\mu$ and standard deviation $\sigma$ is then

$$\phi(x) = \frac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/(2\sigma^2)} \tag{B.8}$$

By completing the square we get the more general form:

$$\int_{-\infty}^{\infty} e^{-(ax^2+bx+c)}\,\mathrm{d}x = \sqrt{\frac{\pi}{a}}\exp\left(\frac{b^2}{4a} - c\right) \tag{B.9}$$

In general $a$, $b$, and $c$ can be complex. If $x$ is an $n$-vector and $a$ is an $n\times n$ matrix, you can rotate to the diagonal basis where $a = \mathrm{diag}[a_{ii}]$, solve $n$ independent integrals, and then use the fact that $\Pi_i a_{ii} = \mathrm{Det}(a)$ which is invariant under rotation:

$$\int_{\mathbb{R}^n} e^{-x^T ax}\,\mathrm{d}^n x = \sqrt{\frac{\pi^n}{\mathrm{Det}(a)}} \tag{B.10}$$

and more generally,

$$\int_{\mathbb{R}^n} e^{-(x^T ax+bx+c)}\,\mathrm{d}^n x = \sqrt{\frac{\pi^n}{\mathrm{Det}(a)}}\exp\left(\tfrac{1}{4}b^T a^{-1}b - nc\right) \tag{B.11}$$

With the substitutions $a = -iA/2$, $b = -iJ$, and $c = 0$, we get this form useful for field theory:

$$\int_{\mathbb{R}^n} e^{\frac{i}{2}x^T Ax+iJx}\,\mathrm{d}^n x = \sqrt{\frac{(2\pi i)^n}{\mathrm{Det}(A)}}\,e^{-\frac{i}{2}JA^{-1}J} \tag{B.12}$$

If $A$ is a linear differential operator instead of a matrix, then $A^{-1}$ is its Green's function; see Appendix B.3.2.

By differentiating $I$ with respect to $a$ multiple times, we find

$$\int_{-\infty}^{\infty} x^{2n}e^{-ax^2}\mathrm{d}x = \sqrt{\frac{\pi}{a}}\frac{(2n-1)!!}{(2a)^n} \tag{B.13}$$

where the **double factorial** $(2n-1)!! \equiv (2n-1)(2n-3)\cdots 5\cdot 3\cdot 1$.

## B.5 Complex Analysis and Cauchy's Residue Theorem

Not every complex function $f(z)$ is differentiable, but the ones of interest in physics in are usually infinitely differentiable ("analytic") almost everywhere.

A **meromorphic function** has only a finite set of poles, which are points where the function goes to infinity ("blows up"), and I will further restrict the discussion to simple poles $z_0$, near which the function has a Laurent series:

$$f(z) = b/(z - z_0) + a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \dots \tag{B.14}$$

and $b$ is called the "residue" at simple pole $z_0$.

You can integrate a function $f(z)$ from $z_1$ to $z_2$ just like for real numbers. In general, you have to specify the integration path in the complex plane. For everywhere-analytic (or "entire") functions, the integral is path-independent, and therefore the integral along any closed path ($z_2 = z_1$) is zero, because you could shrink that path to a point. For meromorphic functions, a closed path integral depends entirely on which poles are enclosed (counterclockwise by convention). Cauchy's Residue Theorem states that the closed-path integral equals $2\pi i$ times the sum of the residues at the enclosed poles.

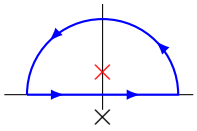
For example, consider $f(z) = 1/z$ which has a simple pole at $z = 0$ with residue 1. The integral of $f(z)$ around any path that encloses the origin (counterclockwise) is therefore $2\pi i$. If the path does not enclose the origin, the integral is 0. You can see this explicitly by choosing the unit circle as your path, along which $f = e^{-i\theta}$ and you run $\theta$ from 0 to $2\pi$.

We can see the power of this theorem by calculating

$$I \equiv \int_{-\infty}^{\infty} \frac{\cos x}{x^2 + a^2}\,\mathrm{d}x \tag{B.15}$$

First we notice that it equals the complex closed contour integral

$$I = \oint \frac{e^{iz}}{z^2 + a^2}\,\mathrm{d}z \qquad \text{over} \tag{B.16}$$



because the imaginary part vanishes by symmetry, and the integral along the large semicircle at $|z| = R$ vanishes as $R \to \infty$. The contour (blue) encloses only the pole (red) at $z = ia$, whose residue is $e^{-a}/(2ia)$, so the result is $I = (\pi/a)e^{-a}$. I challenge you to obtain this answer any other way!

## B.6 Linear Algebra

### B.6.1 States and Bases

A **vector space**[30] $V$ is a set of vectors $|\alpha\rangle$ such that $(a|\alpha\rangle + b|\beta\rangle)$ is also in the set, for any numbers $a$ and $b$. If $a$ and $b$ are complex $(a, b) \in \mathbb{C}$, then $V$ is a complex vector space. The particular choice $a = b = 0$ gives the zero vector $|0\rangle$. If you can define an **inner product** $\langle\alpha|\beta\rangle$ with certain properties, $V$ is a **Hilbert space**. In Dirac's **bra-ket notation**, we call $\langle\alpha|$ a "bra" and $|\alpha\rangle$ a "ket". The **norm** of a vector $|\alpha| = \sqrt{\langle\alpha|\alpha\rangle}$, which is always a positive real number (or zero for $|0\rangle$).

If it takes a **basis** of $n$ linearly independent vectors $|u_i\rangle$ to span the space, so that any vector $|\alpha\rangle$ can be written as $a_1|u_1\rangle + \cdots + a_n|u_n\rangle$, then $V$ is $n$-dimensional. When $V$ is a Hilbert space, you can always find an orthonormal basis such that $\langle u_i|u_j\rangle = \delta_{ij}$. In that basis, you can represent $|\alpha\rangle$ as a column $n$-vector of the coefficients $a_i$, and $\langle\alpha|$ as a row $n$-vector of their complex conjugates $a_i^*$. For example with $n = 2$, ket $|\alpha\rangle$ can be represented as $\binom{a_1}{a_2}$, and its adjoint bra $\langle\alpha|$ as a row vector $(a_1^*, a_2^*)$.

Quantum states $|\psi\rangle$ form a complex Hilbert space. $|0\rangle$ is the "ground" or "vacuum" state. $|\langle\phi|\psi\rangle|^2$ is the probability that a system prepared in state $|\psi\rangle$ will be measured to be in state $|\phi\rangle$ (Born rule), so we usually normalize states $|\langle\psi|\psi\rangle|^2 = 1$. That is, the quantum state of one particle has unit norm, which is preserved by unitary evolution.

### B.6.2 Linear Operators

A **linear operator** $T$ maps any vector in $V$ to another, with the property $T(a|\alpha\rangle + b|\beta\rangle) = aT|\alpha\rangle + bT|\beta\rangle$. Operator multiplication is defined by $(ST)|\alpha\rangle = S(T|\alpha\rangle)$, but it is not necessarily commutative ($ST \neq TS$), and we define the commutator $[S, T] = ST - TS$. In a chosen basis, a linear operator is an $n \times n$ square matrix, and the equation $T|\alpha\rangle = |\beta\rangle$ becomes $n$ equations in $n$ unknowns. For $n = 2$ we have:

$$\begin{aligned} T_{11}a_1 + T_{12}a_2 &= b_1 \\ T_{21}a_1 + T_{22}a_2 &= b_2 \end{aligned} \qquad \text{or} \qquad \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \tag{B.17}$$

(Linear algegra courses often start with this.) If $T$ is non-singular, the solution is $|\beta\rangle = T^{-1}|\alpha\rangle$, where we have introduced the **inverse** satisfying $TT^{-1} = T^{-1}T = I$, and the identity operator (unit matrix) $I$ defined by $I|\alpha\rangle = |\alpha\rangle$.

If $T\langle\alpha| = \langle\beta|$, then $|\beta\rangle = |\alpha\rangle T^\dagger$, thus defining the **Hermitian conjugate** $T^\dagger$. In a given basis, "dagger" is achieved by transposing and taking the complex conjugate, or $T^\dagger = (T^*)^T$. A **Hermitian operator** $H$ is one for which $H = H^\dagger$. Quantum observables are always represented by Hermitian operators, because they are "real" (more accurately, all their eigenvalues are real). Note that if $H$ is Hermitian, then $U = e^{-iH}$ is unitary.

### B.6.3 Eigenvalues, Eigenvectors, Trace and Determinant

If $A|\alpha\rangle = \lambda|\alpha\rangle$, then $|\alpha\rangle$ is an **eigenvector** of $A$, with **eigenvalue** $\lambda$. If a single $\lambda$ is associated with several linearly independent eigenvectors, it is conventional to choose an orthogonal set, and given that convention, all of an operator's eigenvectors are orthogonal and form a basis (the "diagonal" basis). In a particular basis, you find a matrix's eigenvalues by solving $\mathrm{Det}(A - \lambda I) = 0$. In the diagonal basis, $A = \mathrm{diag}[\lambda_i]$, meaning it is all zeroes except for the eigenvalues $\{\lambda_i\}$ along the main diagonal.

The **trace** $\mathrm{Tr}\, A$ of a matrix is the sum of its diagnonal elements. The **determinant** $\mathrm{Det}\, A$ is a more complicated construction I won't detail here, but in the special case that $A$ is diagonal, the determinant is the product of the diagonal elements. Both trace and determinant are invariant under rotations or unitary transformations, so we can calculate them easily in the diagonal basis: $\mathrm{Tr}\, A = \sum_i \lambda_i$ and $\mathrm{Det}\, A = \prod_i \lambda_i$.

### B.6.4 Infinite-Dimensional Hilbert Space

A Hilbert space may be infinite-dimensional. For example, the set of position eigenstates $|x\rangle$ is indexed by the real variable $x$ (which is both infinitely dense and infinite in extent). We may imagine rendering it finite by discretizing and confining the system to a box of length $L$, but that is not necessary. We can replace the column vector of coefficients $a_i = \langle u_i|\alpha\rangle$ by the function $\psi(x) = \langle x|\psi\rangle$. The inner product of states becomes $\langle\phi|\psi\rangle = \int \phi^*(x)\psi(x)\,\mathrm{d}x$. Linear operators can contain derivatives like $\hat{p} = -i\hbar\partial_x$. Inverses become Green's functions.

## B.7 Gamma Matrices

The gamma matrices form a **Clifford algebra** $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$, where $\{A, B\} \equiv AB + BA$ is the anticommutator and $g^{\mu\nu}$ is the flat Minkowski metric. We also[1] define $\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$. The $\gamma^\mu$, and also $\gamma^5$, are Hermitian and traceless.

All $4 \times 4$ spin matrices are spanned by the 16-element set $\Gamma \equiv \{I_4, \gamma^\mu, \sigma^{\mu\nu}, \gamma^\mu\gamma^5, \gamma^5\}$, where $\sigma^{\mu\nu} \equiv \frac{i}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)$. When sandwiched between a cospinor and a spinor $\bar{u}\Gamma u$, they form a Lorentz scalar, vector, tensor, pseudovector, and pseudoscalar, respectively.

The gamma matrices in Eq. 9.21 are the Dirac basis, appropriate for Dirac spinors that look like Eq. 16.18 in the rest frame. There are also degenerate forms with half the degrees of freedom ($\binom{\chi}{\chi}$ instead of $\binom{\chi}{\eta}$): Majorana spinors for a fermion that is its own antiparticle; and Weyl spinors for chirality eigenstates, such as the left-handed neutrino $\nu_L$.

Some useful gamma matrix traces are:

$$\mathrm{Tr}\{\gamma^\mu\gamma^\nu\} = 4g^{\mu\nu} \tag{B.18}$$

$$\mathrm{Tr}\{\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\} = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho}) \tag{B.19}$$

$$\mathrm{Tr}\{\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\} = -4i\varepsilon^{\mu\nu\rho\sigma} \tag{B.20}$$

## B.8 Lie Groups and Lie Algebras

On a real $n$-dimensional vector space, a rotation matrix $R$ is orthogonal $R^TR = RR^T = I$; it preserves the length of vectors and can be thought of as a change of basis. A proper rotation (no mirror flip) has the added requirement $\mathrm{Det}(R) = 1$, and the proper rotations form a Lie group called $SO(n)$. A **group** is a set of objects with an associative multiplication operator ($R_1 \cdot R_2$ is another element of the group), an identity ($I$), and an inverse (here $R^{-1} = R^T$). A **Lie group** is a group characterized by continuous parameters (like $\theta$ and $\phi$).

Analogously on a complex $n$-dimensional vector space, a unitary matrix $U$ satisfying $U^\dagger U = UU^\dagger = I$ preserves the norm of vectors. A special unitary matrix has the added requirement $\mathrm{Det}(U) = 1$, and they form a Lie group called $SU(n)$.

Any element of a Lie group can be written $U = \exp\{i\alpha_k H_k\}$ where the $H_k$ (group generators) are Hermitian matrices, and summation over $k$ is implied. The $H_k$ form the basis for a new vector space. The commutation relations $[H_a, H_b] = if_{abc}H_c$, where $f_{abc}$ are **structure constants**, define the **Lie algebra**.

A group is defined by a set of multiplication rules. A **representation** of that group is a realization of those rules by a particular set of matrices. A representation is **reducible** if it (or an

[1] In the old days, time was component "4" instead of "0", so "5" was kind of the next available index.

equivalent representation) is block-diagonal; we restrict our interest to **irreducible representations**, aka "irreps". A representation's dimension is the dimension of the vector space on which the matrices act. For $SU(n)$, the **fundamental representation** consists of the $U$'s and has dimension $n$, while the **adjoint representation** consists of matrices $T_a$ defined by $(T_a)_{bc} = -if_{abc}$ and has dimension $(n^2-1)$. Other irreducible representations can be found using **Young Tableaux**, which we won't discuss, but we often name them by their dimensionality. For example, $SU(5)$ has a fundamental representation $\underline{\mathbf{5}}$, an adjoint representation $\underline{\mathbf{24}}$, and also a conjugate $\overline{\mathbf{5}}$ for antiparticles, a $\underline{\mathbf{10}}$, and others.

Take $SU(2)_L$ for example, weak isospin symmetry. The group elements are $2 \times 2$ unitary matrices with unit determinant. The three generators are traceless $2 \times 2$ Hermitian matrices, conventionally taken to be $\sigma_k/2$ where $\sigma_k$ are the **Pauli matrices**. The structure constants $f_{abc} = \varepsilon_{abc}$, the **Levi-Civita symbol**. A weak isospin doublet $\binom{\nu}{e}$ forms a $\underline{\mathbf{2}}$ fundamental representation. The three $W$ vector bosons form the $\mathbf{3}$ adjoint representation (after electroweak mixing these become $W^{\pm}$ and $Z^0$).

Take $SU(3)_c$ for example, the color symmetry among quarks. The group elements are $3 \times 3$ unitary matrices with unit determinant. The eight generators are traceless $3\times3$ Hermitian matrices, conventionally taken to be $\lambda_k/2$ where $\lambda_k$ are the **Gell-Mann matrices** (this Wikipedia page also gives the $f_{abc}$'s). There are 3 colors of quarks forming a $\underline{\mathbf{3}}$ fundamental representation. Anti-quarks form a $\overline{\mathbf{3}}$. There are 8 gluons forming the $\mathbf{8}$ adjoint representation.

All simple Lie algebras are either $SO(n)$, $SU(n)$, $Sp(2n)$ (symplectic group), or one of five exceptional groups ($G_2$, $F_4$, $E_6$, $E_7$, or $E_8$). The proof[31] involves Dynkin diagrams which we will not discuss. Groups can be subgroups of others, for example $SU(5) \subset SO(10) \subset E_6$.

# Appendix C

# Proof of Noether's Theorem

I don't love formal proofs, but this one is simple and powerful. We present it in the context of relativistic field theory, and assume the Lagrangian density $\mathcal{L}$ contains only up to first derivatives of $\Phi$. Recall that the classical path is the one where the action $S = \int \mathcal{L} \mathrm{d}^4x$ is invariant under ***any*** small change $\delta\Phi$, which gives us the relativistic Euler-Lagrange equation 10.7, repeated here:

$$\frac{\partial \mathcal{L}}{\partial \Phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial \left(\partial_\mu \Phi\right)} = 0 \tag{C.1}$$

Suppose the action is invariant **everywhere** under some **particular** transformation, whose infinitesimal form is $\delta\Phi = \epsilon\,\psi$. Then the Lagrangian must change by a total derivative $\delta\mathcal{L} = \epsilon\,\partial_\mu \Lambda^\mu$ (and often we just have $\Lambda^\mu = 0$). We have

$$\begin{aligned} \delta\mathcal{L} &= \frac{\partial \mathcal{L}}{\partial \Phi}\delta\Phi + \frac{\partial \mathcal{L}}{\partial(\partial^\mu \Phi)}\delta(\partial^\mu \Phi) \\ &= \partial_\mu \left( \frac{\partial \mathcal{L}}{\partial \left(\partial_\mu \Phi\right)} \right) \epsilon\psi + \frac{\partial \mathcal{L}}{\partial(\partial^\mu \Phi)} \partial^\mu \left(\epsilon\psi\right) \\ &= \epsilon\,\partial_\mu \left( \frac{\partial \mathcal{L}}{\partial \left(\partial_\mu \Phi\right)} \psi \right) \end{aligned} \tag{C.2}$$

In the second line we used the Euler-Lagrange equation on the first term, and switched the order $\delta(\partial^\mu\Phi) \to \partial^\mu(\delta\Phi)$ on the second term. The third line combines the two terms using the chain rule. Setting $\delta\mathcal{L} = \epsilon\partial_\mu\Lambda^\mu$ as promised, we find the **continuity equation** (aka **conservation law**):

$$\partial_\mu J^\mu = 0 \quad \text{where} \quad J^\mu \equiv \Lambda^\mu - \frac{\partial \mathcal{L}}{\partial \left(\partial_\mu \Phi\right)} \psi \tag{C.3}$$

and $J^\mu$ is the conserved Noether current.

If you want to reconstruct the symmetries of quantum mechanics, just set $D$=0, so the only derivatives are time derivatives. $\partial_t J = 0$ then just means that $J$ is a constant of the motion. Invariance to time- and space-translations give conservation of energy and momentum, respectively. Rotational invariance gives angular momentum conservation. Invariance to the electron's quantum phase gives conservation of electric charge.

Returning to $D$=3 field theory, translation invariance $\psi = \partial_\nu \Psi$ leads to conservation of the stress-energy tensor $T^\mu_\nu$. Invariance of the quantum phase $\psi = i\Psi$ leads to conservation of charge $J^\mu = ie[(\partial^\mu\Psi)\Psi^*]$.

You may note that Noether's Theorem applies only on-shell, because it uses the classical equation of motion. The quantum analog is the (generalized) **Ward-Takahashi identity**.

Let's recap. Quantum states evolve under a unitary operator $|\psi(t)\rangle = U(t)|\psi(0)\rangle$, and (for a time-independent Hamiltonian) $U = e^{-iHt}$ where the Hamiltonian $H = T + V$. Feynman told us that the amplitude for an initial state $|i\rangle$ to evolve to a final state $|f\rangle$ is $\langle f|e^{-iHt}|i\rangle = \int D[\Phi]\, e^{iS}$ where the action $S = \int \mathcal{L} \mathrm{d}^4x$ and the Lagrangian $\mathcal{L} = T - V$. In the classical regime $S \gg \hbar$, most paths cancel out due to the rapidly varying phase, so the Method of Stationary Phase tells us that the main contribution comes from the **classical path** on which $S$ is a minimum. This Principle of Least Action gives us the Euler-Lagrange equation, Eq. 10.7. A symmetry of the system is any tranformation $\Phi \to \Phi'$ under which the action is invariant, and (by making use of the Euler-Lagrange equation) this defines a conserved current $J^\mu$. Translation, rotation, and phase invariances lead to the conservation of energy, momentum, angular momentum, and electric charge.

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